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31 July 1963

MILITARY STANDARD

STANDARDS FOR LONG HAUL COMMUNICATIONS

EQUIPMENT TECHNICAL DESIGN STANDARDS FOR ANALOG END INSTRUMENTS AND CENTRAL OFFICE ANCILLARY DEVICES

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DEPARTMENT OF DEFENSE
WASHINGTON, DC 20301

STANDARDS FOR LONG HAUL COMMUNICATIONS

EQUIPMENT TECHNICAL DESIGN STANDARDS FOR
ANALOG END INSTRUMENTS AND CENTRAL OFFICE ANCILLARY DEVICES

MIL-STD-188-346

1. This Military Standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Recommended corrections, additions, or deletions should be addressed to Commander, US Army Communications-Electronics Engineering Installation Agency, ATTN: ACCC-CED-STS, Fort Huachuca, Arizona 85613.

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FOREWORD

MIL-STD-188-346 provides technical design standards for analog end instruments and central office ancillary devices used in the terminal area of long haul communications systems.

This standard supersedes paragraphs 3.4.3 through 3.4.3.4 of Defense Communications Agency Circular (DCAC) 330-175-1, DCS Engineering-Installation Standards Manual.

Analog ancillary devices included in this standard which were not in DCAC 330-175-1 are telephone plant dc power supplies and ringing/signaling equipment for dial central offices.

Although facsimile equipment and television equipment fall within the category of analog end instruments, no coverage is provided within this standard because of other dispositions. Standards for meteorological facsimile equipment used in the common long haul and tactical communication system are included in MIL-STD-188-100. Television equipment will be covered in MIL-STD-188-318 when published.

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ANALOG END INSTRUMENTS

1. SCOPE

1.1 *Purpose.* The purpose of this standard is to establish technical design standards for analog end instruments and central office ancillary devices used in long haul communications.

1.2 *Application.* This standard applies to terminal area communications subsystems within the Defense Communications System (DCS) and the National Military Command System (NMCS) requiring the use of telephone and related ancillary equipment. The standards are to be used in the design and installation of new subsystems and equipment and also in authorized upgrading of existing subsystems and equipment, where permissible, within the parameters of existing criteria.

1.3 *Objective.* The objective of this standard is to provide the minimum electrical performance parameters for the design, engineering, and installation of those portions of the DCS and NMCS concerned with analog end instruments.

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2. REFERENCED DOCUMENTS

The following documents form a part of this standard to the extent specified herein.

Military Standards:

MIL-STD-188-100	Common Long Haul and Tactical Communication System Technical Standards, 15 Nov 72
MIL-STD-188-310	Subsystem Design and Engineering Standards for Technical Control Facilities, 2 Aug 71

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3. TERMS AND DEFINITIONS

The common terms and definitions used in the MIL-STD-188 series are contained in Appendix F of MIL-STD-188-100. When published, MIL-STD-188-120 will supersede Appendix F.

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4. ANALOG END INSTRUMENTS DESIGN AND ENGINEERING STANDARDS

4.1 *End instruments-analog.* The standards in this section are applicable to equipment connected to the terminal of a circuit in which the received or transmitted signals are in analog form.

4.2 *Standards for facsimile equipment.* Standards for facsimile equipment for long haul meteorological applications are covered in the common standards document, MIL-STD-188-100. Equipment standards for other applications are under consideration.

4.3 *Standards for video equipment.* Standards for video end instruments (television) will be covered in a forthcoming standards document in the long haul series designated MIL-STD-188-318.

4.4 *Key telephone instruments.* Characteristics for key set telephone instruments are not considered appropriate for standardization at this time.

4.5 *Acoustic couplers.* Various types and brands of commercial devices are available for coupling quasi-analog information by acoustic means into telephone instruments for transmission over standard voice frequency channels. One type is a narrow-band facsimile system wherein tone signals generated from the optically scanned graphic data (writing, sketches, etc.) are coupled into the telephone transmitter for transmission; at the receiver end the tone signals are coupled from the telephone receiver to the equipment which reproduces the graphic data. Signal conversion is accomplished in quasi-analog modems. Another type is used to transmit teletypewriter messages. At this time, standards for acoustic couplers for military use are not considered to be necessary or desirable because of the limited numbers in use and the continued advancements in techniques. Commercial types of equipment should satisfy ordinary requirements.

4.6 *Data sets.*

4.6.1 *General.* The term "data set" is used to signify a modem and associated terminal equipment used to convert analog or digital data to modulated audio frequency signals for transmission over nominal 3-kHz or 4-kHz voice frequency channels, and, conversely, to demodulate the transmission at the receiver end. Other names applied to these devices are modems, line units, and line terminal equipment.

4.6.2 *Modulation.* Modulation may be amplitude, frequency shift keying (FSK), phase, or other types. One modulation scheme in widespread commercial application for interfacing business machines with the subscriber telephone circuit employs FSK carriers.

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4.6.3 *Digital data equipment.* Output from business machines, such as punched card, punched tape, magnetic tape, or direct keyboard entry is in the form of binary digital data and can range up to 9,600 bits per second (in 1973) for a nominal 4-kHz telephone channel.

4.6.4 *Analog data equipment.* Analog data from facsimile, written or typed messages, medical devices (electrocardiograms), etc., is handled by suitable modems. In some instruments acoustic couplers also may be used (see 4.5).

4.6.5 *Standardization for military applications.* The many variables involved in the numerous types of commercial data sets preclude any general standardization for military applications. In addition to several possible types of modulation, transmission may be transparent or coded for some devices while others require timing and slaving techniques and stable oscillators. Other variables are two-wire or four-wire telephone circuit operation, operating speed, requirement of separate channels for control, and multi-channel operation for bandwidth requirements over 4 kHz.

4.7 *Telephone sets.* The characteristics of the telephone sets (i.e., instruments) described herein are based on the type 500, or equivalent, commercial telephone set using the U-1 receiver and the T-1 transmitter. The standards specified herein shall not preclude the use of future sets which may have higher acoustical/electrical conversion efficiencies. The characteristics covered apply to sets operating on a two-wire basis and on a four-wire basis. Suitable wiring shall be incorporated within the sets so that optional terminal strapping and connections will permit the instrument to operate on either a two-wire loop or a four-wire loop.

4.7.1 *Parameters for telephone service.*

4.7.1.1 *Transmitted speech volume.* The talker volume distribution at the line terminals of a telephone set has been found to approximate a normal or Gaussian distribution when expressed in volume units (VU). Based on a talker volume distribution, the transmitted mean speech volume of a two-wire as well as a four-wire telephone set used in a long haul system is assumed to be -10 VU with a standard deviation (sigma) of 5 VU.

4.7.1.2 *Received speech volume.* Subjective tests indicate that received speech volume at the line terminals of the listener telephone set from -21.5 VU to -33.5 VU results in satisfactory service. The mean received speech volume at the line terminals of the listener telephone set should be nominally -28 VU. Based on subjective tests made with a commercial type of telephone in a typical office environment, the data listed below show the relationship between received speech volume and the percentage of calls rated good or better when the associated noise level is acceptably low:

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Received speech volume	VU	Percentage of calls rated good or better
Maximum	-21.5	95
Mean	-28.0	100
Minimum	-33.5	95

4.7.1.3 *Received noise power.* With a mean received speech volume of -28 VU the value of noise power for acceptable service has been set at 44 dBrnc (design objective: 38 dBrnc) at the line terminals of the telephone set (see fig. 1).

4.7.2 *Equipment parameters.*

4.7.2.1 *Level regulation.* The set shall incorporate passive devices to adjust automatically the transmission loss in the set to deliver a relatively constant transmission level at the 0-dBm transmission level point (0-TLP) at the local switchboard or automatic switching center when connected to loops of various lengths. The transmission level regulation shall be controlled by the local loop resistance and local battery current which decreases as the loop length and resistance are increased. Transmitter level regulation and receiver level regulation shall be within the limits shown in figures 2 and 3 respectively when connected to typical No. 26 American Wire Gage (AWG) exchange cable. The older design type 302 sets are less efficient instruments than the type 500, as indicated in figure 4. Where type 302 sets are still in use, the limits shown in figures 2 and 3 may be exceeded by 4 dB at the 10,000-foot length (830 ohms approximately). Although this standard does not cover exchange cable, the loop characteristics of exchange cable must be considered in the design of the telephone set.

4.7.2.2 *Harmonic distortion and amplitude clipping level.* The total harmonic distortion in the receive direction shall be less than 5 percent for all input signals within the band 300 Hz to 3500 Hz with input level at 0 dBm. No amplitude clipping shall occur below a sound pressure corresponding to +3 dBm.

4.7.2.3 *Impedance.* Ideally, the impedance of the telephone set should be the conjugate of the loop impedance when the loop is terminated properly at the switching office. That is, the resistance components of the telephone set and loop are equal in value and the reactance components are equal in value but have opposite signs. Since loops vary in length, a compromise impedance value must be used for telephone sets.

4.7.2.3.1 *Two-wire instruments.* The terminal impedance of two-wire instruments shall be 600 ohms ±20 percent at 1000 Hz. The reactive component shall be inductive and less than 300 ohms at 1000 Hz. Longitudinal balance shall be not less than 40 dB.

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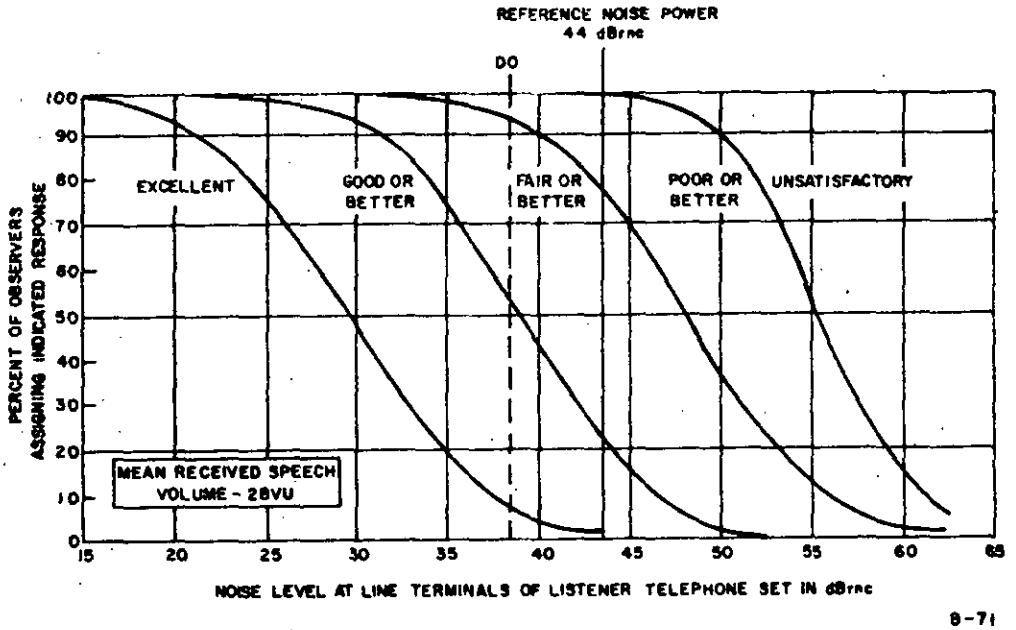


Figure 1. Subjective ratings of received noise power.

4.7.2.3.2 *Four-wire instruments.* The terminal impedance on the transmit and receive sides of a four-wire telephone set shall be 600 ohms ±20 percent at 1000 Hz and a transmitter loop current of 50 milliamperes. The longitudinal balance for the input and output of four-wire instruments shall not be less than 40 dB.

4.7.2.4 *Acoustical/electrical frequency characteristics.* Both the transmitter and receiver shall have substantially flat acoustical/electrical frequency characteristics. Figure 5 shows the expected deviation of a typical set connection normalized to 1000-Hz reference. In figure 5, curve A shows the required characteristics at the transmitter output and the allowable deviation limits; curve B shows similar characteristics required at the output of the subset; and curve C shows the frequency characteristics expected on a typical connection. Figure 6 is a plot of receiver output in VU versus the percent customer reaction.

4.7.2.5 *Supervision.* Basically, all terminal instruments shall have switch hook direct current (dc) supervision. Future systems, however, may require tone supervision. Also, future systems may require a capability within the terminal instrument for loop-back of test signals originated at the local switch. Standardization for other

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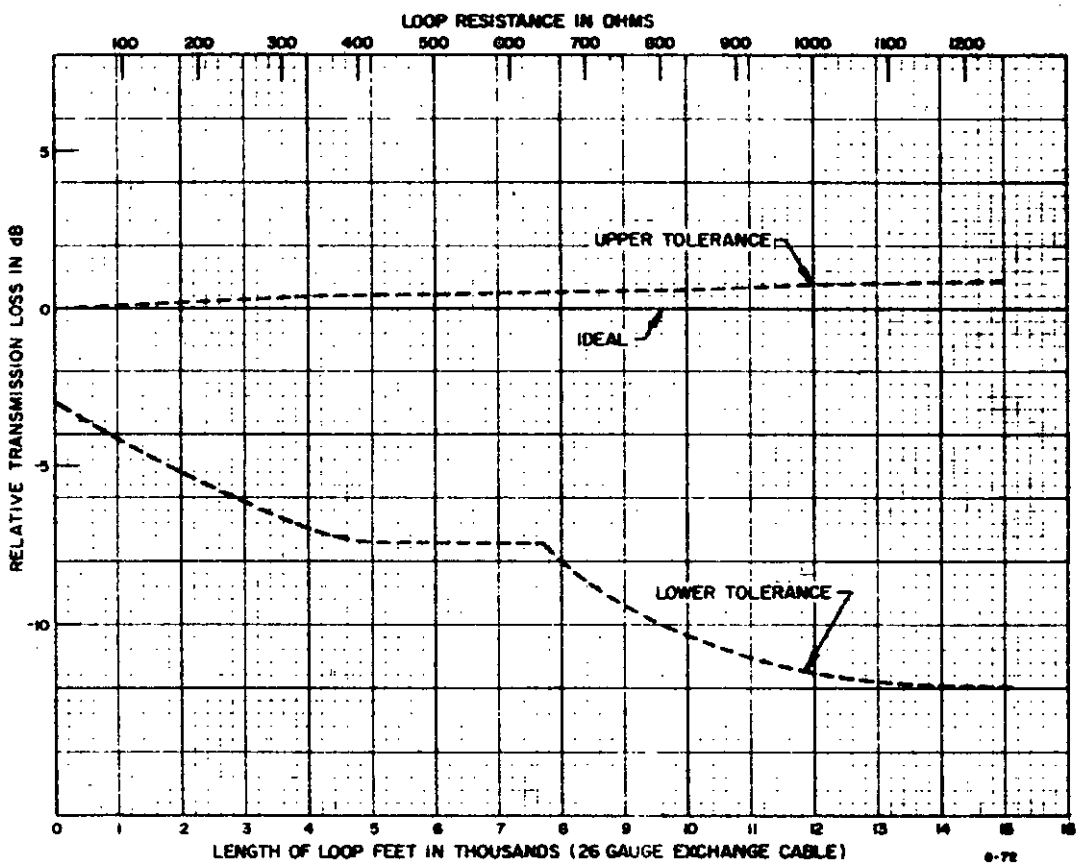


Figure 2. Telephone set transmission level compensation vs loop length.

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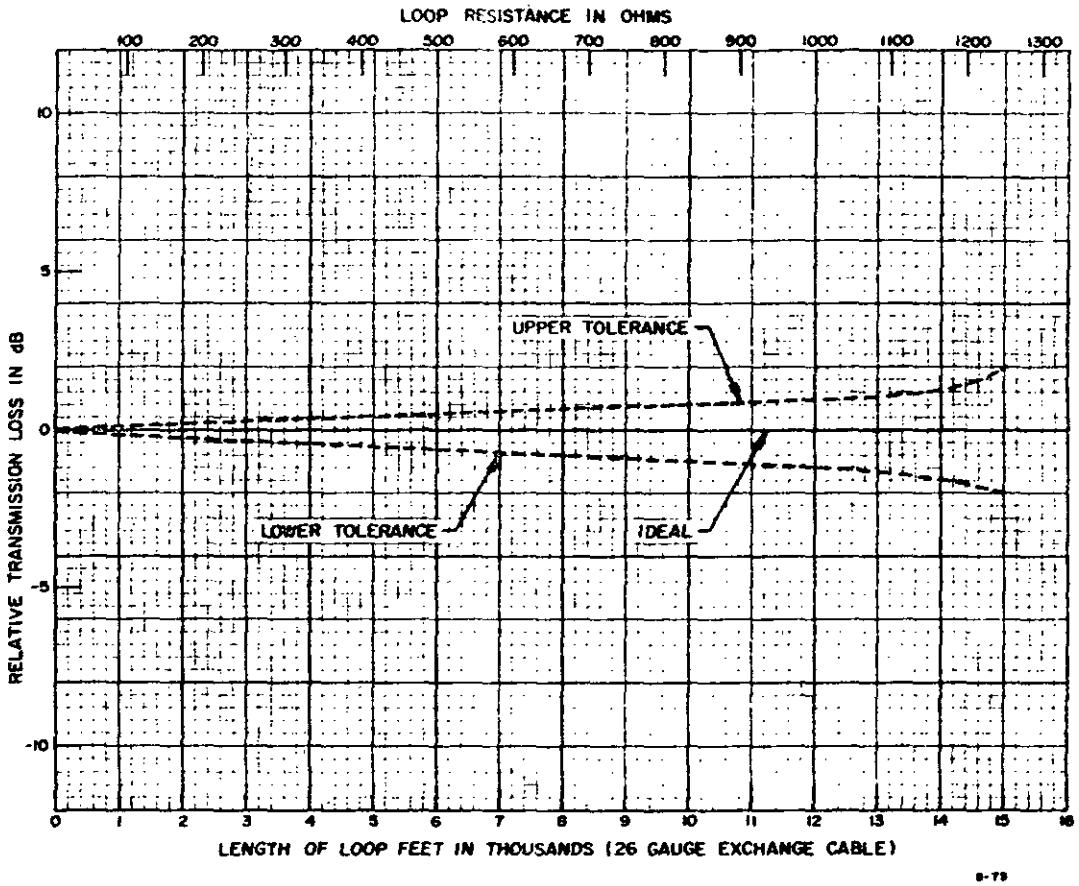


Figure 3. Telephone set receiving level compensation vs loop length.

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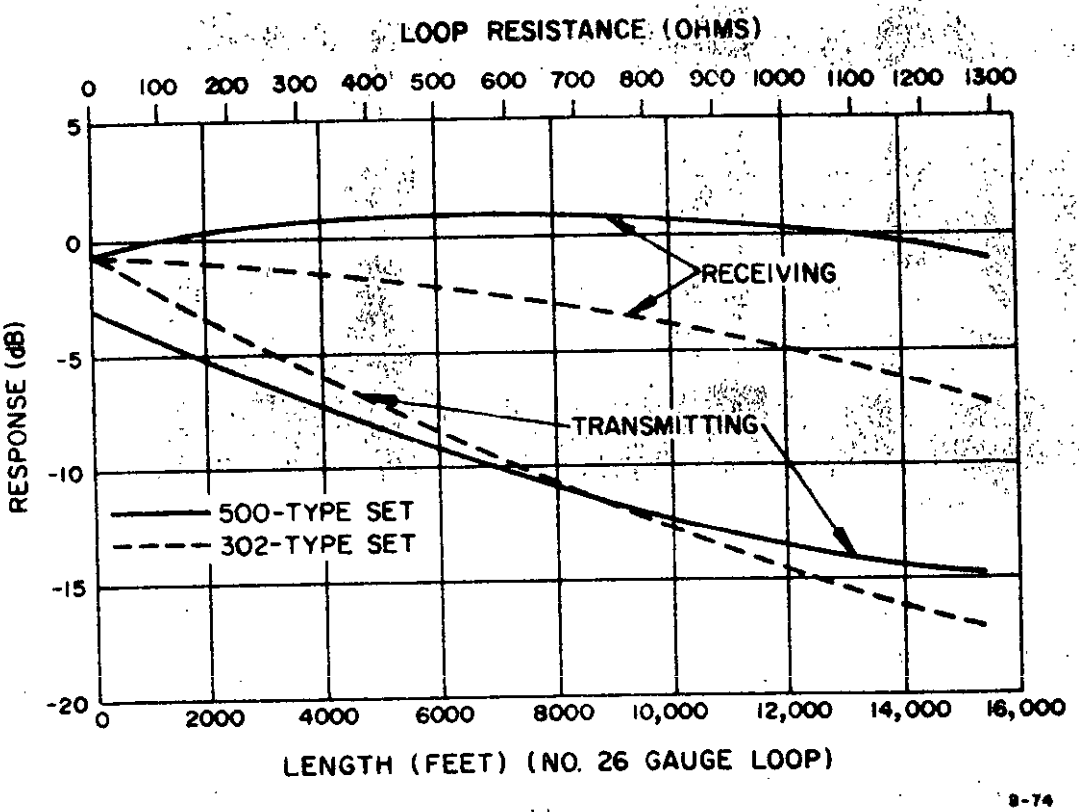


Figure 4. Relative responses of type 302 and type 500 telephone sets.

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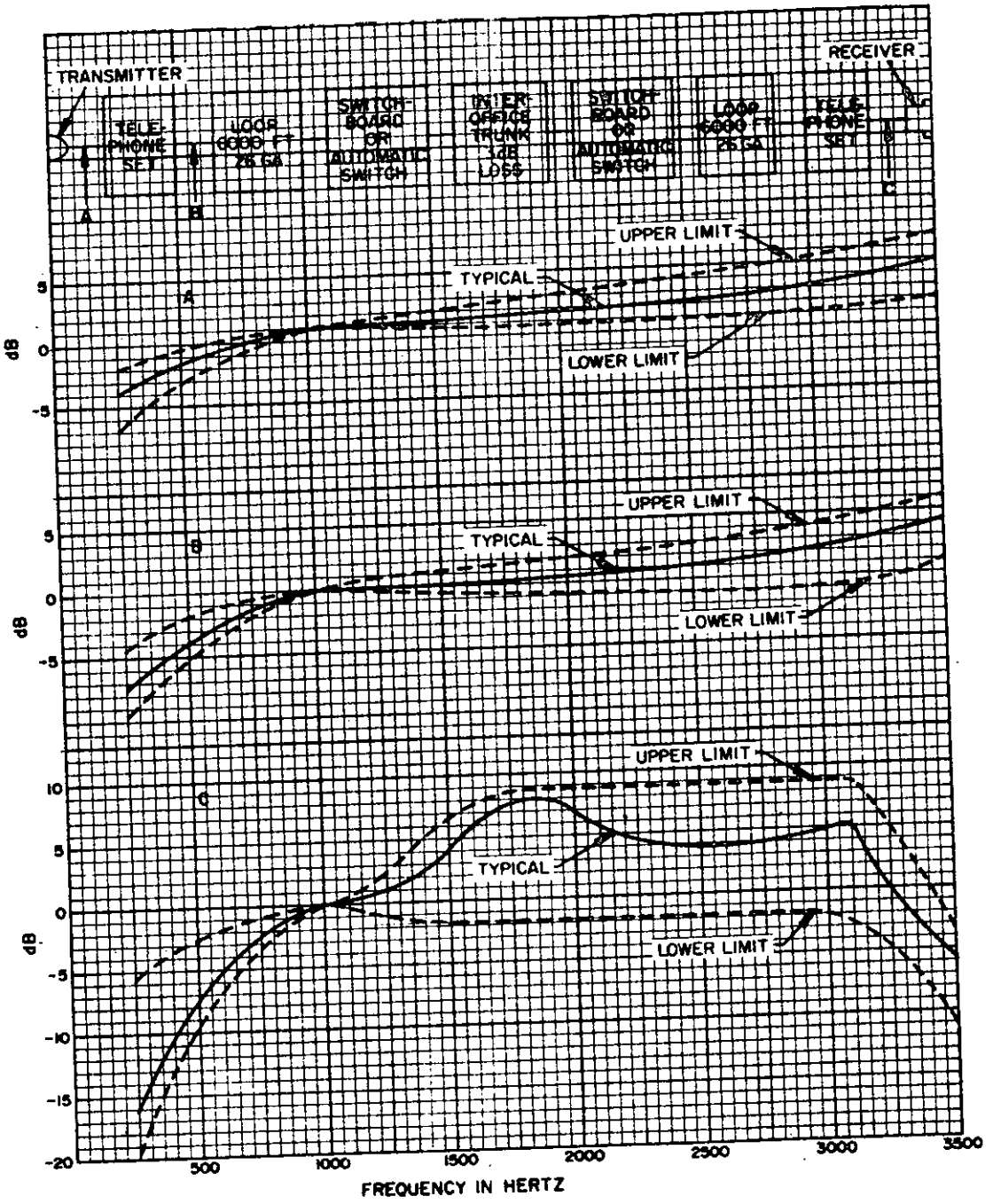


Figure 5. Expected deviation of a typical telephone set connection normalized to 1000-Hz reference.

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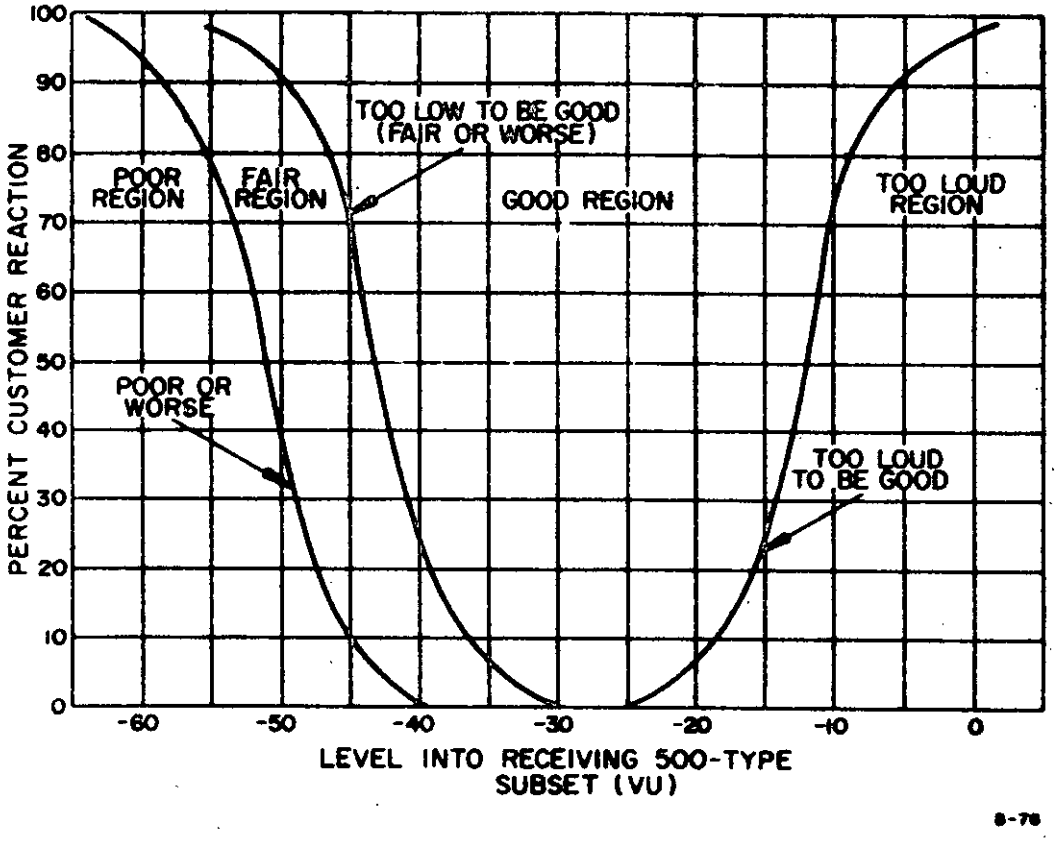


Figure 6. Judgment of grade of telephone conversation vs received volume.

techniques will be deferred until the most desirable techniques have been determined. The line shall be open when the handset is on the instrument, except the ringer shall be bridged on the line side of the hook switch.

4.7.2.6 *Dial pulse speed and break.* The dial telephone subscriber provides signaling to the central office through a series of off-hook, on-hook information signals. On-hook is indicated by a dc open circuit at the subscriber instrument; off-hook is indicated by a dc short circuit. Dial pulsing is accomplished by a series of off-hook, on-hook pulses, termed make and break respectively. The break period of pulses shall be within the limits prescribed below as percentages of the combined make and break periods. Two dial speeds are in general use. Crossbar and panel switching systems normally use a high-speed dial while step-by-step systems require a low-speed dial. The telephone sets shall be equipped with the dial speed

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compatible with the central office equipment to which they are connected. Dial speeds and break periods as manufactured shall be within the limits listed below:

Dial type	Pulses per second		Break period	
	Minimum	Maximum	Minimum	Maximum
High speed	17	19	62%	66%
Low speed	9.5	10.5	58%	64%

NOTE: These figures are adjustment limits and allow for slight aging and switching tolerances.

4.7.2.7 *Dial design.* The finger holes in the dial shall be numbered consecutively and counterclockwise on the stationary portion of the dial starting with numeral 1 and ending with numeral 0. The numbered positions on the dial also shall be identified with letters as follows:

Position	Lettering
1	Blank
2	ABC
3	DEF
4	GHI
5	JKL
6	MNO
7	PRS
8	TUV
9	WXY
0	Operator, or OP, or OPR

4.7.2.8 *Dial operation.* The dial shall rotate counterclockwise at constant speed (governor controlled) when the rotary mechanism is released after the numeral or letter selected has been dialed. The counterclockwise rotation of the dial shall interrupt the line current periodically and cause the central office selection equipment to operate in accordance with the numeral or letter dialed. Operation of the movable part of the dial also shall cause secondary contacts to close and short circuit the receiver during the dialing operation to prevent audible clicks in the receiver. These contacts shall remain closed until the movable dial has returned to its normal resting position. The counterclockwise rotation of the dial, or pulsing operation, shall send out distinctive mark and space signals for each numeral or letter dialed.

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4.7.2.8.1 *Interdigital timing.*

4.7.2.8.1.1 *Crossbar switching centers.* Interdigital timing for crossbar switching centers shall be 0.35 second ±5 percent.

4.7.2.8.1.2 *Step-by-step switching centers.* Interdigital timing for step-by-step switching centers shall be 0.6 second ±5 percent.

4.7.2.9 *Dual tone multifrequency (DTMF) signaling.* Other forms of performing the dial function, such as using pushbuttons, are acceptable. The pushbuttons, or similar arrangements, shall be numbered consecutively from left to right and from top to bottom starting with numeral 1 and ending with numeral 0. DTMF signaling provides for pushbutton operation from a subscriber station using voice transmission paths. The code for this system consists of 16 distinct signals each composed of two voice-band frequencies. These frequencies are selected from two mutually exclusive groups of four frequencies each. The signal frequencies are spaced geometrically and selected to avoid harmonically related frequency combinations.

4.7.2.9.1 *Keying frequencies.* DTMF pushbutton keying shall use tones with the frequencies listed below:

Digit/DTMF key	Frequencies (Hz ±1.3%)
1	697 and 1209
2	697 and 1336
3	697 and 1477
4	770 and 1209
5	770 and 1336
6	770 and 1477
7	852 and 1209
8	852 and 1336
9	852 and 1477
0	941 and 1336

Where precedence service and abbreviated keying is used:

A	End of signaling	941 and 1477
FO	Precedence 0 (flash override)	697 and 1633
F	Precedence 1 (flash)	770 and 1633
I	Precedence 2 (intermediate)	852 and 1633
P	Precedence 3 (priority)	941 and 1633

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4.7.2.9.2 *Signaling output power.* The output power of the DTMF transmitter when terminated in 900 ohms shall conform to the values specified in table I. However, the set also shall operate satisfactorily when terminated in 600 ohms. The tolerances for the values shown in table I are specified in the remainder of this paragraph. Under no conditions shall the output power of any tone in the high group differ from the output power of any tone in the low group by less than 0.5 dB or more than 4.5 dB. In order to satisfy the required range of power differences, the DTMF transmitter output power shall not vary more than ±1 dB from the values specified in table I except in one case. The exception occurs when the power levels of the high group and the low group increase or decrease together in such a manner as to maintain the required range of power differences. In the case of the exception, the DTMF transmitter output power shall not vary by more than ±2.5 dB from the values specified in table I.

Table I. DTMF transmitter output power (dBm) with 900-ohm termination

Tone group	Output power of each frequency Line current 50 mA	Line current 70 mA	Line current 100 mA
Low group (697 - 941 Hz)	-5.7	-7.3	-8.5
High group (1209 - 1633 Hz)	-3.2	-4.8	-6.0

Preferred tolerances for the above values of output power are ±1 dB.

4.7.2.9.3 *Repertory dialers.*

4.7.2.9.3.1 *Tone pulse duration.* The tone pulse output from a repertory DTMF dialer shall be not less than 45 milliseconds in duration with an interdigital interval of not less than 45 milliseconds.

4.7.2.9.3.2 *Keying rate.* The maximum allowable keying rate for a repertory DTMF dialer shall not exceed 10 digits per second.

4.7.2.10 *Ringer.* The telephone set shall have an integral ringer with an arrangement for external manual adjustment of the acoustic ringer level. Provision shall also be made for connecting remote ringing units and visual signals (when required) to the telephone set. Two-wire telephone instruments normally use alternating current (ac) ringers (or dc buzzers for key sets), while four-wire instruments may use ac ringers or tone ringers (dc).

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4.7.2.10.1 *AC ringer.* The ac ringer shall have at least 6,000 ohms impedance bridged across the line throughout the frequency range of from 200 to 3000 Hz. AC ringers shall operate on any frequency between 15 and 25 Hz and shall not respond to frequencies in the 200- to 4000-Hz voice range. The nominal ringing current (from the switchboard) shall be 0.04 ampere, 16 to 20 Hz. The nominal ringing voltage shall be 90 volts.

4.7.2.10.2 *Tone ringer.* Telephone instruments equipped with integral tone ringers shall conform to the following standards:

(a) The tone ringer shall operate from signaling battery supplied on the receive pair. A grille shall be provided as an audio outlet for tones generated by the ringer.

(b) The tone ringer shall consist of an oscillator and a transducer. On an incoming call signal, the oscillator and a transducer shall produce a 2000 ±400 Hz acoustic tone, interrupted 12 ±4 times per second. The output tone level shall be manually adjustable from outside of the instrument. The adjustment, however, shall not completely disable the tone.

(c) The acoustic power output of the tone ringer, when measured in a free field perpendicular to the telephone face plate at a distance of 2 feet with the ringer volume control in the maximum output, shall be greater than +70 dB referred to 2×10^{-4} dynes/cm^2.

4.7.2.11 *Noise.* Random noise generated by the instrument when energized and terminated properly shall not exceed 33 dBrnc at the line terminals. This shall not include room noise picked up by the transmitter.

4.7.3 *Performance.*

4.7.3.1 *Acoustical/electrical conversion efficiency.* The acoustical/electrical conversion efficiency of transmitters and receivers on a typical call connection is shown in figure 7. This is representative of current instruments. For future instruments there is no limit imposed on the maximum efficiency to which the instrument can be designed; however, it shall remain within the frequency characteristic limits. Higher instrument conversion efficiency may permit the use of longer loops without repeaters or the use of smaller gage exchange cable.

4.7.3.1.1 *Transmit direction.* A 1000-Hz signal with a root mean square (rms) sound pressure of 20 dynes per square centimeter at the grid of the transmitter shall produce -6 dBm ±2 dB into a 600-ohm load measured at the two-wire output of the telephone set.

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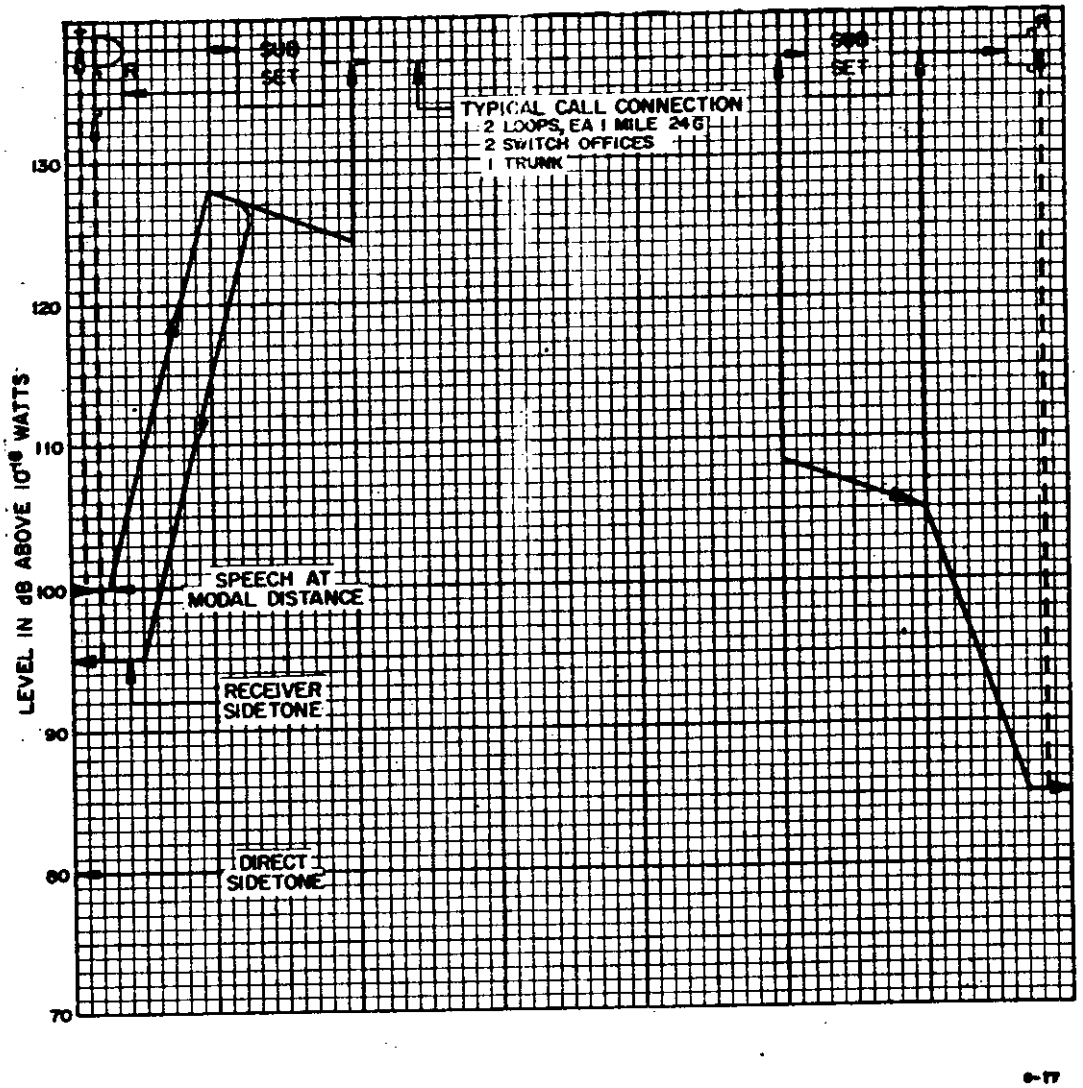


Figure 7. Telephone set acoustical/electrical characteristics.

4.7.3.1.2 *Receive direction.* A 1000-Hz signal of –22 dBm at the input of the telephone set shall produce an rms sound pressure of not less than 3.6 dynes per square centimeter at the receiver output in conjunction with a 6 cubic centimeter coupler. As a design objective, –32 dBm at the two-wire input shall produce 2 dynes per square centimeter at the receiver output.

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4.7.3.1.3 *Sidetone.* A design objective for sidetone coupling loss between the transmitter input and receiver output shall be between 15 and 40 dB and shall be determined by the formula:

$$\text{Sidetone (dB)} = 20 \log_{10} \frac{\text{Transmit rms sound pressure}}{\text{Receiver rms sound pressure}}$$

The actual value is not critical, but sidetone should not result in sound levels greatly in excess of those being received from the distant end nor should they be below audibility.

4.7.3.2 *Loop requirements.* Resistance is a factor limiting loop length. Normally, this is because of the requirements to maintain minimum ringing voltage and transmitter current and to satisfy central office signaling requirements. The present limitation on loop resistance with type 500 telephone sets is 1,200 ohms for a 48-volt system. Current telephone sets with high efficiency and self-level adjusting features shall use loop "resistance design" since this normally is the limiting factor. In any event, the loop loss shall not exceed 7.1 dB at 1000 Hz. Older sets may use "transmission design." The maximum loop characteristics under transmission and resistance design techniques are as follows:

Transmission design

Cable gage AWG	Max loop length in feet	dB loss not loaded	dB loss loaded	Total loop resistance in ohms
26	10,500	5.3		875
24	14,000	6.15		725
22	17,000	5.8		550
19	21,000	5.5	1.4	330

Resistance design

Cable gage AWG	Max loop length in feet	dB loss not loaded	dB loss loaded	Total loop resistance in ohms
26	14,400	7.1		1200
24	22,600		5.2	1200
22	35,500		5.3	1200
19	68.500		5.5	1200

NOTE: Loop losses are given at 1000 Hz.

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4.7.3.3 *Loading.* In order to provide suitable transmission characteristics, cable loops exceeding 18,500 feet in length shall be loaded. Normally this shall be H-88 loading (88 millihenrys at 6000-foot spacing) with properly built-out end sections. Alternatively, intermediate repeaters and equalizers may be used provided sufficient battery current and ringing current are available at the instrument.

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5. DIAL CENTRAL OFFICE EQUIPMENT STANDARDS

5.1 *General.* This section establishes standards for dial central office (DCO) ancillary equipment, which includes telephone plant dc power supplies, ringing/signaling equipment, dial loop to E&M converters, and bridge lifters. Other ancillary equipment, confined principally to technical control installations for conditioning voice frequency circuits and for voice frequency signaling purposes, is covered in MIL-STD-188-310 Appendix A, viz: single frequency signaling equipment, echo suppressors, DX signaling units, four-wire/two-wire terminating sets, line amplifiers, delay equalizers, and amplitude equalizers.

5.2 *Telephone plant power supply.* The direct current required to operate a DCO normally is supplied by rectifiers operating off the primary ac power, with a secondary-type battery floating on the rectifier output. Redundancy shall be provided to prevent system outages in case of a rectifier failure. Figure 8 is a block diagram of a DCO power plant.

5.2.1 *Rectifier.* The rectifier shall convert primary ac power to dc power of suitable voltage and current to operate the DCO equipment and maintain the battery system in fully-charged condition. Duplicate rectifiers shall be provided with automatic switchover in case of unit failure.

5.2.1.1 *Input power.* The rectifier shall operate from a 120-volt single-phase, or 120/208-volt three-phase, 47-63 Hz power source. The rectifier shall deliver the required output under input voltage conditions varying ±10 percent from the nominal value.

5.2.1.2 *Output voltage and current.* The rectifier output voltage shall be adjustable from 48 to 58.5 volts. The rectifier (or rectifier system) shall have a maximum output current rating at least 10 percent greater than the calculated ultimate expanded office busy-hour current requirement.

5.2.1.3 *Line and load regulation.* The rectifier shall maintain the steady state dc output voltage within ±1 percent for any load from no load to full load with ±10 percent variation from nominal input voltage and frequency within allowable limits (i.e., 47-63 Hz), while floating 23 battery cells.

5.2.1.4 *Overvoltage protection.* The rectifier shall use remote sensing to monitor output voltage continuously. If the output voltage should increase to an adjustable preset value (52-58 volts), the output current shall be shut off and an alarm circuit shall be energized.

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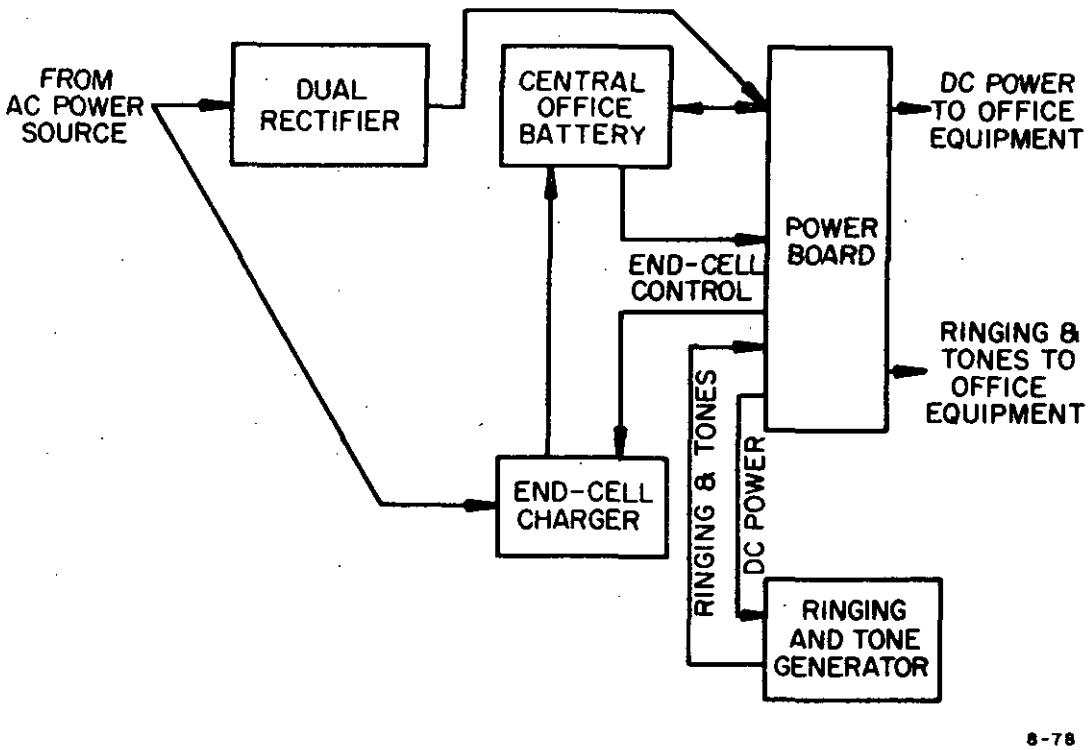


Figure 8. Equipment required for a dial central office power plant.

5.2.2 *End-cell rectifier.* The end-cell rectifier shall convert primary power to power of suitable voltage and current to maintain up to three (1-3) telephone office battery cells (end-cells) in a charged condition.

5.2.2.1 *Input power.* The end-cell rectifier shall operate from the same primary source as the main rectifier and deliver the required output under conditions of ±10.0 percent variation from the nominal input voltage.

5.2.2.2 *Output voltage.* The output voltage shall be adjustable from 2.1 to 7.1 volts to accommodate 1- to 3-cell end-cell systems.

5.2.2.3 *Line and load regulation.* The output voltage shall vary not more than ±2 percent for load variation from no load to full load with ±10 percent variation from nominal input voltage and frequency variation within allowable limits, while floating the end cells.

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5.2.2.4 *Overvoltage protection.* The rectifier shall use remote sensing to monitor output voltage continuously. If the output voltage exceeds an adjustable preset value (2.1 to 7.1 volts), the output current shall shut off and an alarm shall be activated.

5.2.3 *Battery.* The central office battery shall be used as an emergency power source in case of commercial power failure and as a filter to reduce ripple on the rectifier output. The battery normally shall consist of 26 identical cells, including 23 main cells and three end-cells. With a primary power failure, the 23 main cells shall support the office until low battery voltage is detected, at which time the three end-cells shall be connected automatically in series with the main cells to provide additional voltage. In small installations, one or two end-cells may suffice.

5.2.3.1 *Capacity.* The required battery capacity shall be not less than 10 percent greater than eight times the calculated expanded office busy-hour current requirement (rated in ampere-hours).

5.2.3.2 *Life.* The minimum expected cell life shall be 20 years when used in a floating battery application and in an average annual ambient temperature not exceeding 80 °F.

5.2.3.3 *Construction.* A lead-calcium battery incorporating heavy duty construction and explosion protection shall be used. The cells shall have transparent walls to permit inspection of plates.

5.2.4 *Counter electromotive force cells.* In small office (under 4,000 lines) power plants, counter electromotive force (CEMF) cells may be used instead of end-cells to provide additional voltage when the battery voltage reaches a lower limit. In such cases, the battery shall consist of 24 cells. This technique eliminates two cells and an end-cell rectifier from the battery system described in 5.2.3.

5.2.4.1 *Current rating.* The current rating of the CEMF cells shall be not less than 10 percent greater than the calculated expanded office busy-hour current requirement.

5.2.4.2 *Construction.* The CEMF cells shall be silicon dry-cell construction.

5.3 *Ringing/signaling equipment.*

5.3.1 *Signaling tone generator.* Equipment to generate ringback and signaling tones shall be included as part of the DCO plant. The information signals shall have the characteristics indicated in 5.3.1.1 or 5.3.1.2. The tone generation equipment shall be redundant. The generator output shall be strappable to provide 1/8, 1/4, 1/2, 1, 2, 4, or 10 volts.

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5.3.1.1 *Standard information tones.* Information tones shall be as follows:

Signal	Frequency (Hz ±0.5%)	Interruption	Power per frequency (dBmO ±3 dB)
Dial tone	350 + 440 (mixed)	Continuous	-13
Station busy	480 + 620 (mixed)	60 IPM ±2 IPM (50% duty cycle)	-24
Trunk busy	480 + 620 (mixed)	120 IPM ±2 IPM (50% duty cycle)	-24
Ringback tone	440 + 480 (mixed)	1 s on, 4 s off	-16

5.3.1.2 *Other acceptable tones.* (Addition and modification of existing office only). Frequencies and tone power level measured at the distribution point shall be as follows:

Signal	Frequency (Hz ±0.5%)	Interruption	Tone power (dBmO ±3 dB)
Dial tone	600/120 (modulated)	Continuous	-24
Station busy	600/120 (modulated)	60 IPM ±2 IPM (50% duty cycle)	-16
Trunk busy	600/120 (modulated)	120 IPM ±2 IPM (50% duty cycle)	-16
Ringback tone	400/40 (modulated)	1 s on, 4 s off or 1 s on, 5 s off or 2 s on, 4 s off	-24

5.3.2 *Ringing machine.* The ringing machine, either solid state or motor type, shall have characteristics as follows:

(a) Input power: 46 to 52 volts dc.

(b) Output voltage (no load to full load): 90 to 110 volts ac (rms) adjustable.

(c) Output frequency: 20 ±1 Hz.

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(d) Output current: 5 percent greater than 0.01 ampere times the total number of expanded office connectors and attendant consoles in the office (with a 5-phase balanced distribution of ringing current).

5.3.2.1 *Preferred sizes of ringing machines.* A standard 50-watt unit ringing machine is suitable for systems of up to 4,000 lines. Additional capacity is obtained by using multiple units:

Number of on-line ringing machines	Power rating of ringing machines (watts)	Size of expanded office in lines
1	50	Up to 4,000
2	50	4,000 to 8,000
3	50	8,000 to 12,000

5.4 *Dial loop to E&M converter.* Dial loop to E&M converters are used, when required, to convert subscriber instrument signaling information to E&M lead conditions for application to DX or SF signaling equipment. The two types of dial loop to E&M converters are a subscriber unit and a central office unit, used in pairs as a set and located at the subscriber terminal and the central office respectively.

5.4.1 *Subscriber unit.* The subscriber dial loop to E&M converter causes battery voltage (-48 volts ±5 volts) to appear on the M-lead when an off-hook condition (electrical short circuit) exists at the subscriber instrument. The converter causes 20-Hz ringing voltage to be placed on the subscriber line upon receipt of an open circuit condition of the E-lead and provides termination of the voice path toward the central office with a 600-ohm ±10 percent resistor in series with a fixed 1- to 2-microfarad capacitor. During subscriber on-hook conditions (dc open circuit), the converter will place a ground condition on the M-lead. Upon receipt of a ground on the E-lead, the converter causes the subscriber instrument to be connected to the voice path. During dialing the converter causes alternately battery voltage and ground to be placed on the M-lead.

5.4.2 *Central office unit.* This converter causes an off-hook (dc short circuit) to appear at the central office end upon receipt of a ground on the E-lead. The converter causes on-hook (dc open circuit) to appear at the central office end upon receipt of an open circuit on the E-lead. When the converter receives 20-Hz ringing from the central office, it causes a ground to appear on the M-lead, and terminates the voice path toward the subscriber with a 600-ohm ±10 percent resistor in series with a fixed 1- to 2-microfarad capacitor. In the absence of 20-Hz ringing, battery voltage is applied to the M-lead. During dialing the E-lead receives alternately ground and open circuit.

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5.4.3 *Electrical requirements.*

5.4.3.1 *Dial pulses.* Both types of loop to E&M converters shall accept dial pulsing information at from 9 to 12 pulses per second with from 47 to 67 percent break and transmit pulses whose duration is within 4 milliseconds of the duration of the pulses received.

5.4.3.2 *Insertion loss.* The insertion loss shall be 0 ±0.1 dB at 1000 Hz. The insertion loss at any frequency from 300 to 3400 Hz shall not differ from the insertion loss at 1000 Hz by more than ±0.1 dB.

5.4.3.3 *Return loss.* The return loss shall be no less than 26 dB when compared to 600-ohm ±1 percent nonreactive resistor termination.

5.4.3.4 *Longitudinal balance.* The longitudinal balance shall be not less than 40 dB from 50 to 4000 Hz for longitudinal potentials up to 50 volts rms.

5.4.3.5 *Crosstalk.* The crosstalk coupling loss between adjacent VF channels within the equipment or in adjacent units shall be not less than 70 dB at frequencies between 300 and 3400 Hz with units terminated in 600 ohms and test tone level of 0 dBm.

5.4.3.6 *Noise.* The total noise at the voice channel outputs shall not exceed -70 dBm flat weighting between 300 and 3400 Hz with voice channel inputs and outputs terminated in 600 ohms.

5.4.3.7 *Frequency response.* Frequency response shall be flat within ±0.1 dB from 300 to 3400 Hz.

5.4.3.8 *Harmonic distortion.* Harmonic distortion in the voice channels shall not exceed 1 percent from 300 to 3400 Hz with levels up to +10 dBm.

5.5 *Bridge lifter.*

5.5.1 *Application.* A bridge lifter is used to prevent transmission impairment by inactive parties telephones when one party is using a multiparty telephone line. The bridge lifter uses a saturable reactor which has high impedance when dc is not present and low impedance when dc is present.

5.5.2 *Requirements.*

5.5.2.1 *Resistance.* The maximum dc resistance shall be not more than 20 ohms.

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5.5.2.2 *Inductance.*

5.5.2.2.1 ***Maximum inductance.*** The maximum inductance shall be not less than 80 millihenrys at 10 milliamperes dc when measured at 1000 Hz.

5.5.2.2.2 *Minimum inductance.* The minimum inductance shall be not more than 10 millihenrys at 30 milliamperes dc when measured at 1000 Hz.

5.5.2.3 *Crosstalk.* Crosstalk loss between inductors in multiunit assemblies shall be at least 90 dB at 1000 Hz.

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6. TEST PROCEDURES

6.1 *General.* This section sets forth procedures for testing the equipment performance parameters specified in sections 4 and 5. The methods prescribed are to be used as guidelines for testing. A listing of the test methods cross-referenced to the applicable paragraphs in sections 4 and 5 follows:

Test method number	Subject	Paragraph no. in standard
6.2.1	Telephone set level regulation	4.7.2.1
6.2.2	Telephone set and ringer impedance	
	Ringer impedance	4.7.2.10.1
	Ringer operation	4.7.2.10.2
	Telephone set impedance	4.7.2.3
6.2.3	Telephone set acoustical/electrical frequency characteristics	4.7.2.4
6.2.4	Telephone set dial pulse speed and break	4.7.2.6
6.2.5	Dual tone multifrequency tones	4.7.2.9
	Frequency measurement	4.7.2.9.1
	Signaling output power measurement	4.7.2.9.2
	Tone pulse duration (repertory dialer only)	4.7.2.9.3.1
	Keying rate (repertory dialer only)	4.7.2.9.3.2
6.2.6	Telephone set noise	4.7.2.11
6.2.7	Acoustical/electrical conversion efficiency	4.7.3.1
	Transmitter rating without loop loss	4.7.3.1.1
	Transmitter rating with loop loss	4.7.3.1.1
	Receiver rating without loop loss	4.7.3.1.2
	Receiver rating with loop loss	4.7.3.1.2
6.2.8	Dial central office rectifiers	
	Rectifier	
	Input voltage and frequency	5.2.1.1
	Output voltage	5.2.1.2
	Output current	5.2.1.2
	Line and load regulation	5.2.1.3
	Overvoltage protection	5.2.1.4

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Test method number	Subject	Paragraph no. in standard
	End-cell rectifier	
	Input voltage and frequency	5.2.2.1
	Output voltage	5.2.2.2
	Line and load regulation	5.2.2.3
	Overvoltage protection	5.2.2.4
6.2.9	Ringing/signaling tone generator	
	Mixed tones	5.3.1.1
	Modulated tones	5.3.1.2
	Interruption of tones	5.3.1.1-2
6.2.10	Dial loop to E&M converter	
	Signal conversion	
	E&M to dc, dc to E&M	5.4.1-2
	Pulses per second	5.4.3.1
	20-Hz ringing	5.4.1-2
	E-lead -- off hook	
	M-lead -- off hook	
	Return loss	5.4.3.3
	Longitudinal balance	5.4.3.4
	Crosstalk and Noise	5.4.3.5-6
	Frequency response	5.4.3.7
	Harmonic distortion	5.4.3.8
6.2.11	Bridge lifter	
	Resistance	5.5.2.1
	Inductance	5.5.2.2
	Crosstalk	5.5.2.3
6.2.12	Ringing machines output frequency	5.3.2

6.2 *Test methods.* The test methods provide step-by-step procedures to be followed to measure performance of analog end instruments and central office equipment. Test setups are given and each item of test equipment required to conduct the tests is assigned a reference number. Test equipment performance requirements correlated to the test item reference number are listed in the appendix. In general, the test methods do not repeat the standards values specified in the referenced paragraphs of sections 4 and 5.

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6.2.1 *Telephone set level regulation.*

6.2.1.1 *Scope.* This method is used to measure level regulation in telephone sets.

6.2.1.2 *Applicability.* The parameters tested shall conform to the requirements specified in the paragraph cited below:

Item	Paragraph
Level Regulation	4.7.2.1

6.2.1.3 *Required test equipment.*

Test unit	Schematic reference	Item no. in appendix
AF Oscillator*	A	126
Artificial Mouth Equalizer*	B	935**
Variable Attenuator*	C	600
Amplifier*	D	509
Artificial Mouth*	E	935**
Condenser Microphone	F	510
Microphone Amplifier	G	511
Vacuum Tube Voltmeter (VTVM), rms	H	18
Milliammeter, dc, 0-100 mA	I	8
Artificial Line	J	939
Feeding Bridge	K	940
Resistor, 750-ohm ±1%	L	435
Resistor, 900-ohm ±1%	M	436

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Test unit	Schematic reference	Item no. in appendix
Electro-Acoustic Test Set*		342

*Electro-Acoustic Test item no. 342, may be used in place of items 126, 935, 600, and 509.

**Schematic references B and E are halves of the same unit.

6.2.1.4 *Procedures.*

6.2.1.4.1 *Measurement of telephone set transmitter level regulation.*

Step 1. Connect equipment as shown in figure 9.

NOTE. Items A through E constitute the sound source and items F, G, and H constitute a means for calibrating the sound source.

Step 2. Locate condenser microphone F in correct position with respect to guard ring of artificial mouth as shown in diagram (a) of figure 10.

Step 3. Set attenuator C to at least 6 dB of attenuation.

Step 4. Increase gain of amplifier D from minimum value until VTVM H reads (C + K + 26) dBV across resistor L (750 ohms),

where:

C is calibration constant of condenser microphone and its associated amplifier in dBV relative to 1 microbar for pressure measurements

K is condenser microphone correction factor in dB for measurements made in free field

(C and K will be known parameters of the condenser microphone

NOTES.

1. For voltmeters not calibrated in dBV the required voltage corresponding to:

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(C + K + 26) dBV will be X volts where $\log_{10}$ X (volts) =

$$\frac{(C + K + 26)}{20} \text{ dBV.}$$

2. When this reading is obtained, the artificial mouth E will be delivering the preferred sound level of 100 dB relative to reference acoustic pressure (R.A.P.), namely 100 dB relative to 0.0002 microbar.

NOTE. Attainment of the exact level of 100 dB is not critical but it is desirable that a value approximating 100 dB is achieved.

Step 5. Set artificial line J to simulate zero loop length.

Step 6. Adjust feeding bridge K until telephone feed current reaches 100 mA.

Step 7. Remove condenser microphone F from in front of artificial mouth.

Step 8. Mount handset of telephone set under test in vertical plane such that plane of transmitter is 45 degrees to horizontal and at a distance of 1.5 cm as shown in diagram (b) of figure 10.

Step 9. Slowly rotate handset while in its holding fixture, through 300 degrees of arc (as shown in diagram (b) of figure 10) three times without jarring.

Step 10. Return handset to position of step 8.

Step 11. Momentarily (i.e., for 1-2 seconds) decrease attenuator C attenuation setting by 6 dB.

Step 12. Record VTVM H reading as V_{OT}, immediately on reinserting 6 dB attenuation in step 11 above.

NOTE. This reading should be taken before packing of the carbon grandules can occur, i.e., within 5 seconds of the conditioning effected in step 9 above.

Step 13. Repeat steps 7 through 12 for progressively increased simulated loop length of artificial line J in steps up to about 12,000 feet.

Step 14. Convert voltages recorded in steps 12 and 13 to dB (with respect to V_{OT}).

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NOTE. $dBx = 20 \log_{10} \frac{V_x}{V_{OT}}$

where:

V_{OT} is rms voltage measured with zero feet of artificial line inserted; and

V_x is rms voltage measured with X feet of artificial line inserted.

Step 15. Plot response of output level (dB) versus simulated loop length.

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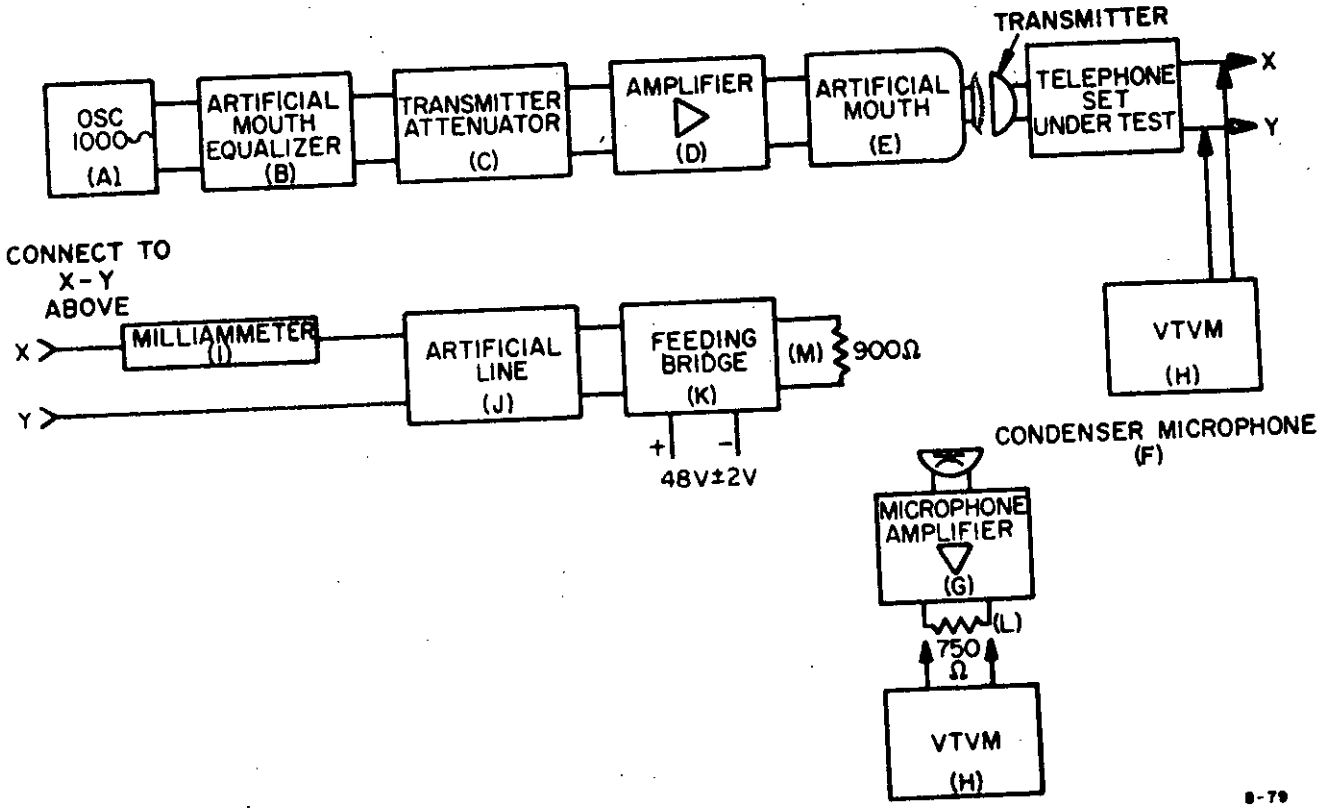
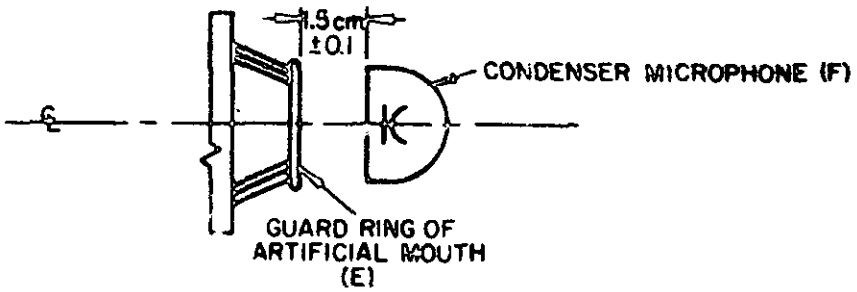
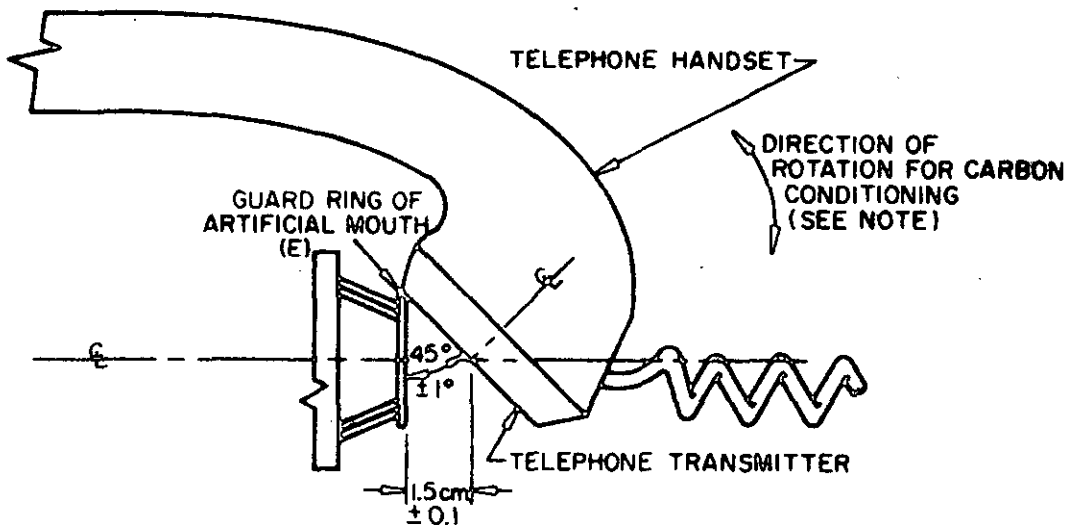


Figure 9. Measurement of telephone set transmitter level regulation.

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(a) CALIBRATE POSITION OF CONDENSER MICROPHONE



(b) MODEL POSITION OF TELEPHONE HANDSET

NOTE: TELEPHONE HANDSET SHALL BE ROTATED THROUGH A 300° ARC (IN THE PLANE OF PAPER) FOR TRANSMITTER CONDITIONING.

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Figure 10. Test positions of artificial mouth.

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6.2.1.4.2 *Measurement of telephone set receiver level regulation.*

Step 1. Connect equipment as shown in figure 11.

Step 2. Set artificial line J to simulate zero length.

Step 3. Adjust output of feeding bridge K to deliver 100 mA to telephone set under test as read on milliammeter I.

Step 4. Increase oscillator A output from zero until a reading of approximately V_{OT} volts (as measured in 6.2.1.4.1, step 1 above) is obtained at the line terminal of the telephone set.

Step 5. Record VTVM H voltage reading as V_{OR}.

Step 6. Progressively increase simulated loop length of artificial line J in steps up to about 12,000 feet.

Step 7. Record VTVM H voltage reading for each loop length of step 6.

Step 8. Convert the reading of steps 5 and 7 to dB values with respect to V_{OR}.

NOTE. $dBx = 20 \log \frac{V_x}{V_{OR}}$

where:

V_{OR} is signal voltage measured at the receiver terminals with zero feet of artificial line inserted; and

V_x is signal voltage measured at the receiver terminals with X feet of artificial line inserted.

Step 9. Plot response curve of receiver level regulation (dB) versus simulated loop length.

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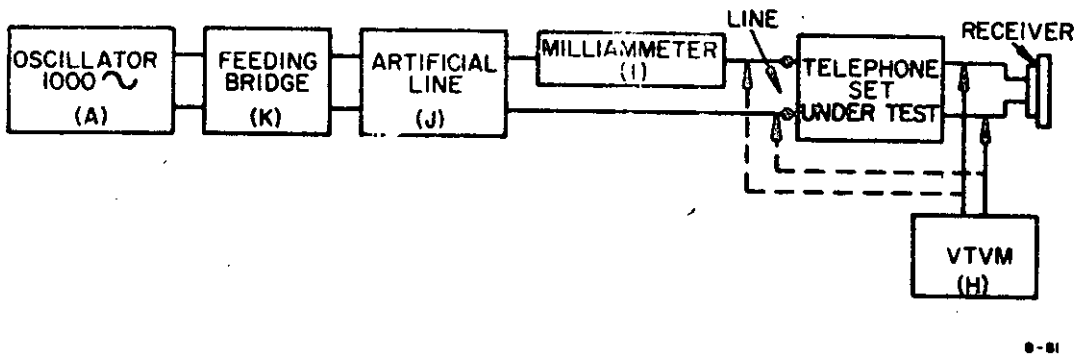


Figure 11. Measurement of telephone set receiver level regulation.

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6.2.2 *Telephone set and ringer impedance.*

6.2.2.1 *Scope.* This method is used to measure the overall telephone set impedance and ringer impedance.

6.2.2.2 *Applicability.* The parameters tested shall conform to the requirements specified in the paragraphs cited below.

Item	Paragraph
Ringer Impedance	4.7.2.10.1
Ringer Operation	4.7.2.10.2
Telephone Set Impedance	4.7.2.3

6.2.2.3 *Required test equipment.*

Test unit	Schematic reference	Item no. in appendix
AF Oscillator	A	174
Impedance Bridge	B	904
Detector Amplifier	C	503
AC Voltmeter, 0-100 V	D	37
Power Supply, 0-100 V dc	E	565
Inductor	F	437
Milliammeter, dc, 0-100 mA	G	8
Switch	H	942
AF Oscillator	I	173
Impedance Bridge	J	941

NOTES

1. Oscillator A (item no. 174) is not required if a 1000-Hz oscillator is included within the bridge.

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2. A detector-amplifier for indicating bridge unbalance will not be required if a detecting device is included in the impedance bridge. If a detecting device is not included within the impedance bridge and a detector-amplifier is not available, a pair of headphones may be used in place of the detector-amplifier for determining bridge balance.

6.2.2.4 *Procedures.*

6.2.2.4.1 *Measurement of ringer impedance.*

Step 1. Connect equipment as shown in figure 12.

NOTE. Do not remove the handset from its cradle as in this condition the ringer is in shunt across the line terminals.

Step 2. Set up impedance bridge B in accordance with operating instructions.

Step 3. Adjust oscillator A to output frequency of 200 Hz.

NOTE. The output level is not critical and should be such as to give a convenient value at detector amplifier C.

Step 4. Balance impedance bridge B.

Step 5. Record impedance bridge B dial settings.

Step 6. Calculate impedance of telephone set ringer from following formula:

$$|Z| = \sqrt{R^2 + (X_L - X_C)^2}$$

where:

R is resistance in ohms

X_L is inductive reactance = $2\pi fL$

X_C is capacitive reactance = $\frac{1}{2\pi fC}$

f is frequency setting of oscillator A in hertz

L is inductance in henrys

C is capacitance in farads

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Step 7. Repeat steps 4, 5, and 6 using appropriate oscillator frequencies between 200 and 3000 Hz, such as 400 Hz, 600 Hz, etc.

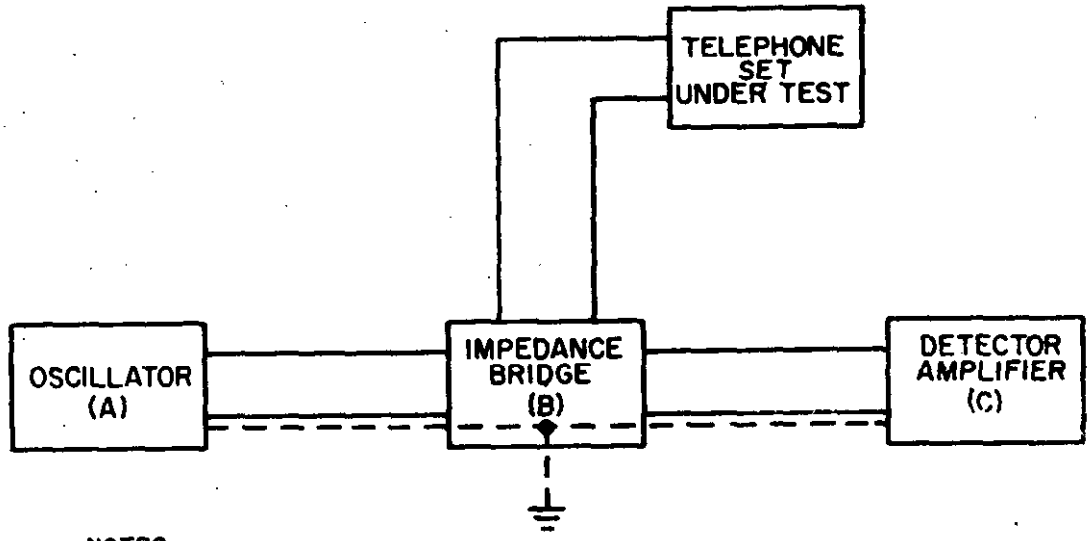


Figure 12. Measurement of ringer impedance.

6.2.2.4.2 *Test of ac ringer operation.*

Step 1. Connect equipment as shown in figure 13.

NOTE. Do not remove the handset of the telephone set under test from its cradle.

Step 2. Adjust oscillator 1 to an output of 15 Hz at 38 volts rms (as read on ac voltmeter D).

Step 3. Check ringer response.

NOTE. A uniformly steady ring is the criterion of satisfactory operation.

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Step 4. Adjust oscillator I frequency in appropriate steps, such as 1 or 2 Hz, to deliver frequencies between 15 and 25 Hz, maintaining 38 volts at all times, determining whether or not ringer operates satisfactorily.

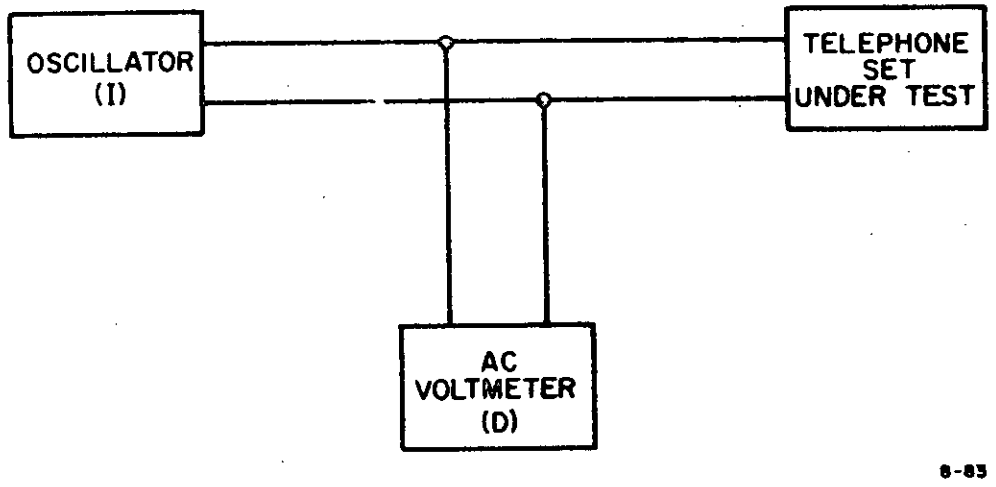


Figure 13. Test of ac ringer operation.

6.2.2.4.3 *Measurement of telephone set impedance.*

6.2.2.4.3.1 *Method 1.*

This method is used when an impedance bridge is available that will make measurements with dc flowing in the bridge. This method is preferred because of ease of measurement.

Step 1. Connect equipment as shown in figure 14.

NOTE. The impedance of the telephone set is to be determined for normal operation, i.e., with dc of the correct nominal value flowing through the instrument. Power supply E will furnish the required current.

Step 2. Set up impedance bridge J in accordance with operating instructions.

Step 3. Remove handset from cradle.

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Step 4. Adjust power supply E to deliver 74 mA read on milliammeter G.

Step 5. Set oscillator A to deliver 1000 Hz.

Step 6. Balance bridge J.

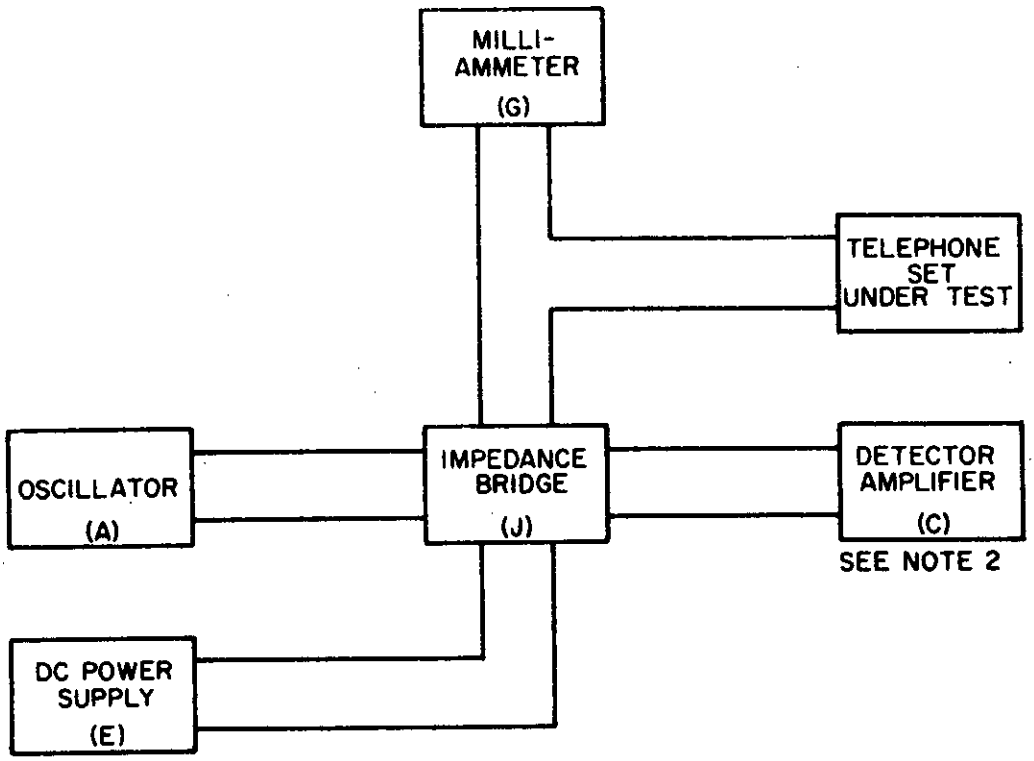
Step 7. Record bridge J dial setting.

Step 8. Calculate impedance of telephone set under test from following formulas:

$$Z = R + j\,(X_L - X_C);\ \text{or}\ |Z| = \sqrt{R^2 + (X_L - X_C)^2}$$

NOTE. X_L and X_C are defined in 6.2.2.4.1.

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NOTES:
1. TEST LEADS SHOULD BE TWISTED PAIR.
2. OSCILLATOR AND DETECTOR AMPLIFIER ARE NOT REQUIRED IF IMPEDANCE BRIDGE HAS INTERNAL 1000 Hz OSCILLATOR AND DETECTOR. HEADPHONES MAY BE USED IN PLACE OF DETECTOR AMPLIFIER.

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Figure 14. Measurement of telephone set impedance, method 1.

6.2.2.4.3.2 *Method 2.*

This method is to be used when the bridge used in method 1 is not available, and blocking is required to prevent dc from entering the impedance bridge.

Step 1. Connect equipment as shown in figure 15.

NOTE. The impedance of the telephone set is to be determined under normal operating conditions, i.e., with dc flowing through the instrument. Power supply E furnishes the required current. Capacitors C_1 and C_2 are used to block the dc

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from entering impedance bridge B. Inductor F adds a high ac impedance to the dc power supply. The high impedance power supply network, which is in shunt with the telephone set as seen by the impedance bridge, has negligible effect on the telephone set impedance measurement.

Step 2. Set up impedance bridge B in accordance with operating instructions.

Step 3. Place switch H in position 1.

Step 4. Adjust oscillator A to deliver a frequency of 1000 Hz.

Step 5. Balance impedance bridge B.

Step 6. Record value of resistance and capacitance of C_1 - C_2 network as read on dials of bridge.

Step 7. Turn switch H from position 1 to position 2.

Step 8. Remove handset from its cradle.

Step 9. Adjust power supply E to deliver 74 mA as read on milliammeter G.

Step 10. Set oscillator A to 1000 Hz.

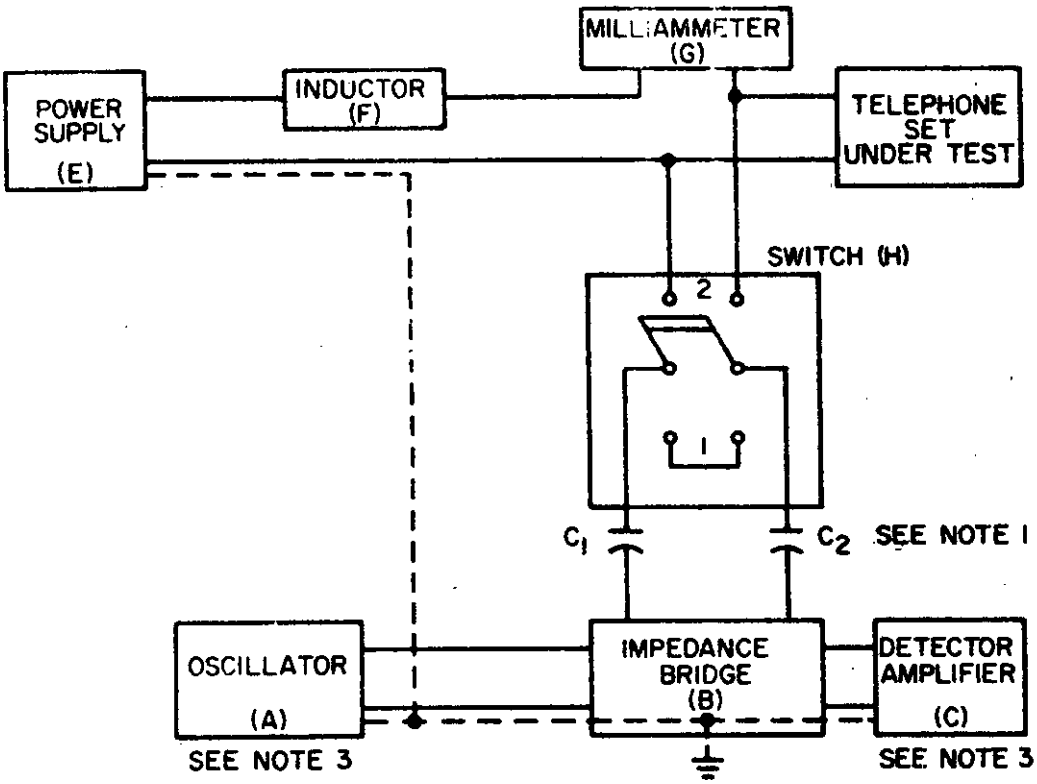
Step 11. Balance impedance bridge B.

Step 12. Read impedance components R_2, L, C from dials of the bridge.

Step 13. Calculate impedance as follows:

$$Z = R_2 + j\,(2\pi fL - \frac{1}{2\pi fC}); \text{ or } |Z| = \sqrt{R_2^2 + (2\pi fL - \frac{1}{2\pi fC})^2}$$

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NOTES:

1. CAPACITORS C_1 AND C_2 = APPROX 2 MICROFARADS EACH (100 VOLTS). ACTUAL VALUE IS NOT CRITICAL.
2. TEST LEADS SHOULD BE TWISTED PAIR.
3. OSCILLATOR AND DETECTOR AMPLIFIER ARE NOT REQUIRED IF IMPEDANCE BRIDGE HAS INTERNAL 1000 Hz OSCILLATOR AND DETECTOR. HEADPHONES MAY BE USED IN PLACE OF DETECTOR AMPLIFIER (C).

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Figure 15. Measurement of telephone set impedance, method 2.

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6.2.3 *Telephone set acoustical/electrical frequency characteristics.*

6.2.3.1 *Scope.* This method is used to measure the transmitter and receiver acoustical/electrical frequency characteristics of telephone sets.

6.2.3.2 *Applicability.* The parameters tested shall conform to the requirements specified in the paragraph cited below:

Item	Paragraph
Acoustical/Electrical Frequency Characteristics	4.7.2.4

6.2.3.3 *Required test equipment.*

Test unit	Schematic reference	Item no. in appendix
AF Oscillator*	A	126
Artificial Mouth Equalizer*	B	935**
Variable Attenuator*	C	600
Amplifier*	D	509
Artificial Mouth*	E	935**
Condenser Microphone	F	510
Microphone Amplifier	G	511
Vacuum Tube Voltmeter (VTVM), rms	H	18
Resistor, 750-ohm ±1%	I	435
Electro-Acoustic Test Set*		342

*Electro-Acoustic Test, item no. 342, may be used in place of items 126, 935, 600, and 509.

**Schematic references B and E are halves of the same unit.

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6.2.3.4 *Procedure.*

Step 1. Establish typical telephone connection between two test telephones using loops and interoffice trunk between manual or automatic switchboards as shown in figure 16.

NOTE. The telephone must be "off-hook" throughout the test and arrangements made with the exchange authorities so that disconnections will be prevented. Both telephone sets must be side by side in the test location unless additional test equipment is available for two separate test locations.

Step 2. Connect equipment as shown in figure 16.

Step 3. Locate condenser microphone F in correct position with respect to guard ring of artificial mouth E as shown in diagram (a) of figure 10.

Step 4. Set oscillator A to output frequency of 1000 Hz.

Step 5. Set attenuator C for at least 6 dB of attenuation.

Step 6. Increase gain of amplifier D from zero until VTVM H reads (C + K + 26) dBV across resistor I (750 ohms).

where:

C is calibration constant of condenser microphone and its associated amplifier in dBV relative to 1 microbar for pressure measurements

K is condenser microphone correction factor in dB for measurements made in a free field

(C and K will be known parameters of the condenser microphone used).

For voltmeters not calibrated in dBV, the required voltage corresponding to (C + K + 26) dBV will be X volts where:

$$\log_{10} X \text{ (volts)} = \frac{(C + K + 26)}{20} \text{ dBV}$$

NOTE. When this reading is obtained, artificial mouth E will be delivering the preferred sound level of 100 dB relative to reference acoustic pressure, namely, 100 dB relative to 0.0002 microbar.

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Step 7. Remove condenser microphone F from in front of artificial mouth.

Step 8. Mount handset of telephone set under test in vertical plane such that plane of transmitter is 45 degrees to horizontal and at a distance of 1.5 cm from the guard ring of the artificial mouth as shown in diagram (b) of figure 10.

Step 9. Condition carbon granules of transmitter by slowly rotating the handset while in its holding fixture, through 300 degrees of arc (as shown in diagram (b) of figure 10) three times without jarring.

Step 10. Return handset to position of step 8.

Step 11. Momentarily (i.e., for 1-2 seconds) reduce attenuator C attenuation by 6 dB.

Step 12. Record VTVM H readings at points Ⓐ , Ⓑ , and Ⓒ of figure 16 as V_{OA}, V_{OB}, and V_{OC}.

Step 13. Repeat measurements in steps 4 through 12 for other frequencies in appropriate steps (for example each 200 Hz) within band of from 200 to 3800 Hz.

NOTE. If the receiver has a pronounced principal resonance, smaller frequency intervals may be necessary.

Step 14. Convert voltage ratios of steps 13 and 14 to decibels using following formula:

$$dB = 20 \log_{10} \frac{V_X}{V_O}$$

where:

V_X is voltage measured at points Ⓐ , Ⓑ , or Ⓒ at frequency "X"

V_O is voltage measured at points Ⓐ , Ⓑ , or Ⓒ at 1000 Hz

Step 15. Plot response of telephone sets versus frequency.

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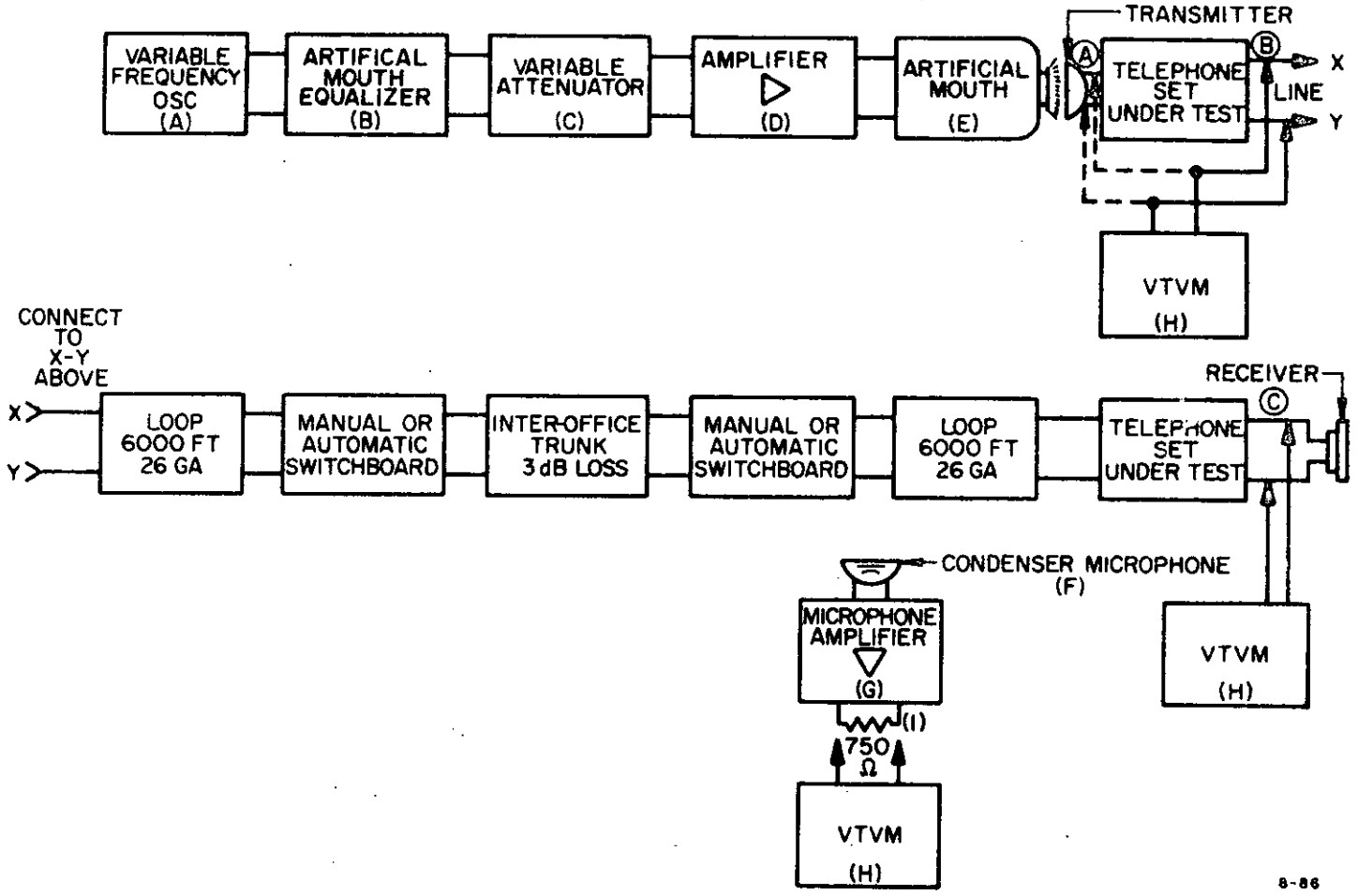


Figure 16. Measurement of telephone set acoustical/electrical frequency characteristics.

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6.2.4 *Telephone set dial pulse speed and break.*

6.2.4.1 *Scope.* This method is used to measure the dial pulse speed and dial pulse break percentage of telephone sets.

6.2.4.2 *Applicability.* The parameters tested shall conform to the requirements specified in the paragraph cited below:

Item	Paragraph
Dial Pulse Speed and Break	4.7.2.6

6.2.4.3 *Required test equipment.*

Test unit	Schematic reference	Item no. in appendix
Dial Test Set	A	337

6.2.4.4 *Procedure.*

Step 1. Connect equipment as shown in diagram (a) of figure 17.

NOTE. If the dial under test is mounted in a telephone set, connect the line terminal of the set to dial test set A. If the dial under test is unmounted, provide a bracket of the type mounted to the chassis of a telephone set and fasten it to a mounting board together with two terminal connectors. Mount the dial on the bracket and connect the leads from the pulsing contact of the dial to the connector terminals. Also connect these terminals to dial test set A.

Step 2. Set dial test set A to measure dial speed in accordance with operating instructions for test set.

Step 3. Lift handset from mounting if dial is equipped in telephone set.

Step 4. Dial digit 0.

Step 5. Record dial speed in pulses per second from dial test set A.

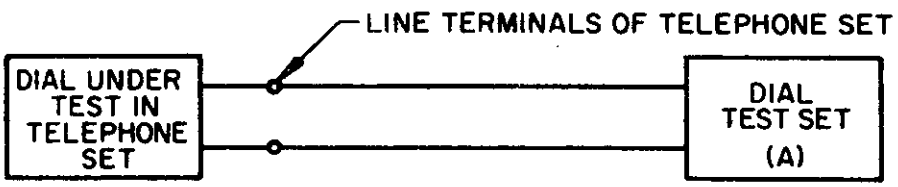
Step 6. Connect equipment as shown in diagram (b) of figure 17 and set dial test set A to measure break percentage.

Step 7. Lift handset from mounting if dial is equipped in telephone set.

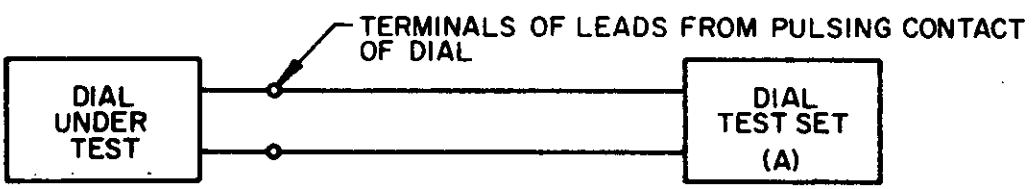
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Step 8. Dial digit 0.

Step 9. Record percent break from dial test set A.



(a) TEST OF DIAL IN TELEPHONE SET



(b) TEST OF SEPARATE DIAL

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Figure 17. Measurement of dial pulse speed and dial pulse break.

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6.2.5 *Dual tone multifrequency (DTMF) tones.*

6.2.5.1 *Scope.* This method is used to measure the frequencies, signaling output power, tone pulse, and keying rate of DTMF telephones.

6.2.5.2 *Applicability.* The parameters tested shall conform to the requirements specified in the paragraphs listed below:

Item	Paragraph
Keying Frequencies	4.7.2.9.1
Signaling Output Power	4.7.2.9.2
Tone Pulse Duration (repertory dials only)	4.7.2.9.3.1
Keying Rate (repertory dials only)	4.7.2.9.3.2

6.2.5.3 *Required test equipment.*

Test unit	Schematic reference	Item no. in appendix
Oscilloscope	A	200
Vacuum Tube Voltmeter (VTVM), rms	B	64 or 80
Frequency Counter	C	201, 211 or 255
Frequency Selective Voltmeter or Wave Analyzer	D	363 366
Milliammeter, dc, 0-100 mA	E	8
Power Supply, 0-100 V dc	F	565
Repeat Coil	G	659
Resistor, 900-ohm ±1%	H	436

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6.2.5.4 *Procedures.*

6.2.5.4.1 *Tone frequency measurement.*

Step 1. Connect equipment as shown in figure 18.

NOTE. Wave analyzer may be used in lieu of frequency selective voltmeter.

Step 2. Adjust power supply for 50 volts dc and 70 mA.

Step 3. Set frequency counter C for frequency range of interest.

Step 4. Set frequency selective voltmeter D bandwidth to 25 Hz (or less if available) and dB switch to an appropriate position.

Step 5. Depress and hold digit 1 key (697 and 1209 Hz) on the DTMF subset.

Step 6. Tune in 697-Hz component for peak reading on frequency selective voltmeter D and record voltage reading. Record frequency counter reading.

Step 7. Tune in 1209-Hz component for peak reading on frequency selective voltmeter D and record voltage reading. Record frequency counter reading.

Step 8. Release digit 1 key; depress and hold digit 2 key (697 and 1336 Hz).

Step 9. Tune in 1336-Hz component for peak reading on frequency selective voltmeter D and record voltage reading. Record frequency counter reading.

Step 10. Release digit 2 key; depress and hold digit 3 key (697 and 1477 Hz).

Step 11. Tune in 1477-Hz component for peak reading on frequency selective voltmeter D and record voltage reading. Record frequency counter reading.

Step 12. Release digit 3 key; depress and hold digit 4 key (770 and 1209 Hz).

Step 13. Tune in 770-Hz component for peak reading on frequency selective voltmeter D and record voltage reading. Record frequency counter reading.

Step 14. Release digit 4 key; depress and hold digit 7 key (852 and 1209 Hz).

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Step 15. Tune in 852-Hz component for peak reading on frequency selective voltmeter D and record voltage reading. Record frequency counter reading.

Step 16. Release digit 7 key; depress and hold 0 key (941 and 1336 Hz).

Step 17. Tune in 941-Hz component for peak reading on frequency selective voltmeter D and record voltage reading. Record frequency counter reading.

Step 18. Release digit 0 key; depress and hold FO key (697 and 1633 Hz).

Step 19. Tune in 1633-Hz component for peak reading on frequency selective voltmeter D and record voltage reading. Record frequency counter reading.

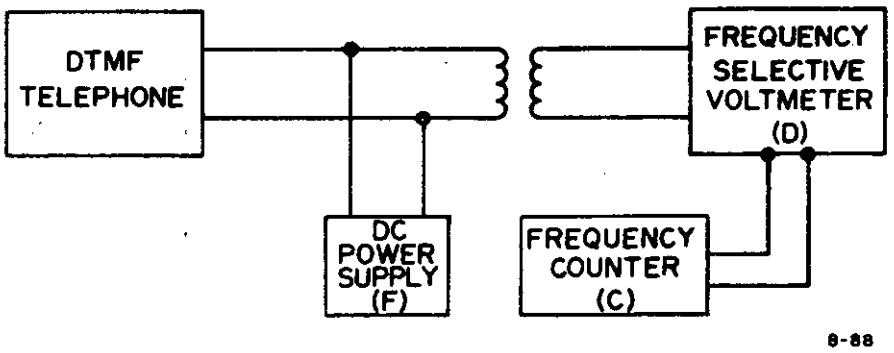


Figure 18. Measurement of tone frequencies, dual tone multifrequency telephone.

6.2.5.4.2 *Signaling output power tone level measurement.*

Step 1. Connect equipment as shown in figure 19.

Step 2. Adjust power supply F for 50 volts dc and 50 mA.

Step 3. Open subset to repeat coil at any convenient point and insert milliammeter E. Measure loop current with the subset off-hook. This should be 50 mA.

Step 4. Remove milliammeter E and restore the loop.

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Step 5. Press pairs of keys on subset simultaneously as shown below for single tone outputs. Measure the level of each tone on the VTVM B and record readings.

Keys	Tone output (Hz)
1 and 2	697
4 and 5	770
7 and 8	852
0 and A	941
1 and 4	1209
2 and 5	1336
3 and 6	1477
I and P	1633

Step 6. Repeat steps 2 through 5 for line loop currents of 70 mA and 100 mA.

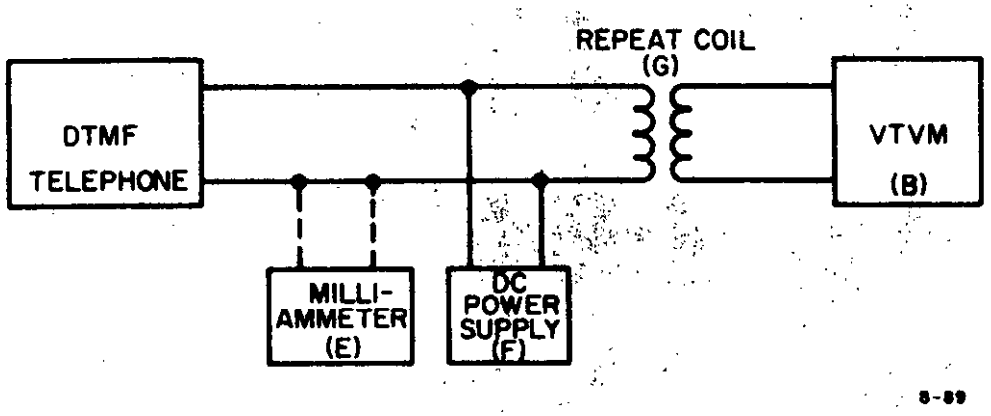


Figure 19. Measurement of tone level, dual tone multifrequency telephone.

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6.2.5.4.3 *Tone pulse duration (length) measurement (repertory dialer only).*

Step 1. Connect equipment as shown in figure 20.

Step 2. Adjust power supply F to 50 volts dc and 50 mA.

Step 3. Set oscilloscope A MODE switch to DELAY SWEEP, time/division to 50 milliseconds, and delay time range to 1.15 seconds.

Step 4. Adjust volts/cm for convenient vertical gain and make other adjustments as required to produce a good trace.

Step 5. From repertory dialer, key a seven-digit number (e.g., 650-0200).

Step 6. Observe display on the oscilloscope A. A train of seven tone pulses should be displayed. Each digit tone pulse should have a duration of not less than 45 milliseconds, with an interdigital tone pulse duration of not less than 45 milliseconds.

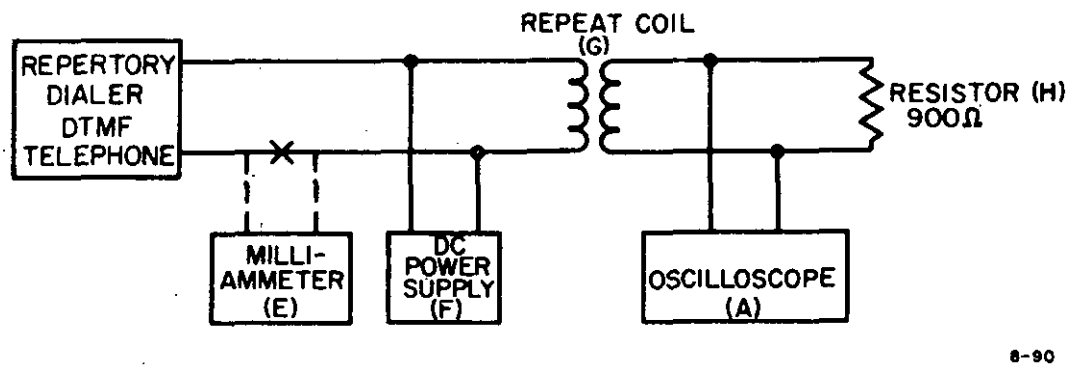


Figure 20. Measurement of keying speed and tone duration, dual tone multifrequency telephone with repertory dialer.

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6.2.5.4.4 *Keying rate measurement (repertory dialer only).*

Step 1. Connect equipment as shown in figure 20.

Step 2. Adjust power supply F for 50 volts dc and 50 mA.

Step 3. Set oscilloscope A MODE switch to DELAY SWEEP, time/division to 0.5 seconds, and delay time range to 1 second.

Step 4. Adjust volts/cm for convenient vertical gain and make other adjustments as required to produce a good trace.

Step 5. From repertory dialer, key a seven-digit number (e.g., 650-0200).

Step 6. Observe display on the oscilloscope A. A train of seven tone pulses at a rate not to exceed 10 pps should be displayed.

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6.2.6 *Telephone set noise.*

6.2.6.1 *Scope.* This method is used to measure the random noise generated within a telephone set.

6.2.6.2 *Applicability.* The parameters tested shall conform to the requirements specified in the paragraph cited below:

Item	Paragraph
Noise	4.7.2.11

6.2.6.3 *Required test equipment.*

Test unit	Schematic reference	Item no. in appendix
Noise Measuring Set	A	306
Oscilloscope	B	200
Feeding Bridge	C	940
Isolation Transformer	D	659
Milliammeter, dc, 0-100 mA	E	8
Resistor, 900-ohm ±1%	F	436

6.2.6.4 *Procedure.*

Step 1. Connect equipment as shown in figure 21.

Step 2. Adjust noise measuring set A so that it will measure with weighting network appropriate for use with telephone set under test.

Step 3. Remove telephone handset from cradle on telephone instrument.

Step 4. Isolate handset from ambient noise and mechanical vibration by covering transmitter with close fitting rubber cap or with other sound absorbing material and placing handset on sound and vibration absorbing surface.

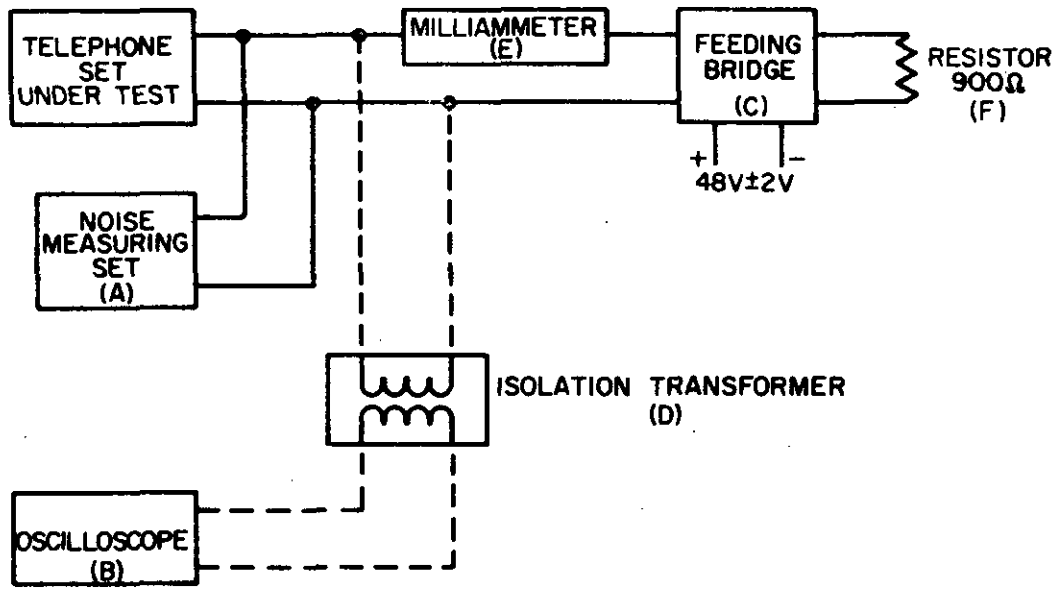
Step 5. Adjust potentiometer of feeding bridge C to supply maximum current to telephone set under test as observed on milliammeter E.

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Step 6. Calibrate noise measuring set A in accordance with manufacturer's operating instructions.

Step 7. Listen with high impedance headphone (at oscilloscope B input connected as shown in figure 21) to be certain that noise does not possess cyclic or impulse characteristic.

Step 8. Record weighted noise level.



NOTE: WHEN OSCILLOSCOPE (B) IS USED, ISOLATION TRANSFORMER (D) MUST BE INSERTED BETWEEN THE CIRCUIT UNDER TEST AND THE OSCILLOSCOPE.

8-91

Figure 21. Measurement of random noise generated within a telephone set.

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6.2.7 *Acoustical/electrical conversion efficiency.*

6.2.7.1 *Scope.* This method is used to measure the acoustical/electrical conversion efficiency of the transmitter and receiver of a telephone set.

6.2.7.2 *Applicability.* The parameters shall conform to the requirements specified in the paragraph cited below:

Item	Paragraph
Acoustical/Electrical Conversion Efficiency	4.7.3.1

6.2.7.3 *Required test equipment.*

Test unit	Schematic reference	Item no. in appendix
AF Oscillator*	A	126 or 1106
Artificial Mouth Equalizer*	B	935**
Variable Attenuator*	C	600
Amplifier*	D	509
Artificial Mouth	E	935**
Condenser Microphone	F	510
Microphone Amplifier	G	511
Vacuum Tube Voltmeter (VTVM), rms	H	18
Milliammeter, dc, 0-50/100 mA	I	48
Artificial Line	J	939
Feeding Bridge	K	940
Resistor, 750-ohm ±1%	L	435
Resistor, 900-ohm ±1%	M	436

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Test unit	Schematic reference	Item no. in appendix
6 cc Coupler	N	
Precision Sound-Level Meter	O	998

*Electro-Acoustic Test, item no. 342, may be used in place of the above items

**Schematic references B and E are halves of the same unit.

6.2.7.4 *Procedures.*

6.2.7.4.1 *Transmitter.*

6.2.7.4.1.1 *Transmitter rating without loop loss.*

Step 1. Connect equipment as shown in figure 22.

NOTE. Items A through E constitute the sound source and items F, G, and H constitute a means for calibrating the sound source. Frequency 1000 Hz (125, 250, 500, and 2000 Hz).

Step 2. Locate condenser microphone F in correct position with respect to guard ring of artificial mouth as shown in diagram (a) of figure 10.

Step 3. Set attenuator C to at least 6 dB of attenuation.

Step 4. Increase gain of amplifier D from minimum value until VTVM H reads (C + K + 26) dBV across resistor L (750 ohms).

where:

C is calibration constant of condenser microphone and its associated amplifier in dBV relative to 1 microbar for pressure measurements

K is condenser microphone correction factor in dB for measurements made in free field

(C and K will be known parameters of the condenser microphone used).

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NOTES.

1. For voltmeters not calibrated in dBV the required voltage corresponding to (C + K + 26) dBV will be X volts where $\log_{10}$ X (volts) =

$$\frac{(C + K + 26)}{20} \text{ dBV.}$$

2. When this reading is obtained, artificial mouth E will be delivering the preferred sound level of 100 dB relative to reference acoustic pressure, namely 100 dB relative to 0.0002 microbar.

3. Attainment of the exact level of 100 dB is not critical but it is desirable that a value approximating 100 dB be achieved.

Step 5. Adjust feeding bridge K until telephone feed current reaches 100 mA.

Step 6. Remove condenser microphone F from in front of artificial mouth (refer to diagram (a) of figure 10).

Step 7. Mount handset of telephone under test in the vertical plane such that plane of transmitter is 45 degrees to or from horizontal and at a distance of 1.5 cm from the guard ring of the artificial mouth as shown in diagram (b) of figure 10.

Step 8. Slowly rotate handset while in its holding fixture through 300 degrees of arc (as shown in diagram (b) of figure 10) three times without jarring.

Step 9. Return handset to position of step 7.

Step 10. Momentarily (i.e., for 1-2 seconds) decrease attenuator C attenuation setting by 6 dB.

Step 11. Record VTVM H reading as V_{TS}, immediately on reinserting 6-dB attenuation as directed in step 10 above.

NOTE. This reading should be taken before packing of the carbon granules can occur, i.e., within 5 seconds.

Step 12. Repeat steps 1 through 11 for the following frequencies: 125 Hz, 240 Hz, 500 Hz, and 2000 Hz.

Step 13. Determine transmitter rating for each of the frequencies with the form [illegible] below:

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$$R_T = 20 \log_{10} \frac{V_{TS}}{S_T}$$

where:

R_T is rating in dB

V_{TS} is voltage across 900-ohm resistor

S_T is sound pressure reference, 0.0002 microbar

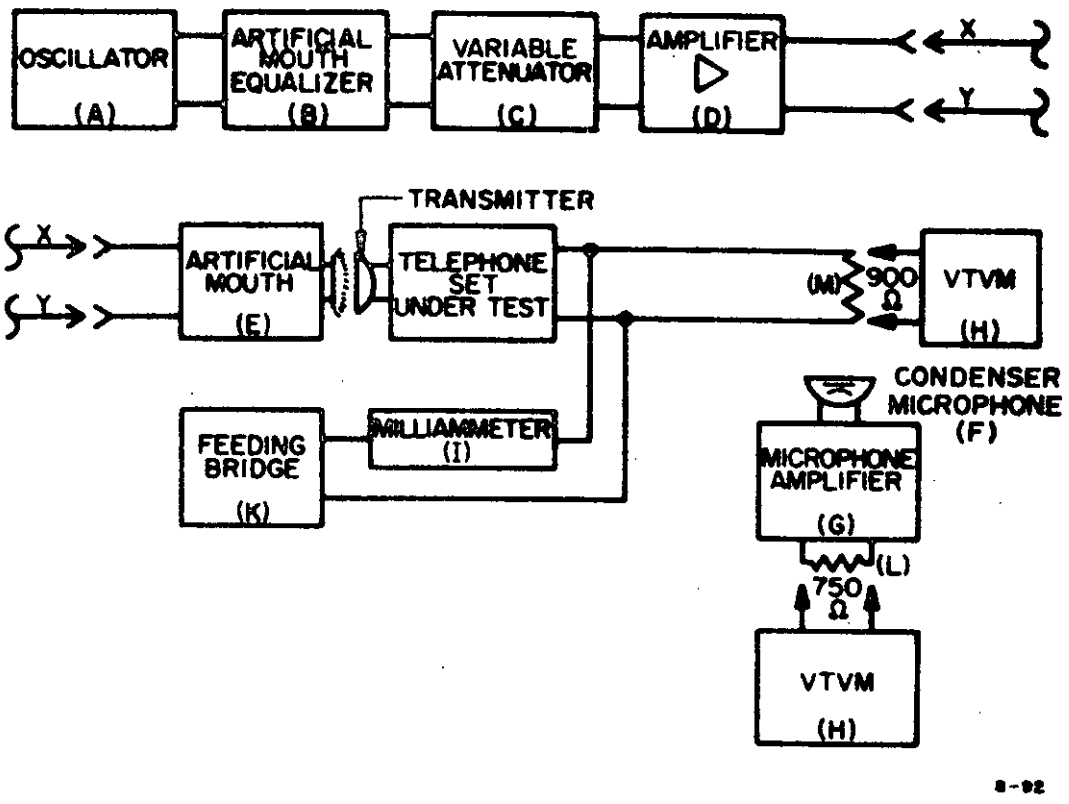


Figure 22. Measurement of acoustical/electrical conversion efficiency of telephone set transmitter without loop loss.

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6.2.7.4.1.2 *Transmitter rating with loop loss.*

Step 1. Connect equipment as shown in figure 23.

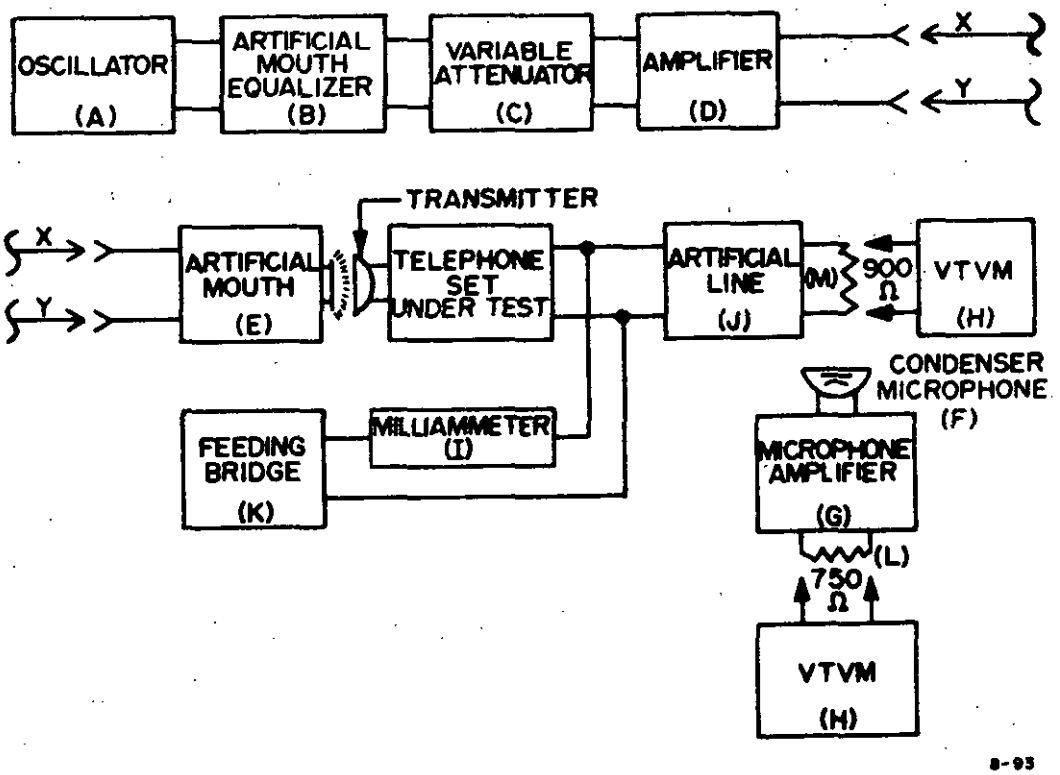


Figure 23. Measurement of acoustical/electrical conversion efficiency of telephone set transmitter with loop loss.

Step 2. Set artificial line J to simulate 10,560-foot loop length.

Step 3. Repeat steps 1 through 12 in 6.2.7.4.1.1 above.

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Step 4. Determine transmitter loop loss rating for each of the frequencies with the formula below:

$$L_T = 20 \log_{10} \frac{V_{TL}}{S_T}$$

where:

L_T is loop loss in dB

V_{TL} is voltage across 900-ohm resistor M

S_T is sound pressure reference, 0.0002 microbar

6.2.7.4.2 *Receiver.*

6.2.7.4.2.1 *Receiver rating without loop loss.*

Step 1. Connect equipment as shown in figure 24.

Step 2. Adjust feeding bridge K until telephone feed current reaches 100 mA.

Step 3. Adjust oscillator A output for 1000 Hz, 0 dB.

Step 4. Measure voltage across the 900-ohm resistor M and record.

Step 5. With the oscillator A output of 1000 Hz, 0 dB, measure sound output pressure on the sound level meter O and record.

Step 6. Repeat steps 1 through 5 for the frequencies of 125 Hz, 250 Hz, 500 Hz, and 2000 Hz.

Step 7. Determine receiver rating for each frequency with the formula listed below:

$$R_L = 20 \log_{10} \frac{S_{LS}}{\frac{V_W}{2}}$$

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where:

R_L is receiver rating in dB

S_{LS} is sound pressure in millions of microbars

$\frac{V_W}{2}$ is voltage across 900-ohm resistor

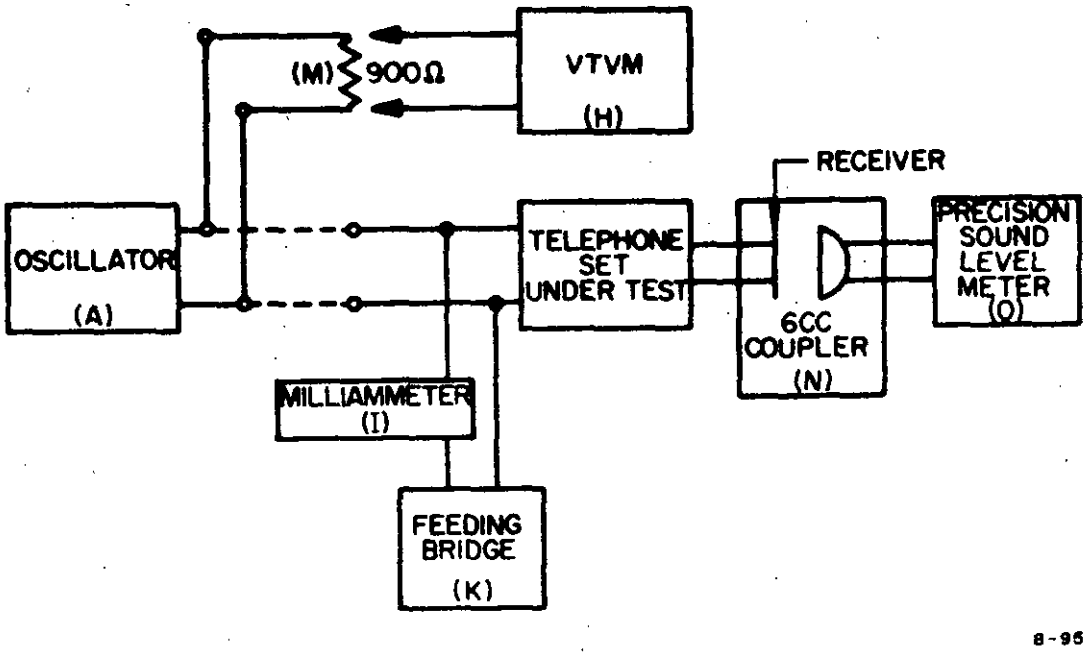


Figure 24. Measurement of acoustical/electrical conversion efficiency of telephone set receiver without loop loss.

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6.2.7.4.2.2 *Receiver rating with loop loss.*

Step 1. Connect equipment as shown in figure 25.

Step 2. Set artificial line J to simulate 10,560 foot loop length.

Step 3. Repeat steps 2 through 7 in 6.2.7.4.2.1 above.

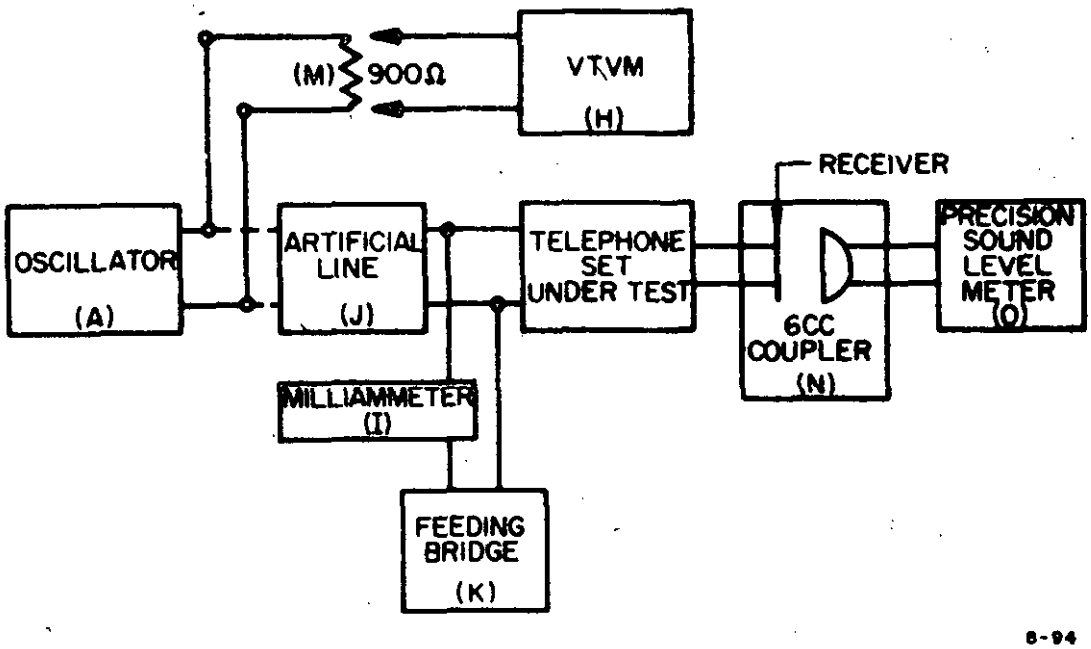


Figure 25. Measurement of acoustical/electrical conversion efficiency of telephone set receiver with loop loss.

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6.2.8 *Dial central office rectifiers.*

6.2.8.1 *Scope.* This method is used to measure the operating performance of the dial central office rectifiers.

6.2.8.2 *Applicability.* The parameters tested shall conform to the requirements specified in the paragraphs cited below:

Item	Paragraph
Rectifier	
Input Power	5.2.1.1
Output Voltage and Current	5.2.1.2
Line and Load Regulation	5.2.1.3
Overvoltage Protection	5.2.1.4
End-Cell Rectifier	
Input Power	5.2.2.1
Output Voltage	5.2.2.2
Line and Load Regulation	5.2.2.3
Overvoltage Protection	5.2.2.4

6.2.8.3 *Required test equipment.*

Test unit	Schematic reference	Item no. in appendix
Variable Frequency Power Supply, 100-150 V ac, 47-63 Hz, kVA rating as required	A	
Variable Auto-Transformer, 100-150 V ac, kVA rating as required	B	
Voltmeter, 0-150 V ac	C	38

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Test unit	Schematic reference	Item no. in appendix
Voltmeter, 0-150 V dc	D	54
Frequency Meter	E	39
Ammeter, dc, current rating as required	F	
Dummy Load, current rating as required	G	
Switch, SPST, current rating as required	H	

NOTE. Dangerous voltages exist on equipment being tested. Consult manufacturers instructions prior to conducting any test.

6.2.8.4 *Procedures.*

6.2.8.4.1 *Rectifier.*

6.2.8.4.1.1 *Input voltage and input frequency.*

Step 1. Connect equipment as shown in figure 26.

Step 2. Adjust variable frequency power A supply for an output of 120 volts, 60 Hz.

Step 3. Adjust variable auto-transformer B for an output of 120 volts.

Step 4. Adjust rectifier for an output of 50 volts.

Step 5. Adjust dummy load G for full-load condition.

Step 6. Vary variable auto-transformer B output in 2-volt steps over the range of from 114 to 126 volts.

Step 7. At each 2-volt step change, vary frequency output of the variable frequency power supply A over the range of from 47 to 63 Hz.

Step 8. With each variation of voltage input, measure and record rectifier output voltage on the voltmeter D.

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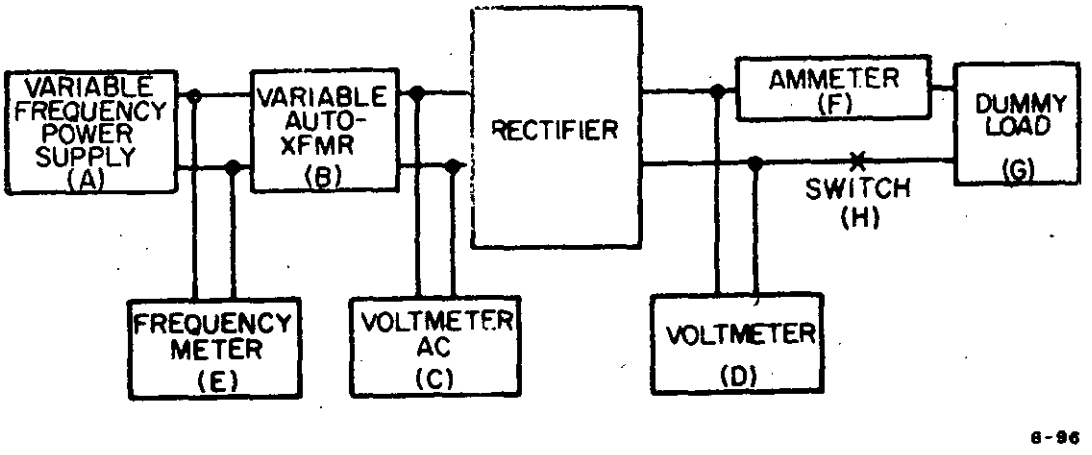


Figure 26. Measurement of rectifier output voltage and current vs input voltage and frequency, together with overload protection.

6.2.8.4.1.2 *Output current.*

Step 1. Connect equipment as shown in figure 26.

Step 2. Adjust variable frequency power supply A, variable auto-transformer B, and rectifier as stated in 6.2.8.4.1.1, steps 2, 3, and 4 respectively.

Step 3. Adjust dummy load G for the calculated expected load of the dial central office.

NOTE. This load is the 48 volts dc current requirement of the dial central office (e.g., 30 A, 80 A, 200 A, etc).

Step 4. Measure and record the rectifier voltage and current output.

Step 5. Vary variable auto-transformer B output in 2-volt steps over the range of from 114 to 126 volts.

Step 6. At each 2-volt step change, vary frequency output of the variable frequency power supply A over the range of from 47 to 63 Hz.

Step 7. With each variation of voltage and frequency input to the rectifier, measure and record voltage and current output of the rectifier.

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6.2.8.4.1.3 *Overvoltage protection.*

Step 1. Connect equipment as shown in figure 26.

Step 2. Adjust rectifier overvoltage protection sensing to a preset value between 52 and 58.5 volts.

Step 3. Adjust voltage input to rectifier to 120 volts, 60 Hz.

Step 4. Adjust rectifier output to 50 volts.

Step 5. Increase rectifier output to overvoltage preset sensing value.

Step 6. When overvoltage preset value is reached, rectifier output voltage should drop to zero, as indicated by the voltmeter D.

6.2.8.4.1.4 *Output voltage.*

Step 1. Connect equipment as shown in figure 26.

Step 2. Adjust variable frequency power supply A for an output of 120 volts, 60 Hz.

Step 3. Adjust variable auto-transformer B for an output of 120 volts.

Step 4. Adjust rectifier for an output of 50 volts.

Step 5. Adjust dummy load G for full-load condition.

Step 6. Vary rectifier voltage output from 48 to 58.5 volts, and switch dummy load G on and off at various output voltages throughout the range.

Step 7. Verify that rectifier output can be varied over the range in step 6 by observing the voltmeter D.

Step 8. Readjust rectifier for an output of 50 volts.

6.2.8.4.1.5 *Line and load regulation.*

Step 1. Connect equipment as shown in figure 27.

Step 2. Adjust variable frequency power supply A, variable auto-transformer B and rectifier as stated in 6.2.8.4.1.1, steps 2, 3, and 4 respectively.

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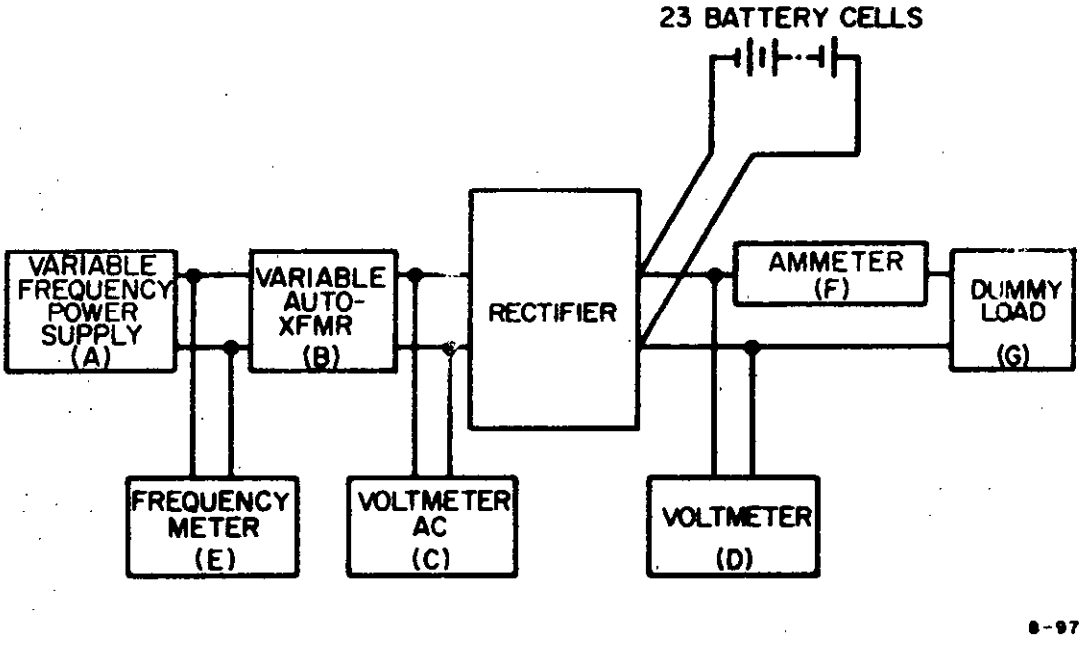


Figure 27. Measurement of rectifier line and load regulation.

Step 3. Adjust dummy load G for no-load condition.

Step 4. Measure and record the output of the rectifier.

Step 5. Measure and record the voltage and frequency input to the rectifier.

Step 6. Adjust dummy load G for the calculated expected load of the dial central office.

NOTE. This load is the 48 volts dc current requirement of the dial central office (e.g., 30 A, 80 A, 200 A, etc.).

Step 7. Measure and record the voltage output of the rectifier.

Step 8. Measure and record the voltage and frequency input to the rectifier.

Step 9. Repeat steps 3 through 8, varying the voltage input in 2-volt steps over the range of from 114 to 126 volts. At each 2-volt step vary the frequency over the range of from 47 to 63 Hz.

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NOTE. Rectifier output is at 50 volts.

Step 10. Measure and record the voltage output of the rectifier and the ac voltage and frequency input to the rectifier at each 2-volt interval.

6.2.8.4.2 *End-cell rectifier.*

6.2.8.4.2.1 *Output voltage.*

Step 1. Connect equipment as shown in figure 28.

Step 2. Adjust variable frequency power supply A for an output of 120 volts, 60 Hz.

Step 3. Adjust variable auto-transformer B for an output of 120 volts.

Step 4. Adjust end-cell rectifier for an output of 4 volts.

Step 5. Adjust dummy load G for full-load current condition.

Step 6. Vary end-cell rectifier output from 2 to 7.1 volts.

Step 7. Verify that end-cell rectifier output can be varied over the voltage range in step 6 by observing the voltmeter D.

Step 8. Readjust end-cell rectifier output to 4 volts.

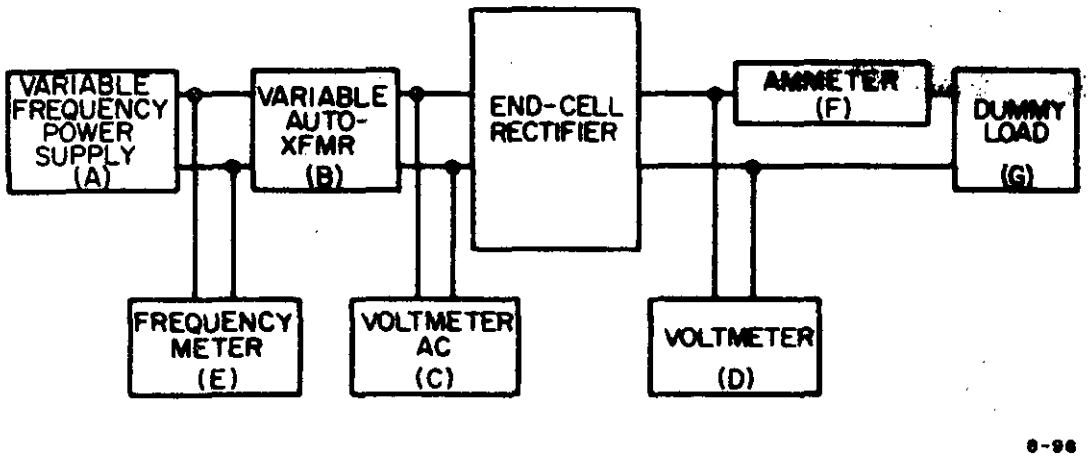


Figure 28. Measurement of end-cell rectifier output voltage and current vs input voltage and frequency, overload protection, and line and load regulation.

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6.2.8.4.2.2 *Input voltage and input frequency.*

Step 1. Connect equipment as shown in figure 27.

Step 2. Adjust variable frequency power supply A for an output of 120 volts, 60 Hz.

Step 3. Adjust variable auto-transformer B for an output of 120 volts.

Step 4. Adjust end-cell rectifier for an output of 4 volts.

Step 5. Adjust dummy load G for full-load current condition.

Step 6. Vary variable auto-transformer B output in 2-volt steps over the range of from 114 to 126 volts.

Step 7. At each 2-volt step change, vary the frequency output of the variable frequency power supply A over the range of from 47 to 63 Hz.

Step 8. With each variation of voltage input, measure and record the end-cell rectifier output voltage on the voltmeter D.

6.2.8.4.2.3 *Line and load regulation.*

Step 1. Connect equipment as shown in figure 28.

Step 2. Adjust variable frequency power supply A, variable auto-transformer B as stated in 6.2.8.4.2.1, steps 2 and 3 respectively.

Step 3. Adjust end-cell rectifier output for 4 volts.

Step 4. Adjust dummy load G for no-load condition.

Step 5. Measure and record the voltage output of the rectifier, and the input voltage and frequency to the rectifier.

Step 6. Adjust dummy load G for full-load current.

Step 7. Measure and record the voltage and frequency input to the rectifier, and voltage output of the rectifier.

Step 8. Repeat steps 3 through 7, varying the voltage input in 2-volt steps over the range of from 114 to 126 volts. At each 2-volt step vary the frequency

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over the range from 47 to 63 Hz, and dummy load G for no-load and full-load current conditions.

6.2.8.4.2.4 *Overvoltage protection.*

Step 1. Connect equipment as shown in figure 28.

Step 2. Adjust end-cell rectifier overvoltage protection sensing to a preset value between 2.1 and 7.5 volts.

Step 3. Adjust voltage input to end-cell rectifier to 120 volts, 60 Hz.

Step 4. Adjust end-cell rectifier output voltage to a value below the overvoltage protection preset value.

Step 5. Increase end-cell rectifier output voltage to overvoltage preset value.

Step 6. When overvoltage preset value is reached, end-cell rectifier output voltage should drop to zero, as indicated by voltmeter D.

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6.2.9 *Ringing/signaling equipment.*

6.2.9.1 *Scope.* This method is used to measure the output tones and interruptions of a ring and signaling tone generator.

6.2.9.2 *Applicability.* The parameters tested shall conform to the requirements specified in the paragraphs cited below:

Item	Paragraph
Mixed Tones	5.3.1.1
Modulated Tones	5.3.1.2
Interruption of Tones	5.3.1.1-2

6.2.9.3 *Required test equipment.*

Test unit	Schematic reference	Item no. in appendix
Frequency Selective Voltmeter	A	363
Wave Analyzer	B	366
Oscilloscope	C	256
Frequency Counter	D	280

6.2.9.4 *Procedures.*

NOTE. Tones are to be measured at the output terminals of the ring and tone generator. The measurement of the interruptions are to be made at the output terminals of the interrupter.

6.2.9.4.1 *Mixed tones.*

Step 1. Connect equipment as shown in figure 29.

Step 2. Tune selective frequency meter A to measure the levels of the individual frequencies listed.

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Mixed tones	Individual frequencies (Hz)
Dial Tone, 350 Hz + 440 Hz	350, 440
Station Busy, 480 Hz + 620 Hz	480, 620
Trunks Busy, 480 Hz + 620 Hz	480, 620
Ringback Tone 440 Hz + 480 Hz	440, 480

Step 3. Observe and record the level of the individual frequency being measured and the frequency as indicated on the frequency counter D.

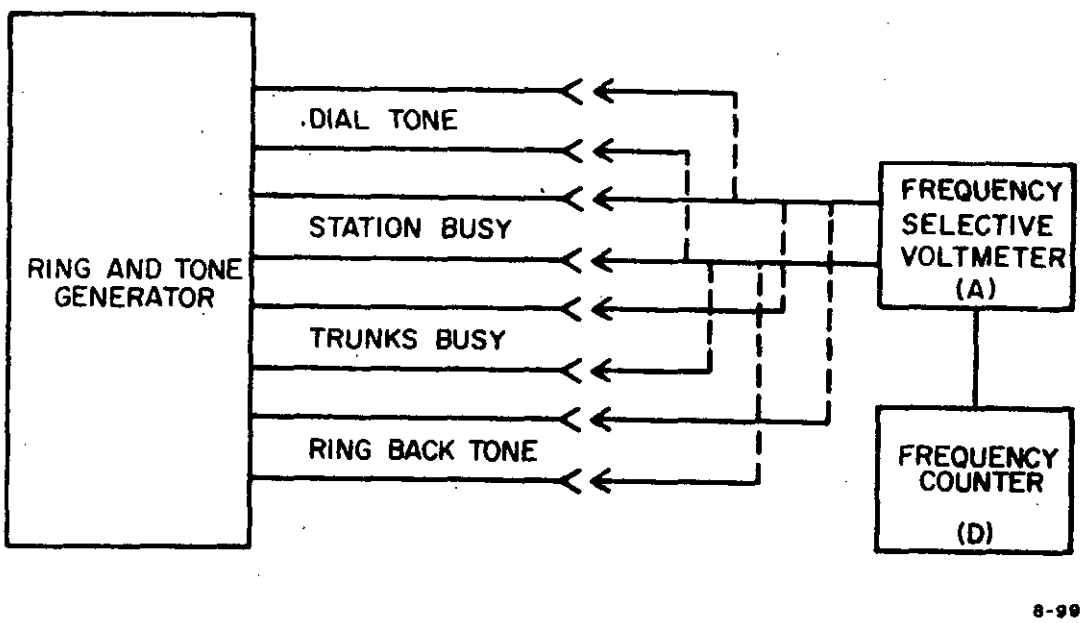


Figure 29. Measurement of mixed tone of ring and tone generator.

6.2.9.4.2 *Modulated tones.*

Step 1. Connect equipment as shown in figure 30.

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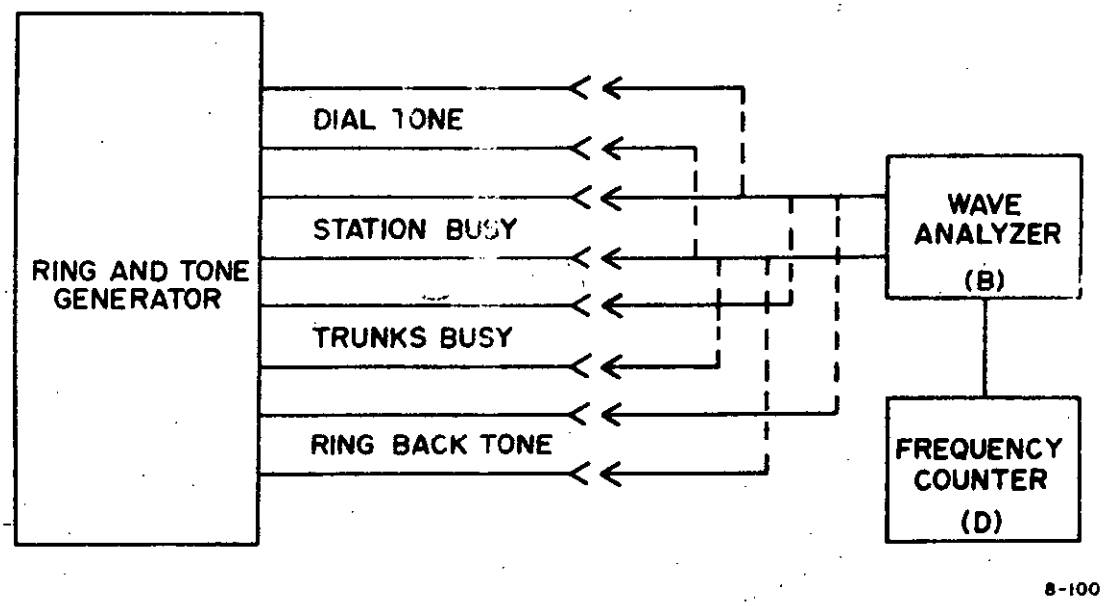


Figure 30. Measurement of modulated tones of ring and tone generator.

Step 2. Tune wave analyzer B to measure the levels of the individual frequencies listed, using suitably narrow bandwidth setting.

Modulated tones	Individual frequencies (Hz)
Dial Tone, 600 Hz/120 Hz	600, 120
Station Busy, 600 Hz/120 Hz	600, 120
Ringback Tone, 400 Hz/40 Hz (420 Hz/40 Hz)	400 (420), 40

Step 3. Observe and record the level and frequency of the individual frequencies being measured.

Step 4. With the wave analyzer D set for broad bandwidth, record the level of the modulated tones.

6.2.9.4.3 *Interruption of tones.*

Step 1. Connect equipment as shown in figure 31.

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Step 2. Adjust oscilloscope C to measure the interruptions listed below.

Signal	Interruptions	Tone on	Tone off
Dial Tone	Continuous		
Station Busy	60 IPM	0.5 s	0.5 s
Trunks Busy	120 IPM	0.25 s	0.25 s
Ringback Tone	10 IPM	1 s	4 s

Step 3. Observe and record the interruptions of the tones.

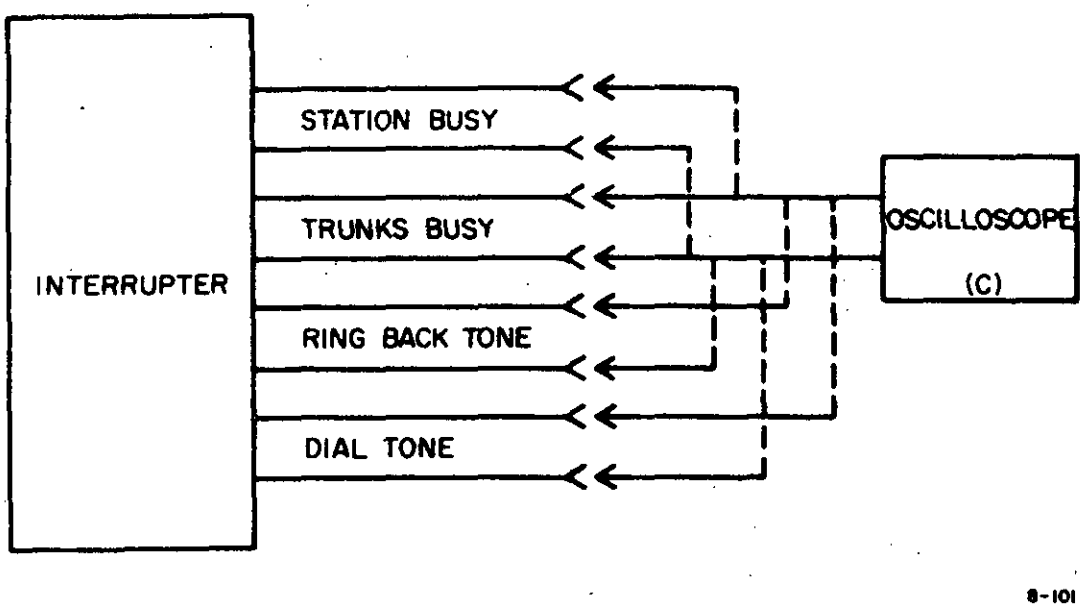


Figure 31. Measurement of interrupted tones of ring and tone generator.

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6.2.10 *Dial loop to E&M converter.*

6.2.10.1 *Signal conversion.*

6.2.10.1.1 *Scope.* This method is used to determine whether the converter will convert E&M signals to dc signals and dc signals to E&M signals.

6.2.10.1.2 *Applicability.* The parameters tested shall conform to the requirements specified in the paragraphs cited below:

Item	Paragraph
Dial Loop to E&M Converter	
Subscriber Unit	5.4.1
Central Office Unit	5.4.2

6.2.10.1.3 *Required test equipment.*

Test unit	Schematic reference	Item no. in appendix
Ohmmeter	A	12
Voltmeter	B	7
Pulse Signaling Test Set	C	121
Power Supply, 0-100 V dc	D	565
20-Hz, 90-volt Ringing Source	E	
Frequency Counter	F	211

6.2.10.1.4 *Procedures.*

6.2.10.1.4.1 *E&M to dc/dc to E&M.*

NOTE. Consult manufacturers specifications for strapping options for individual modes or operation of converter units.

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6.2.10.1.4.1.1 *M-lead condition -- on-hook/off-hook.*

Step 1. Connect equipment as shown in figure 32.

Step 2. Adjust power source D for -48 volt dc.

Step 3. Apply a ground to the ST-lead.

NOTE. On-hook indication is indicated by a zero reading on the voltmeter B (ground on M-lead).

Step 4. Remove ground from the ST-lead.

NOTE. Off-hook indication is indicated by a -48 volts dc reading on the voltmeter B (-48 volts dc on M-lead).

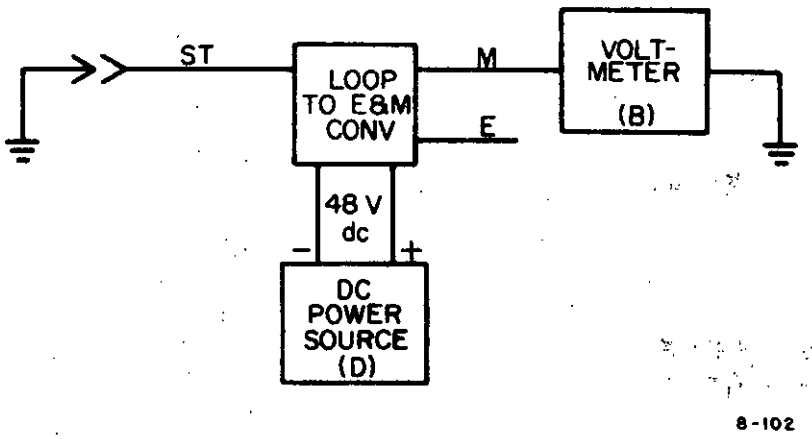


Figure 32. Dial loop to E&M converter operational check, M-lead condition -- on-hook/off-hook.

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6.2.10.1.4.1.2 *E-lead condition -- on-hook/off-hook.*

Step 1. Connect equipment as shown in figure 33.

Step 2. With no ground applied to the E-lead, there should be an infinite reading on the ohmmeter A.

NOTE. This infinite reading indicates an off-hook condition (ground on ST-lead).

Step 3. Apply a ground on the E-lead. There should be a zero reading on the ohmmeter A.

NOTE. Zero reading indicates an on-hook condition (open, no potential on G-lead).

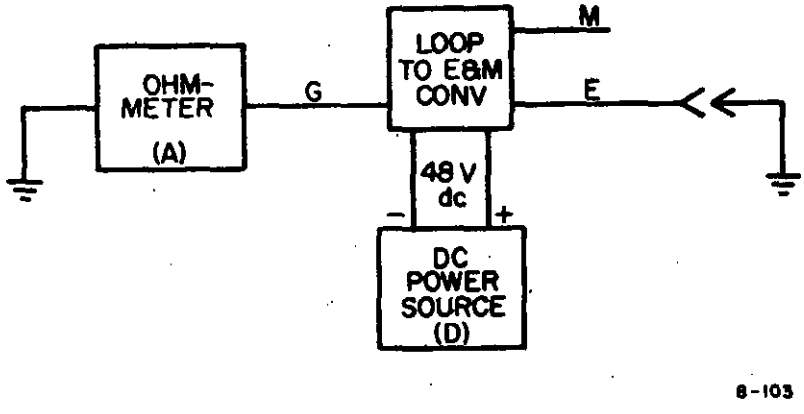


Figure 33. Dial loop to E&M converter operational check, E-lead condition -- on-hook/off-hook.

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6.2.10.1.4.2 *Pulses per second.*

6.2.10.1.4.2.1 *M-lead.*

Step 1. Connect equipment as shown in figure 34.

Step 2. Adjust pulse signaling test set C for the following output signals:

PPS	% break
9	58
10	58
11	58
12	58

Step 3. Observe the pulses and break percentage for each of the above output signals.

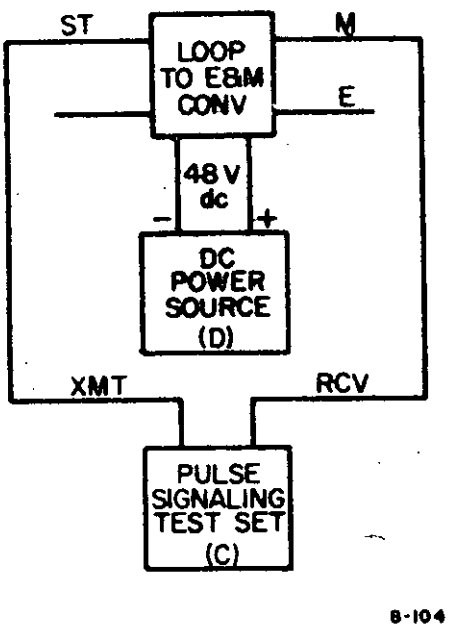


Figure 34. Dial loop to E&M converter operational check, M-lead condition -- pulses per second.

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6.2.10.1.4.2.2 *E-lead.*

Step 1. Connect equipment as shown in figure 35.

Step 2. Repeat steps 2 and 3 of 6.2.10.1.4.2.1.

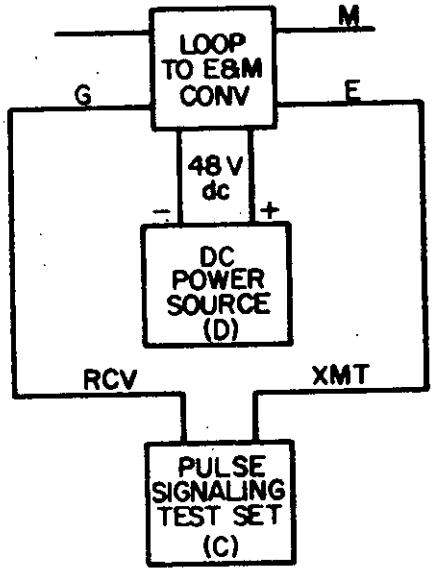


Figure 35. Dial loop to E&M converter operational check, E-lead condition -- pulses per second.

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6.2.10.1.4.3 *20-Hz ringing.*

6.2.10.1.4.3.1 *E-lead condition -- off-hook.*

Step 1. Connect equipment as shown in figure 36.

Step 2. Establish an on-hook E-lead condition by applying ground to the E-lead.

NOTE. With an E-lead on-hook condition, both voltmeter B and frequency counter F should read 0.

Step 3. Remove the ground from the E-lead for an E-lead off-hook condition.

NOTE. 20 Hz, 90 volts should appear on the tip and ring of the converter.

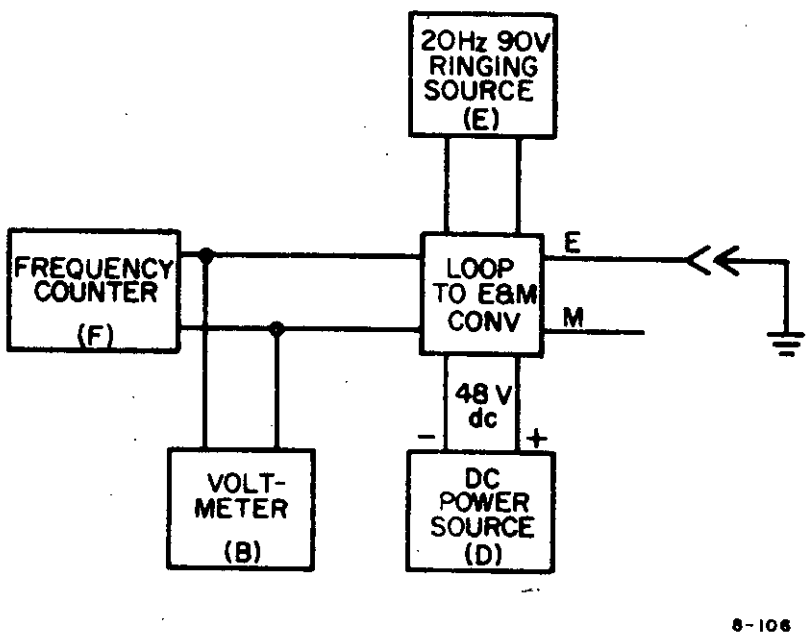


Figure 36. Dial loop to E&M converter operational check, 20-Hz ringing, E-lead condition -- off-hook.

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6.2.10.1.4.3.2 *M-lead condition -- off-hook.*

Step 1. Connect equipment as shown in figure 37.

NOTE. E-lead condition is on-hook. M-lead on-hook is indicated by -48 volts dc on the voltmeter B.

Step 2. Apply 20 Hz, 90 volts to the converter's tip and ring leads.

NOTE. M-lead off-hook is indicated by a zero reading on the voltmeter B.

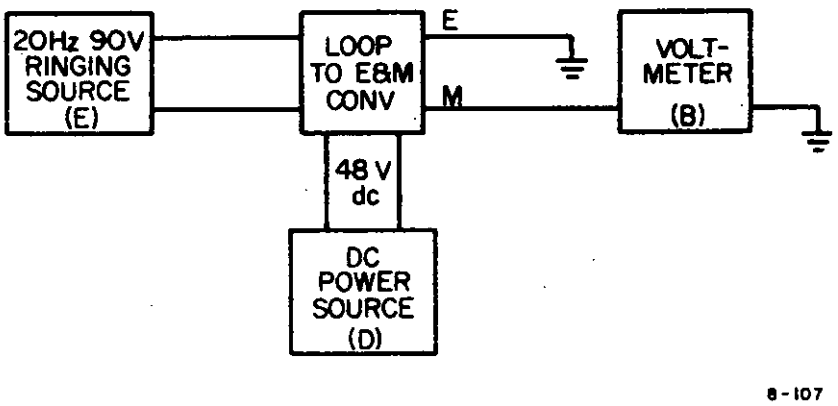


Figure 37. Dial loop to E&M converter operational check, 20-Hz ringing, M-lead condition -- off-hook.

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6.2.10.2 *Return loss.*

6.2.10.2.1 *Scope.* This method is used to measure the return loss of loop to E&M converters.

6.2.10.2.2 *Applicability.* The parameters tested shall conform to the requirements specified in the paragraph cited below:

Item	Paragraph
Return Loss	5.4.3.3

6.2.10.2.3 *Required test equipment.*

Test unit	Schematic reference	Item no. in appendix
Resistor, Precision Type Termination, Noninductive, ±0.5%	R, R_1	(see below)
AF Oscillator	A	100
Level Measuring Set	B	301
Hybrid Transformer	C	656
Level Measuring Set	D	301
Switch	E	902

Details of precision resistors

Nominal system impedance (ohms)	R (ohms)	R_1 (ohms)
600	600	672
900	900	1,008

6.2.10.2.4 *Procedure.*

Step 1. Connect equipment as shown in figure 38.

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Step 2. Connect the test equipment to hybrid transformer and to the relevant test point of the system under test. The output circuit from oscillator A should be in a balanced condition and should present an impedance R to the primary winding of hybrid transformer C. Level measuring set B should present a high impedance to the hybrid transformer (about 20 times R).

Step 3. Operate switch E to connect hybrid transformer C to the system under test (terminated correctly) and connect level measuring set D to the same test point. The input impedance of level measuring set D should be high compared to the nominal resistance R.

Step 4. Adjust oscillator A output level so that the correct relative test-signal level is applied to the system under test. The oscillator frequency should be within the normal operating frequency range of the system at the point of measurement.

Step 5. Remove level measuring set D from the system test point and connect it across the output terminals of oscillator A. Note the oscillator signal level.

Step 6. Operate switch E to connect hybrid transformer C to resistor R_1, and readjust the oscillator output level, if necessary, to maintain the signal level noted in step 5.

Step 7. Adjust the sensitivity of level measuring set B (tuning it in addition, if it is a selective level measuring device) to obtain a convenient reading of the signal level on the scale of its indicating device. Record the signal level in dB.

Step 8. Operate switch E to reconnect hybrid transformer C to the system under test, and readjust the oscillator output level, if necessary, to maintain the signal level noted in step 5, as measured by level measuring set D across the output terminals of oscillator A. Measure and record the new signal level in dB as indicated on level measuring set B.

Step 9. The return loss of the system impedance against the nominal resistance R is given by (X + Y) dB, where X is the calculated return loss of R_1 against R, and Y is the difference between the signal levels recorded in steps 7 and 8.

NOTE. If the value of resistor R_1 has been appropriately selected, then the return loss is given by (25 + Y) dB.

Step 10. Repeat the above steps for as many frequencies over the band of interest at the specific point of measurement as deemed necessary in any particular case.

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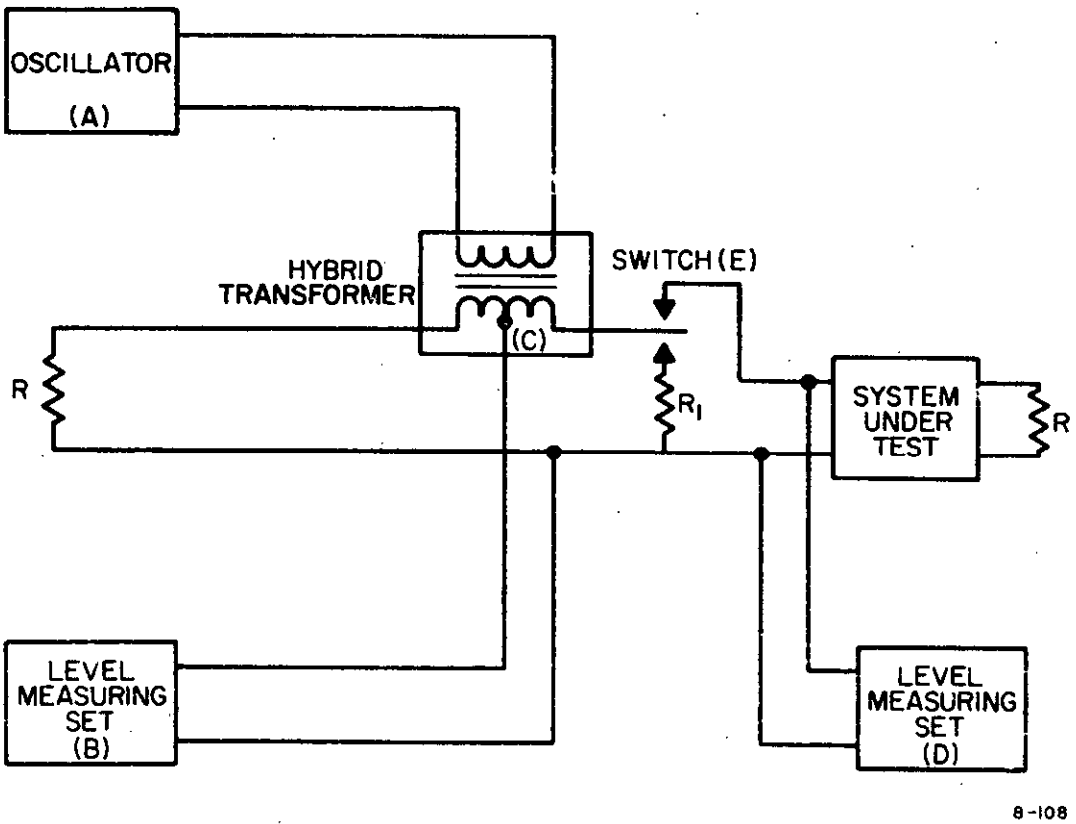


Figure 38. Measurement of return loss, dial loop to E&M converter.

6.2.10.3 *Longitudinal balance.*

6.2.10.3.1 *Scope.* This method is used to measure the longitudinal balance of dial loop to E&M converters.

6.2.10.3.2 *Applicability.* The parameters tested shall conform to the requirements specified in the paragraph cited below:

Item	Paragraph
Longitudinal Balance	5.4.3.4

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6.2.10.3.3 *Required test equipment.*

Test unit	Schematic reference	Item no. in appendix
Vacuum Tube Voltmeter (VTVM), rms (2 required)	A	64 or 80
AF Oscillator	B	1106
Resistor, 600-ohm ±1%	C	410
Resistor, 300-ohm ±1% (2 required)	D	406
Resistor, 150-ohm ±5%	E	448
Transformer, 600 ohms : 600 ohms C. T.	F	658
Power Supply, 48 V dc	G	

6.2.10.3.4 *Procedure.*

Step 1. Connect equipment as shown in figure 39.

Step 2. Adjust oscillator B for 3200 Hz (600-ohm output impedance) and a signal level adequate to obtain 0.775 volts (0 dBm) as read on the VTVM (V_1) at the line input. Record the voltage (V_2) across the 150-ohm resistor. Calculate the line input longitudinal balance (LB) in dB using the expression:

$$LB = 20 \log_{10} \frac{V_1}{V_2}$$

Step 3. Terminate the equipment in a test configuration as shown in figure 40.

Step 4. Adjust oscillator B for 3200 Hz (600-ohm output impedance) and a signal level adequate to obtain an output voltage of 0.775 volts as read on the VTVM (V_1) at the drop output. Record the voltage (V_2) across the 150-ohm resistor. Calculate the drop output longitudinal balance using the expression of step 2 above.

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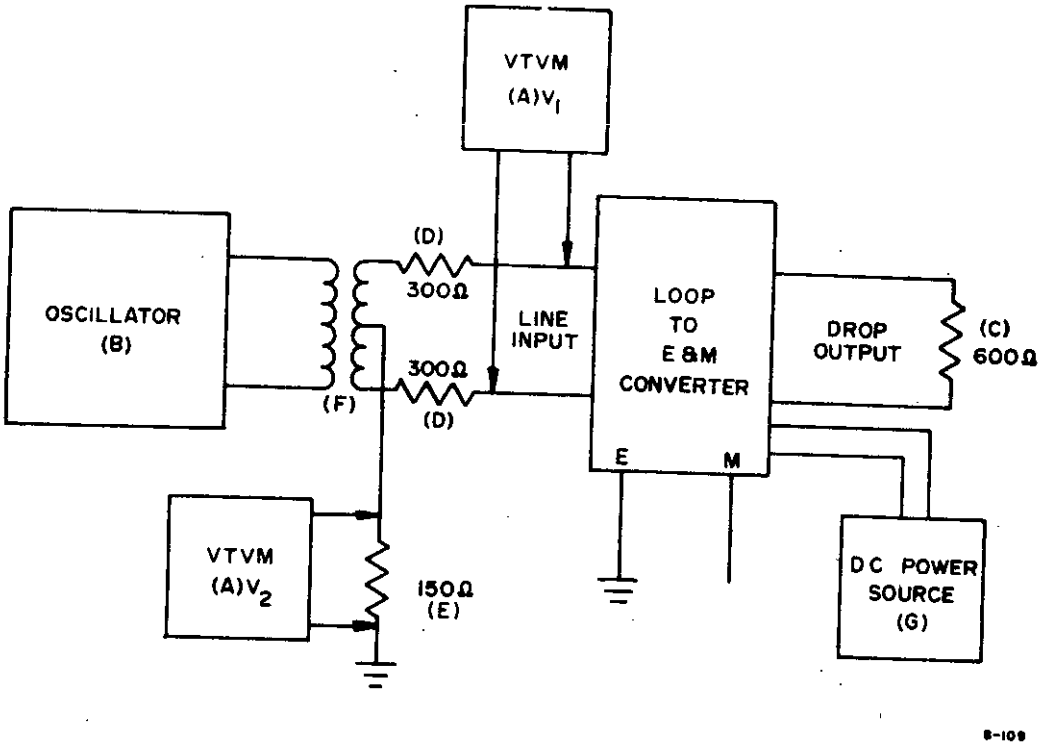


Figure 39. Longitudinal balance test, line input, dial loop to E&M converter.

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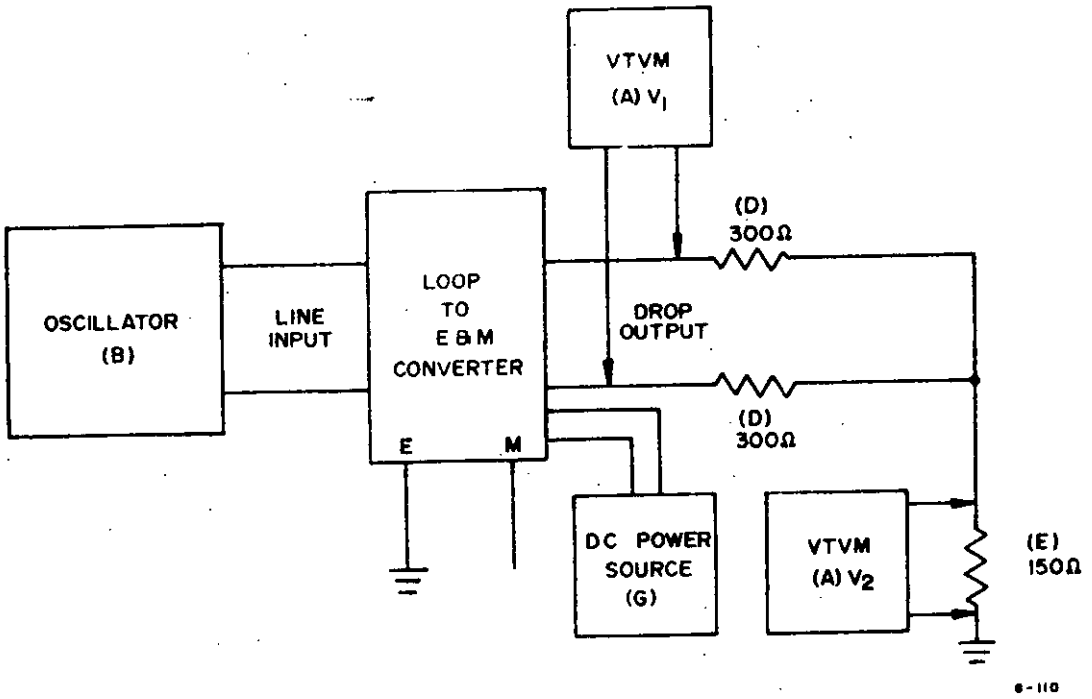


Figure 40. Longitudinal balance test, drop output, dial loop to E&M converter.

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6.2.10.4 *Noise and crosstalk.*

6.2.10.4.1 *Scope.* These methods are used to measure crosstalk and random noise between adjacent units of dial loop to E&M converters.

6.2.10.4.2 *Applicability.* The parameters tested shall conform to the requirements specified in the paragraphs cited below:

Item	Paragraph
Crosstalk	5.4.3.5
Noise	5.4.3.6

6.2.10.4.3 *Required test equipment.*

Test unit	Schematic reference	Item no. in appendix
AF Oscillator	A	174
Vacuum Tube Voltmeter (VTVM), rms	B	18
Noise Measuring Set	C	306
Power Supply, 0-100 V dc (see note)	D	565
Resistor, 600-ohm ±1% (8 required)	E	410

NOTE. If dial central office power is available, power supply D will not be required.

6.2.10.4.4 *Procedure.*

6.2.10.4.4.1 *Crosstalk.*

Step 1. Connect equipment as shown in figure 41.

Step 2. Adjust oscillator A for 1-kHz, 0-dBm output.

Step 3. Measure on VTVM B the amount of crosstalk on the adjacent unit (No. 2).

Step 4. Record this reading.

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Step 5. Remove VTVM B from unit No. 2 and terminate the unit with a 600-ohm resistor.

Step 6. Connect VTVM B to unit No. 3 terminated in 600 ohms.

Step 7. Recheck oscillator A output for 1-kHz, 0-dBm output.

Step 8. Measure and record reading on the VTVM A.

Step 9. Repeat steps 5 through 8 for crosstalk measurement of the remaining units.

6.2.10.4.4.2 *Random noise.*

Step 1. Connect equipment as shown in figure 42. Noise measuring set C should be set for flat weighting.

Step 2. Record the noise power in dBm, as read on the noise meter.

Step 3. Remove the noise measuring set C from unit No. 1 and terminate unit No. 1 in 600 ohms.

Step 4. Connect noise measuring set C to unit No. 2 and record the noise power in dBm as read on the noise meter.

Step 5. Repeat the above steps for noise readings on the remaining units.

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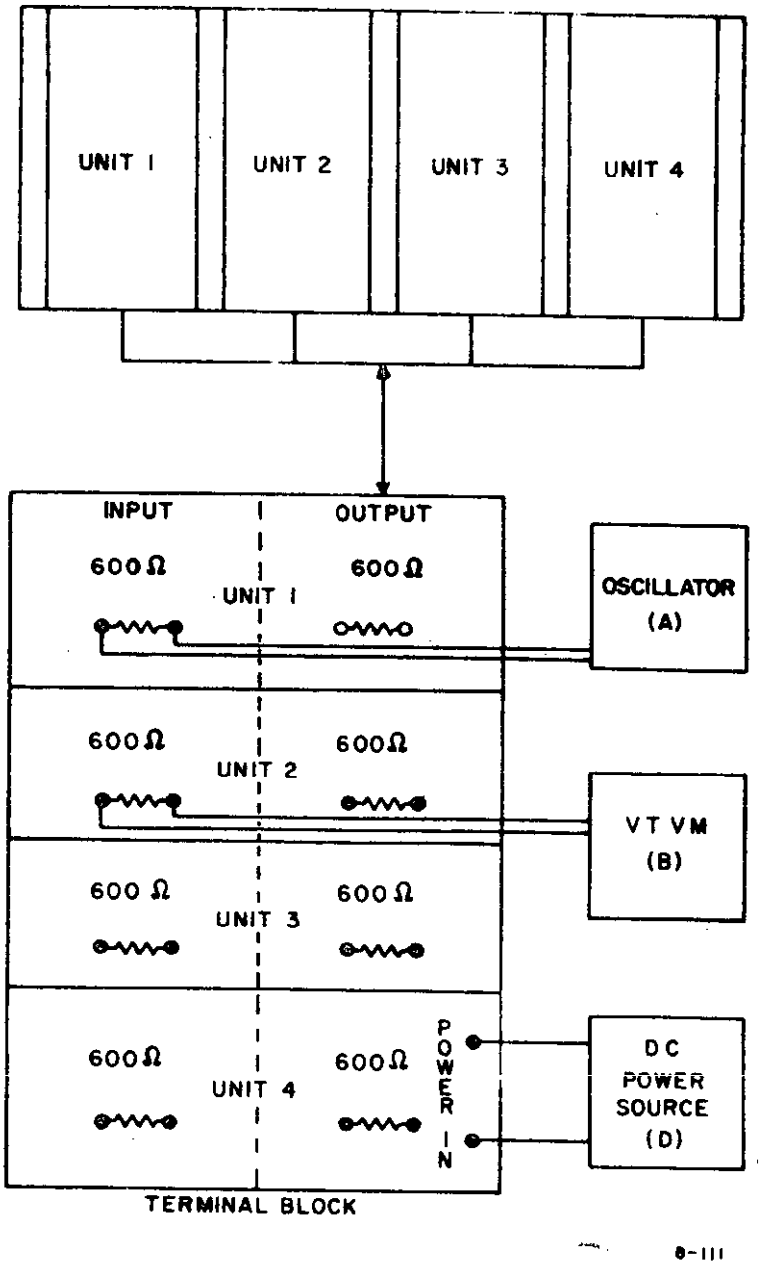


Figure 41. Measurement of crosstalk, dial loop to E&M converter.

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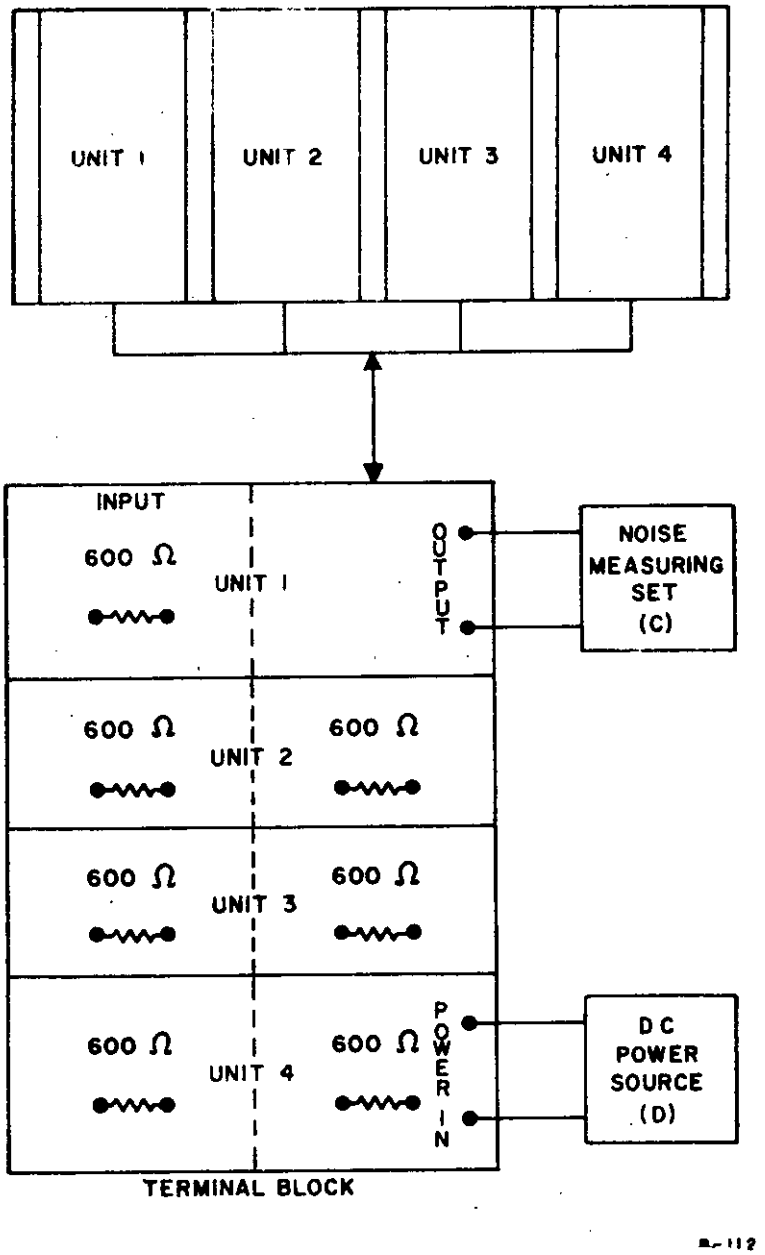


Figure 42. Measurement of random noise, dial loop to E&M converter.

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6.2.10.5 *Frequency response.*

6.2.10.5.1 *Scope.* This method is used to measure frequency response of dial loop to E&M converters.

6.2.10.5.2 *Applicability.* The parameters tested shall conform to the requirements specified in the paragraph cited below:

Item	Paragraph
Frequency Response	5.4.3.7

6.2.10.5.3 *Required test equipment.*

Test unit	Schematic reference	Item no. in appendix
AF Oscillator	A	100
Attenuator	B	600
Level Measuring Set	C	301
Level Measuring Set	D	301
Level Measuring Set	E	301
Sending Unit	F	101
Terminating Resistors (as required)		

6.2.10.5.4 *Procedure.*

Step 1. Connect equipment as shown in figure 43.

Step 2. Adjust output of sending unit F at 1000 Hz so that the correct test-tone level is applied to the appropriate audio test point of the system under test.

NOTE. Where a sending unit is not available, adjust output of oscillator A at 1000 Hz to obtain a test-tone level of 0 dBm at the input terminals of attenuator B. If level measuring set C (input impedance high compared to R_2) is used to check

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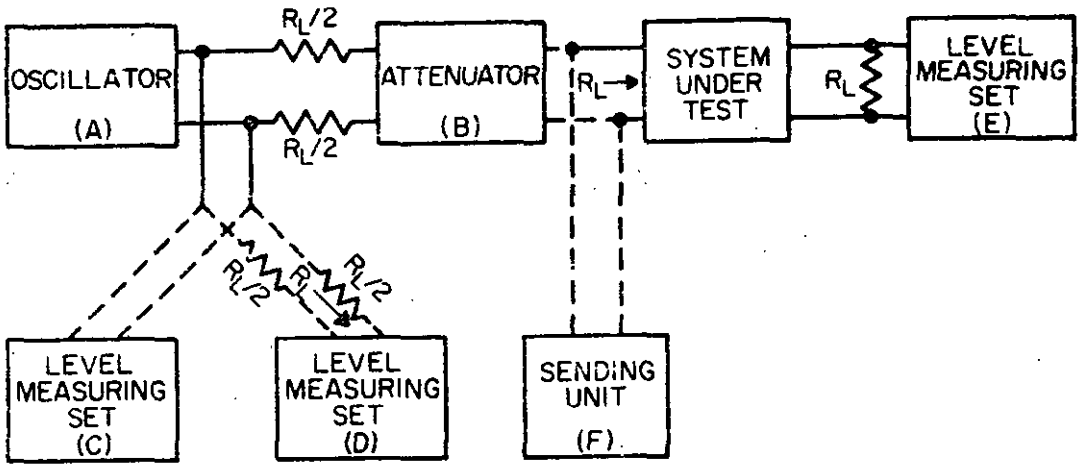
levels, the oscillator output should be adjusted to give a reading of ±6 dBm on level measuring set D. Attenuator B should then be set to obtain the correct relative test-tone level at the input to the system under test.

Step 3. Adjust sensitivity of level measuring set E to obtain a convenient reading on the scale of its indicating instruments. Record the reading.

Step 4. Vary the frequency of sending unit F or oscillator A in convenient steps, e.g., 200-Hz intervals over the range of interest (e.g., 300 to 3400 Hz for a nominal 4-kHz voice circuit) checking each step to ensure that the input level remains constant. If necessary, readjust the sending unit or oscillator signal amplitude to maintain the initial sending level.

Step 5. Record relative changes, in dB, from the reading at 1000 Hz as indicated on level measuring set E at the system output.

NOTE. If, when plotted, the level measuring set readings show significant departures from a smooth curve, these departures should be explored by taking measurements at close frequency intervals.



NOTES:

1. BROKEN LINES REPRESENT ALTERNATIVE CONNECTIONS, I.E. (C) AND (D) ARE ALTERNATIVE TEST UNITS: (C) IS HIGH IMPEDANCE (D) IS 600Ω, AND (F) CAN REPLACE THE OSCILLATOR, ATTENUATOR, AND LEVEL MEASURING SET.
2. WHERE R_L IS AN IMPEDANCE OTHER THAN 600Ω, SUITABLE IMPEDANCE MATCHING TRANSFORMERS SHOULD BE INCLUDED IF NECESSARY.

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Figure 43. Measurement of frequency distortion, dial loop to E&M converter.

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6.2.10.6 *Harmonic distortion.*

6.2.10.6.1 *Scope.* This method is used to measure harmonic distortion of dial loop to E&M converters.

6.2.10.6.2 *Applicability.* The parameters shall conform to the requirements specified in the paragraph cited below:

Item	Paragraph
Harmonic Distortion	5.4.3.8

6.2.10.6.3 *Required test equipment.*

Test unit	Schematic reference	Item no. in appendix
Audio Oscillator (low distortion)	A	185
Variable Attenuator	B	600
Level Measuring Set	C	301
Level Measuring Set	D	301
Sending Unit (low distortion)	F	106
Selective Level Measuring Set	G	310
Resistors, Terminating		406, 410, etc.

6.2.10.6.4 *Procedure.*

Step 1. Connect equipment as shown in figure 44.

Step 2. Adjust audio oscillator A output at 1000 Hz to obtain a test tone level of 0 dBm at the input terminals of the attenuator B. If level measuring set C (input impedance high compared to R_L) is employed, the audio oscillator output should be adjusted to give a reading of 0 dBm on level measuring set D.

Step 3. Set variable attenuator B so that the correct relative test tone level is applied to the system under test, e.g., -16 dBm at the input to the relevant

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channel modulator. If sending unit F is being used, its control should be adjusted to provide the proper sending level.

Step 4. Connect selective level measuring set G to the audio output point of the relevant channel at the far end of the system and terminate the line with a noninductive terminating resistor R_L. In certain instances, the terminating resistor may be contained in the measuring equipment itself; otherwise the latter should be set to its high input impedance condition.

Step 5. Tune selective level measuring set G about the 1000-Hz frequency setting, assuming that all transmission levels through the system have been set to their correct relative levels to obtain a maximum reading on its indicating instrument. Adjust the sensitivity controls in the process so that the meter pointer is set to a reference scale mark. Note the setting of the sensitivity controls (usually given in volts or decibels).

Step 6. Retune selective level measuring set G to obtain a peak reading on its indicating instrument at (or about) a frequency of 2000 Hz (second harmonic), and adjust the sensitivity controls so that the pointer of the instrument reads to the same reference scale mark. Note the new setting of the sensitivity control.

NOTE. It may be necessary to apply a correction to the setting of the sensitivity controls for the second harmonic, to allow for variations in the gain vs frequency response characteristic of the selective-level measuring set G. (Check manufacturer's specifications).

Step 7. Compute percentage of distortion at the relevant frequency if the readings of the selective measuring set are in volts. The ratio of the two readings gives the second harmonic distortion at a signal frequency of 1000 Hz. Percent distortion can be converted to distortion ratio in dB using figure 45.

Step 8. Retune selective level measuring set G to 3000 Hz and repeat the measurement procedure. The ratio of the readings at 1000 Hz and 3000 Hz gives the third harmonic distortion at a signal frequency of 1000 Hz.

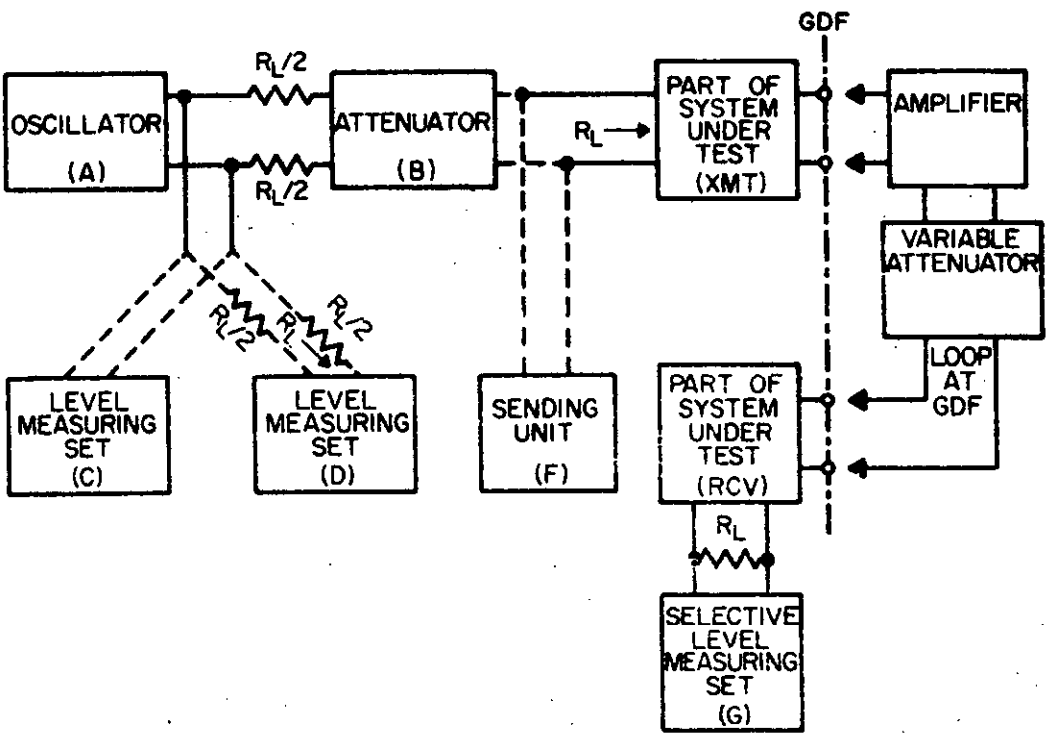
Step 9. Measurement of harmonics beyond the third is normally not required. However, the same procedure can be used to measure higher harmonics.

Step 10. Increase insertion loss of the variable attenuator B in 1-dB steps, with the fundamental test tone frequency maintained at 1000 Hz, and make sure that the level of the second harmonic received on the selective level measuring set changes by approximately 2-dB steps and that the level of the third harmonic changes by approximately 3-dB steps.

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Step 11. If the overall procedure is now repeated at different signal frequencies over the frequency band of interest, then characteristics of single harmonic distortion vs signal frequency can be determined for each of the measured harmonics.

NOTE. Total harmonic distortion can be computed from measurements of single harmonic distortion. If "a" and "b" are the individual percentage distortion figures for the second and third harmonic frequencies respectively, then the percentage total harmonic distortion is given by $\sqrt{a^2 + b^2}$.



NOTES:

1. BROKEN LINES REPRESENT ALTERNATIVE CONNECTIONS, I.E. (C) AND (D) ARE ALTERNATIVE TEST UNITS, AND (F) CAN REPLACE THE OSCILLATOR, ATTENUATOR, AND LEVEL MEASURING SET.
2. WHERE R IS AN IMPEDANCE OF OTHER THAN 600Ω, SUITABLE IMPEDANCE MATCHING TRANSFORMERS SHOULD BE INCLUDED IF NECESSARY.
3. ATTENUATOR AND AMPLIFIER OF SUITABLE FREQUENCY RESPONSE.

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Figure 44. Measurement of harmonic distortion, dial loop to E&M converter.

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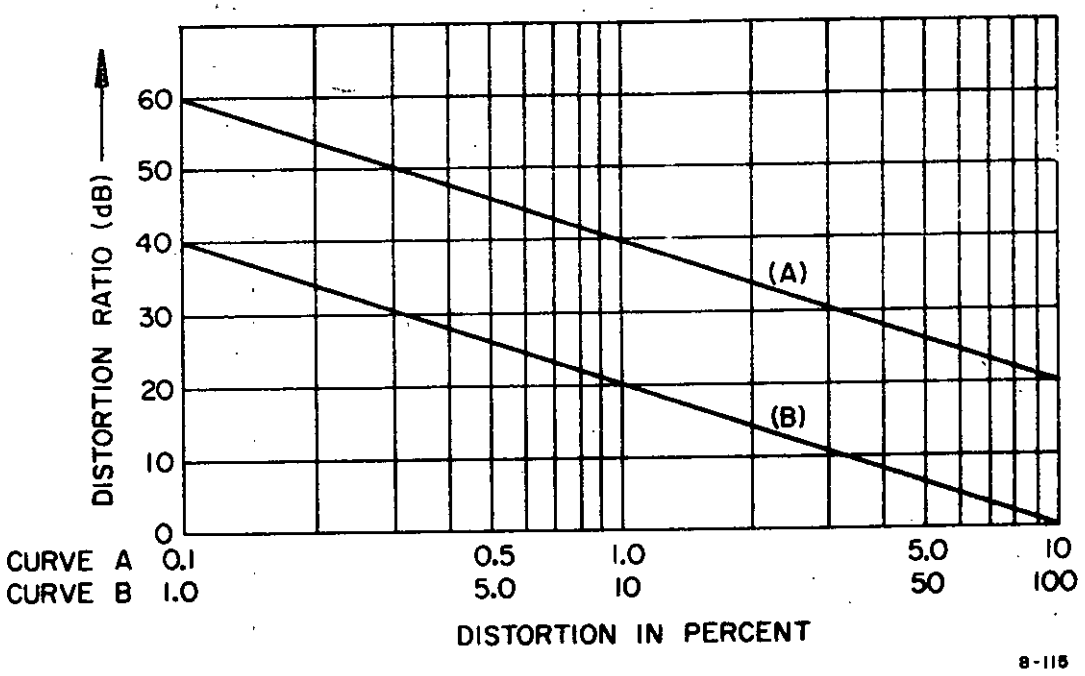


Figure 45. Relationship between distortion in percent and dB.

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6.2.11 *Bridge lifter.*

6.2.11.1 *Scope.* This method is used to test bridge lifters.

6.2.11.2 *Applicability.* The parameters tested shall conform to the requirements specified in the paragraphs cited below:

Item	Paragraph
Resistance	5.5.2.1
Inductance	5.5.2.2
Crosstalk	5.5.2.3

6.2.11.3 *Required test equipment.*

Test unit	Schematic reference	Item no. in appendix
Impedance Bridge	A	904
Impedance Bridge	B	952
AF Oscillator	C	100
AC Electronic Voltmeter (VTVM)	D	81
Resistor, 600-ohm ±1% (3 required)	E	410

6.2.11.4 *Procedures.*

6.2.11.4.1 *Resistance.*

Step 1. Terminate bridge lifter in a test configuration as shown in figure 46.

Step 2. Using impedance bridge A (or B), measure the dc resistance.

6.2.11.4.2 *Inductance.*

Step 1. Terminate bridge lifter in a test configuration as shown in figure 47.

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Step 2. Adjust impedance bridge B for 10-mA dc bias and measure the inductance. Record the inductance value.

Step 3. Repeat the procedure in steps 1 and 2, except adjust the impedance bridge for 30-mA dc bias instead of 10-mA bias.

6.2.11.4.3 *Crosstalk.*

Step 1. Terminate bridge lifter in test configuration as shown in figure 48.

Step 2. Adjust oscillator for an output of 600 ohms, 1000 Hz, and 0 dBm. Terminate all other inputs and outputs with 600-ohm ±1 percent noninductive resistances. Measure and record the signal amplitude at all inputs and outputs.

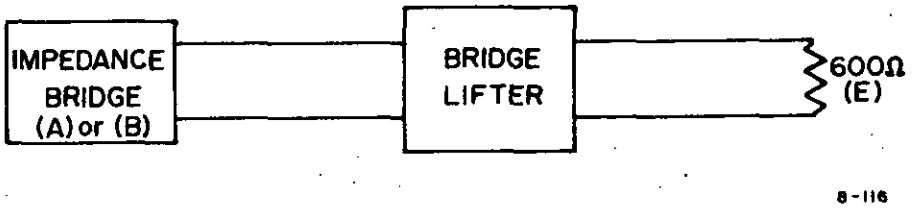


Figure 46. Measurement of resistance, bridge lifter.

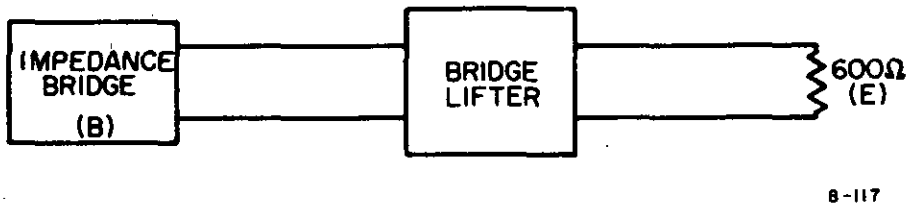


Figure 47. Measurement of inductance, bridge lifter.

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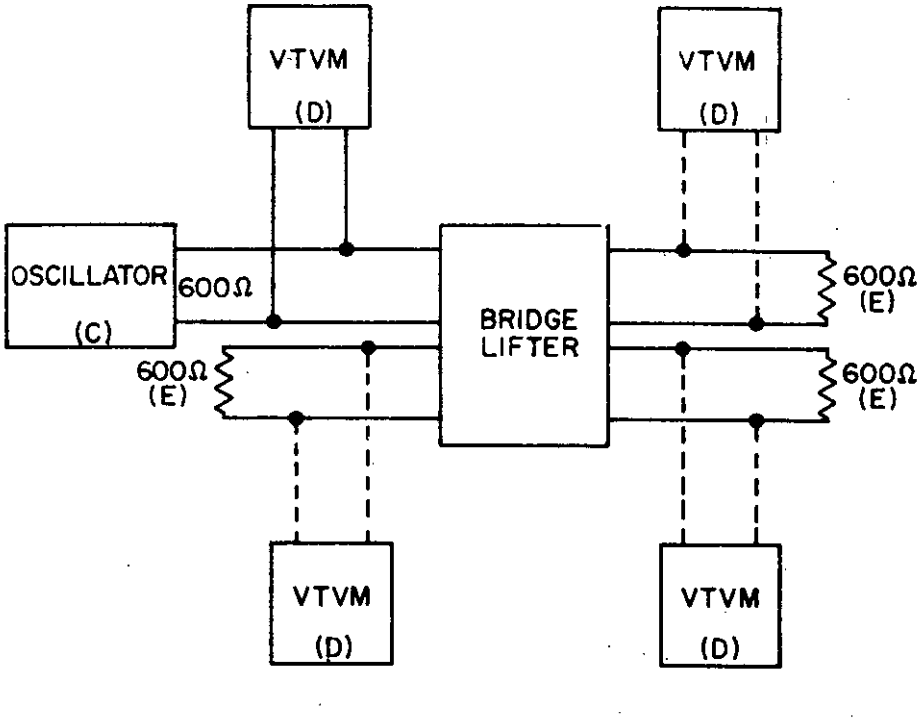


Figure 48. Measurement of crosstalk, bridge lifter.

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6.2.12 *Ringing machines output frequency.*

6.2.12.1 *Scope.* This method is used to measure the output frequency of ringing machines.

6.2.12.2 *Applicability.* The parameters tested shall conform to the requirements specified in the paragraph cited below:

Item	Paragraph
Ringing Machine Output Frequency	5.3.2

6.2.12.3 *Required test equipment.*

Test unit	Schematic reference	Item no. in appendix
Power Supply, 0-100 V dc	A	565
Frequency Counter	B	279
Vacuum Tube Voltmeter (VTVM), ac	C	64
Voltmeter, dc	D	73

6.2.12.4 *Procedure.*

Step 1. Connect equipment as shown in figure 49.

Step 2. Adjust dc power source A for an output of 48 volts dc.

Step 3. Measure ringing machine output frequency and rms voltage with the frequency counter B and VTVM C respectively.

Step 4. Vary dc power source A output from 46 volts dc to 52 volts dc in 2-volt steps.

Step 5. At each step change measure the frequency and rms voltage.

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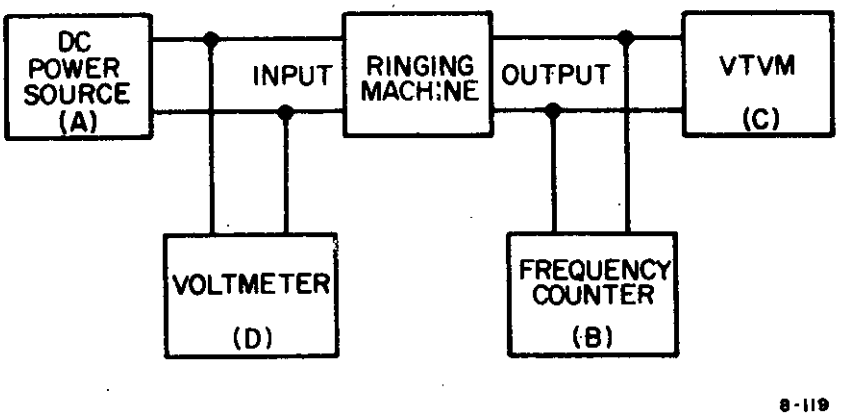


Figure 49. Measurement of output frequency, ringing machine.

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CUSTODIANS

Army - SC
Navy - EC
Air Force - 17

REVIEW ACTIVITIES

Army - SC
Navy - EC
Air Force - 17
NSA - NS
DCA - DC

PREPARING ACTIVITY

Army - SC

Project SLHC 0016

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APPENDIX

TEST EQUIPMENT LIST

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PREFACE

The purpose of this appendix is to present supplementary information concerning the test equipment referenced in section 6 of the standard. This information is presented in columnar format as follows:

a. Item no. This column lists the index numbers assigned to each item of test equipment. The index numbers are used for cross referencing between sections 5 and 6 and the appendix.

b. Test Unit. This column lists each item of test equipment by its common nomenclature.

c. Minimum Performance Requirements. This column lists the required operating characteristics for each item of test equipment.

The test equipment generally used in engineering and installing telecommunications systems and subsystems is divided into 14 categories, grouped by basic type for ease in finding a particular item as well as adding new items required for future measurement techniques. The equipment categories and item numbers are as follows:

A.	Meters	1-99; 0100-0199
B.	Generators	100-199; 1100-1199
C.	Indicators, Recorders, and Detectors	200-299; 2100-2199
D.	Testers, Analyzers and Measuring Sets	300-399; 3100-3199
E.	Circuit Components (Resistors, Capacitors & Diodes)	400-499; 4100-4199
F.	Amplifiers	500-549; 5100-5149
G.	Power Sources	550-599; 5150-5199
H.	Attenuators	600-649; 6100-6149
I.	Transformers	650-699; 6150-6199
J.	Waveshaping Circuits	700-749; 7100-7149
K.	Unassigned	750-799; 7150-7199
L.	Unassigned	800-849; 8100-8149
M.	Unassigned	850-899; 8150-8199
N.	Miscellaneous	900-999; 9100-9199

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A. Test equipment for measurements, meters (items 1-99; 0100-0199)

Item no.	Test unit	Minimum performance requirements
7	Voltmeter	Range of voltage - - - - - - - 2 to 300 V (multirange) Frequency - - - - - - - - - - DC Sensitivity - - - - - - - - - - 1,000 ohms/V or better Accuracy - - - - - - - - - - - Within 2%
8	Milliammeter	Range of current - - - - - - - 0 to 100 mA Frequency - - - - - - - - - - DC Impedance - - - - - - - - - - 10 ohms or less Accuracy - - - - - - - - - - - Within 5%
12	Ohmmeter	Range of resistance - - - - - - 10 to 1,000 ohms Test voltage - - - - - - - - - Between 1 and 2 V Midscale indication - - - - - - - Between 100 and 300 ohms Accuracy - - - - - - - - - - - ±2%
18	Vacuum Tube Voltmeter	Voltage range - - - - - - - - - 1.0 mV to 10 V full scale Accuracy - - - - - - - - - - - 1% of full scale 50 Hz to 500 kHz Calibration voltage scale - - - - Logarithmic indicating rms value of sine wave. dB scale - LINEAR based on 0 dB = 1 mW in 600 ohms
37	AC Voltmeter	Voltage range - - - - - - - - - 0 to 100 V Accuracy - - - - - - - - - - - ±5% Frequency - - - - - - - - - - 10 Hz to 6 kHz Impedance - - - - - - - - - - 1,000 ohms per V

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A. Test equipment for measurements, meters (items 1-99; 0100-0199) – Continued

Item no.	Test unit	Minimum performance requirements	
38	Voltmeter	Voltage range	0 to 150 V (75 scale divisions minimum)
		Accuracy	±3/4% of full scale
		Frequency	25 to 125 Hz
		Approximate burden	1.2 VA
39	Frequency Meter	Frequency range	40 to 70 Hz (60 scale divisions)
		Approximate burden	3.0 VA
		Accuracy	0.34 Hz
		Voltage rating	100-130 V
48	Milliammeter	DC–dual range	0 to 100 mA 0 to 50 mA
		Accuracy	±1%
54	Voltmeter	DC	0 to 150 V
		Accuracy	±0.5%
64	True RMS Voltmeter	Voltage range	100 μV to 1 V rms
		Accuracy	2% of full scale
		Frequency range	100 Hz to 6 MHz
		Input impedance	Greater than 10,000 ohms
73	DC Voltmeter	Voltage range	1 to 150/300 V (150 scale divisions)
		Accuracy	1/2%

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Item no.	Test unit	Minimum performance requirements	
80	True RMS Voltmeter	Voltage ranges	0 to 0.001, 0.003, 0.01, 0.03, 0.1, 0.3 and 1 V rms
		Accuracy	±3% full scale
		Input impedance	100,000 ohms minimum
		Frequency range	10 Hz to 7 kHz
81	AC Electronic Voltmeter	Voltage range	100 mV to 300 V
		Decibel range	-80 to +50 dB (0 dB = 0 dBm or 1 mV in 600 ohms)
		Accuracy	2%
		Frequency range	40 Hz to 2.54 MHz
		Calibration	Linear dB scale

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B. Test equipment for measurements, generators (items 100-199; 1100-1199)

Item no.	Test unit	Minimum performance requirements	
100	Oscillator	Frequency range	30 Hz to 10 kHz
		Calibration accuracy	±2%
		Output to 600 ohms	+8 dBm maximum
		Source impedance	600 ohms balanced; 900 ohms balanced
		Maximum distortion	1%
		Maximum hum and noise	0.1%
101	Sending Unit	Frequency range	30 Hz to 10 kHz
		Calibration accuracy	±2%
		Sending level range	+8 to −40 dBm
		Sending impedance	600 ohms balanced; 900 ohms balanced
		Send level accuracy	Better than ±0.5 dB
106	Low Distortion Sending Unit:	Frequency range	30 Hz to 10 kHz
		Calibration accuracy	±2%
		Output	+15 dBm minimum
		Output impedance	600 ohms balanced; 900 ohms balanced
		Distortion	Less than 0.3%, 30 Hz to 5 kHz
		Hum level	At least 70 dB below 0-dBm level
		Output meter	Calibrated in dBm
		Output attenuation	1 dB step attenuator, 40-dB range
121	Pulsing System	Pulse duration	Adjustable, 15 to 25 ms
		Pulse rate	Adjustable, 1 to 10 pps

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B. Test equipment for measurements, generators (items 100-199; 1100-1199) – Continued

Item no.	Test unit	Minimum performance requirements	
121 (continued)		Output	Semiconductor or relay switching, 120 V, 20 W (maximum)
126	Oscillator	Frequency range	5 Hz to 600 kHz
		Output	10 V into 600 ohms
173	Oscillator	Frequency range	15 to 4000 Hz
		Amplitude range	40 V
		Accuracy	±2%
		Power output	2 W
174	Oscillator	Frequency range	200 to 4000 Hz
		Output	10 V into 600 ohms
185	Low Distortion Oscillator	Frequency range	300 Hz to 7.5 kHz
		Accuracy (freq)	±2%
		Output	18 mW into 600 ohms
		Output impedance	600 ohms balanced
		Distortion, hum and spurious	0.1%
1106	Audio Signal Generator	Frequency range	20 Hz to 20 kHz
		Accuracy	2%
		Output level	-60 dBm to +5 dBm

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B. Test equipment for measurements, generators (items 100-199; 1100-1199) – Continued

Item no.	Test unit	Minimum performance requirements
1106 (continued)		Distortion - - - - - - - - - - - -65 dB maximum Attenuator range - - - - - - - - 65 dB minimum Output impedance - - - - - - - 600 ohms

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C. Test equipment for measurements, indicators, recorders and detectors (items 200-299; 2100-2199)

Item no.	Test unit	Minimum performance requirements
200	Oscilloscope	Tube diameter - - - - - - - - 5 in. or larger Function - - - - - - - - - - - General purpose Vertical sensitivity - - - - - - - 1 mV per centimeter to 10 V per centimeter
201	Frequency Counter	Count capacity - - - - - - - - 5 digits Frequency range - - - - - - - - 200 to 4000 Hz Measuring accuracy - - - - - - ±1 count ± stability of oscillator Oscillator stability - - - - - - ±3 parts in 10^6 per week Sensitivity - - - - - - - - - - 100 mV minimum Input impedance - - - - - - - Greater than 40,000 ohms
211	Counter with frequency converter plug-in units up to 500 MHz	Count capacity - - - - - - - - 7 digits Measurement range - - - - - - - Sine wave or pulse frequency 10 Hz to 11.5 MHz Time interval - - - - - - - - - 0.1 μs to 10^7 s Accuracy - - - - - - - - - - - 3 parts in 10^7 ±1 count
255	Frequency Counter	Frequency range - - - - - - - - 10 Hz to 120 kHz Accuracy - - - - - - - - - - ±1 count Gate time - - - - - - - - - - 0.1 and 1 second minimum Time interval measurement - - 1 μs - 1 s Gating voltage maximum - - - -100 to +100

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C. Test equipment for measurements, indicators, recorders and detectors (items 200-299; 2100-2199) – Continued

Item no.	Test unit	Minimum performance requirements	
255 (continued)		Sensitivity	0.3 V peak to peak 0.25 V rms for sine waves
		Separate start-stop channels, each with trigger level controls	
256	Oscilloscope	Tube diameter	Not less than 5 inches
		Function	General purpose
		Frequency range	DC to 10 kHz minimum
		Vertical sensitivity	1 mV per centimeter minimum
		Sweep rate	At least 100 ms/cm
279	Frequency Counter	Frequency range	10 Hz to 7 kHz
		Accuracy	±3 Hz
		Input impedance	100,000 ohms minimum
		Input level	0.01 to 1 V rms
		Registration	4 digits minimum
280	Frequency Counter	Frequency range	200 Hz to 7 kHz
		Accuracy	±1 count ± time base accuracy
		Input sensitivity	0.1 to 1 V rms
		Input impedance	100,000 ohms minimum
		Gate time	1 to 1,000 seconds
		Registration	5 digits minimum

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Item no.	Test unit	Minimum performance requirements
280 (continued)		External time base input level - - - - - - - - - 1 V rms External time base input impedance - - - - - - - 50 ohms minimum

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D. Test equipment for measurements, testers, analyzers & measuring sets (items 300-399; 3100-3199)

Item no.	Test unit	Minimum performance requirements	
301	Level Measuring Set	Frequency range	30 Hz to 10 kHz
		Measuring range	+20 to –40 dBm across 600 ohms
		Impedance	600 ohms balanced; 900 ohms balanced (terminated). Greater than 40,000 ohms (bridged).
306	Noise Measuring Set	Range	Noise and signal levels to –85 dBm
		Frequency range	50 to 5000 Hz
		Impedance	600 ohms balanced; 900 ohms balanced
		Bridging Impedance	6000 ohms minimum balanced
		Weighting	Combination of "FLAT" plus one of the following: C – MESSAGE FIA CCIF – 1951
310	Selective Level Measuring Set	Measuring range	Down to at least –100 dBm
		Frequency range	30 Hz to 10 kHz
		Frequency calibration	Linear graduation of at least 1 division per 10 Hz
		Calibration accuracy	Approximately ±2%
		Residual modulation products and hum	At least 60 dB down

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D. Test equipment for measurements, testers, analyzers & measuring sets (items 300-399; 3100-3199) – Continued

Item no.	Test unit	Minimum performance requirements
310 (continued)		IF frequency rejection - - - - At least 75 dB Input impedance - - - - - - 600 ohms balanced; 900 ohms balanced (terminated). Greater than 40,000 ohms (bridged).
337	Dial Test Set	Dial speed measuring range - - 9.5 to 19 pps Percent break measuring range - - - - - - - - - - 60 to 67% break Accuracy - - - - - - - - - - Dial speed ±0.01 pps Make/Break ±0.2%
342	Electro-Acoustic Test Set	Same as Item 126 (Oscillator), 509 (Amplifier), 600 (Variable Attenuator) and 935 (Artificial Mouth and Associated Equalizer), combined as one test set.
363	Frequency Selective Voltmeter	Frequency range - - - - - - - 250 Hz to 4 kHz Accuracy - - - - - - - - - - 0.5 dB Input impedance - - - - - - 100,000 ohms Sensitivity - - - - - - - - - -30 dBm to +10 dBm Bandwidth - - - - - - - - - - 25 Hz
366	Wave Analyzer	Frequency range - - - - - - - 10 Hz to 7 kHz Frequency accuracy - - - - - ±1% +5 Hz

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D. Test equipment for measurements, testers, analyzers & measuring sets (items 300-399; 3100-3199) – Continued

Item no.	Test unit	Minimum performance requirements
366 (continued)		Tracking range - - - - - - - - ±100 Hz minimum Input impedance - - - - - - - 100,000 ohms minimum Input level - - - - - - - - - - 0 to 0.003, 0.01, 0.03, 0.1, 0.3, 1.0, and 3.0 V Residual hum and modulation selectivity - - - - - - - - - 60 dB down or better 3 dB bandwidth approx. 7 Hz 50 dB bandwidth approx. 50 Hz

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E. Test equipment for measurements, circuit component (resistors, capacitors & diodes) (items 400-499; 4100-4199)

Item no.	Test unit	Minimum performance requirements
406	Resistor	Type - - - - - - - - - - - - Fixed Resistance - - - - - - - - - - 300 ohms ±1% Wattage rating - - - - - - - - 3 W or more
410	Resistor	Type - - - - - - - - - - - - Fixed Resistance - - - - - - - - - - 600 ohms ±1% Wattage rating - - - - - - - - 1 W or more
435	Resistor	Type - - - - - - - - - - - - Fixed Resistance - - - - - - - - - - 750 ohms ±1% Wattage rating - - - - - - - - 1/4 W
436	Resistor	Type - - - - - - - - - - - - Fixed Resistance - - - - - - - - - - 900 ohms ±1% Wattage rating - - - - - - - - 2.5 W
437	Inductor	Inductance - - - - - - - - - - 10 H at 100 mA Resistance - - - - - - - - - - 400 ohms (nominal)
448	Resistor	Type - - - - - - - - - - - - Fixed Resistance - - - - - - - - - - 150 ohms ±5% Dissipation - - - - - - - - - - 0.25 W

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F. Test equipment for measurements, amplifiers (items 500-549; 5100-5149)

Item no.	Test unit	Minimum performance requirements
503	Detector Amplifier	Frequency range - - - - - - - - 30 Hz to 10 kHz Input impedance - - - - - - - 75 ohms unbalanced; 600 ohms balanced Sensitivity - - - - - - - - - - To -90 dBm for midscale on meter
509	Amplifier	Frequency range - - - - - - - 100 Hz to 5 kHz Frequency response - - - - - Flat within 0.2 dB over frequency range Input impedance - - - - - - - 600 ohms Output impedance - - - - - - Equal to input impedance of Artificial Mouth, item 935. (15-20 ohm) Noise levels - - - - - - - - - -70 dBV at maximum output Harmonic distortion - - - - - Less than 0.2% total at 1 kHz Gain - - - - - - - - - - - - Sufficient to provide output sound pressure level from artificial mouth of 30 μbars minimum and 40 μbars maximum
510	Condenser Microphone	This should be calibrated in terms of dB relative to 1 V (open circuit) per microbar acoustic pressure. The free field corrections for both 0° and 90° incidence of the field should be known. There should be a protective grid.

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F. Test equipment for measurements, amplifiers (items 500-549; 5100-5149) – Continued

Item no.	Test unit	Minimum performance requirements	
511	Microphone Amplifier	Input impedance	700 megohms
		Input capacity	3 pF
		Output impedance	600 ohms
		Noise level	Approximately 40 μV with the input loaded by 20 pF
		Input voltage	20 V maximum
		Gain	0.9 dB ±0.2 dB

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G. Test equipment for measurements - power sources (items 550-599; 5150-5199)

Item no.	Test unit	Minimum performance requirements
565	Power Supply	Voltage - - - - - - - - - - - 0 to 100 V Current - - - - - - - - - - - 0 to 100 mA

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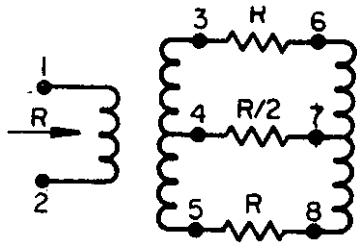
H. Test equipment for measurements, attenuators (items 600-649; 6100-6149)

Item no.	Test unit	Minimum performance requirements
600	Variable Attenuator	Range - - - - - - - - - - - - 0 to 60 dB in 0.1 dB steps Impedance - - - - - - - - - - 600 ohms balanced; 900 ohms balanced Attenuation accuracy - - - - - ±0.1 dB per 10 dB step Frequency range - - - - - - - DC to 10 kHz

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I. Test equipment for measurements, transformers (items 650-699; 6150-6199)

Item no.	Test unit	Minimum performance requirements
656	Hybrid Transformer	Frequency range - - - - - - - - 30 Hz to 10 kHz Turns ratio - - - - - - - - - - (1-2) : (3-5) = $\sqrt{2}$:1 (1-2) : (6-8) = $\sqrt{2}$:1 (3-4) : (4-5) = 1:1 (6-7) : (7-8) = 1:1 Impedance - - - - - - - - - - (1-2) = R = 600 ohms Insertion loss - - - - - - - - - When winding (1-2) is fed from a source impedance R, insertion loss from (1-2) to (3-6) or (1-2) to (5-8) = 4.0 dB ±0.5 dB. Transhybrid loss - - - - - - - - Greater than 50 dB between (1-2) and (4-7) when (1-2) is terminated in 600 ohms.
658	Transformer	Frequency range - - - - - - - - 200 Hz to 4 kHz Impedance ratio - - - - - - - - 1:1 (600 ohms:600 ohms) Secondary winding - - - - - - - Center-tapped with unbalance not greater than 1%
659	Transformer	Frequency range - - - - - - - - 200 to 4000 Hz Impedance ratio - - - - - - - - 1:1 (900 ohms:900 ohms) Secondary winding - - - - - - - Center-tapped with unbalance not greater than 1%



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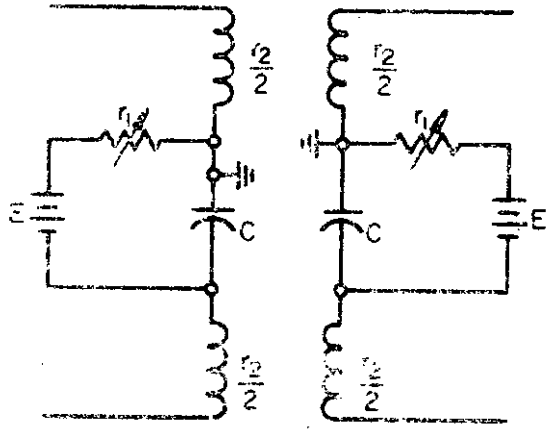
N. Test equipment for measurements, miscellaneous (items 900-999; 9100-9199)

Item no.	Test unit	Minimum performance requirements	
902	Switch	Function	2-pole, 2-position
		Frequency range	30 Hz to 120 kHz
904	Impedance Bridge	Frequency range	30 Hz to 10 kHz
		Parallel capacitance range	0 to 1 μF
		Parallel resistance range	10 to 2,000 ohms
		Measuring accuracy	±2%
935	Artificial Mouth and Associated Artificial Mouth Equalizer	Combined response	Flat-free field response for flat electrical input within ±2 dB relative to 1000 Hz over the frequency range 200/7000 Hz
		Impedance of equalizer	600 ohms
		Mouth output	40 μbars maximum
		Distortion	Less than 1% at 1 kHz and at level below maximum
		Other features	Acoustic output of artificial mouth should faithfully reproduce source voltage and there should be no spurious frequencies, rattles, or other evidence of overloading.

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N. Test equipment for measurements, miscellaneous (items 900-999; 9100-9199) - Continued

Item no.	Test unit	Minimum performance requirements
939	Artificial Line	Network simulating impedance characteristics of a 26-gauge cable pair - adjustable in steps not exceeding equivalent of 2000-foot conductor loop Minimum range - - - - - - - - 1500 feet
940	Feeding Bridge	Transmission circuit simulating switchboard cord or automatic switch circuit Voltage - - - - - - - - - - - - 48 ±2 V Current adjustment - - - - - - Rheostat to adjust line current to 100 mA on "zero" loop



E = 48V
C = 4μF
$r_1 + r_2 = 400\Omega$

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N. Test equipment for measurements, miscellaneous (items 900-999; 9100-9199) -- Continued

Item no.	Test unit	Minimum performance requirements	
941	Impedance Bridge	Frequency	1000 Hz
		Measurement range resistance	0 to 500 ohms
		Capacitance	0 to 2 μF
		Inductance	0 to 1,000 mH
		Special features	Must be capable of measuring impedance with dc flowing through impedance-dc power must be applied internally or externally
942	Switch (Manual Double Pole-Double Throw)	Voltage rating	120 V
		Current rating	100 mA
952	Impedance Bridge	Range of resistance	10 to 10,000 ohms
		Range of capacitance	100 pF to 1 μF
		Range of inductance	1 mH to 1 H
		Accuracy	±1% for R, C, and L
		Frequency range	50 Hz to 10 kHz
998	Sound Level Meter	Sound level range	24 to 150 dB (re 0.0002 μbar)
		Frequency characteristics	Weighting A, B, C, or 20 kHz
		Calibration	±1 dB at 400 Hz
		Meter range	16 dB

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